APPLICATION FOR FEDERAL ASSISTANCE
SF 424 (R&R)

3. DATE RECEIVED BY STATE | **State Application Identifier**

1. TYPE OF SUBMISSION*

❍ Pre-application ● Application ❍ Changed/Corrected Application

4.a. Federal Identifier

b. Agency Routing Number

2. DATE SUBMITTED
2020-10-13

Application Identifier

c. Previous Grants.gov Tracking Number

5. APPLICANT INFORMATION **Organizational DUNS*:** 608195277

Legal Name*: University of North Carolina at Chapel Hill
Department:
Division:
Street1*: 104 Airport Drive
Street2: Campus Box 1350
City*: Chapel Hill
County: Orange
State*: NC: North Carolina
Province:
Country*: USA: UNITED STATES
ZIP / Postal Code*: 27599-1350

Person to be contacted on matters involving this application
Prefix: First Name*: Andrea Middle Name: Last Name*: Cogdill Suffix:
Position/Title: Grants Analyst
Street1*: 130 Mason Farm Road
Street2: Bioinformatics Bldg., St. 1140
City*: Chapel Hill
County: Orange
State*: NC: North Carolina
Province:
Country*: USA: UNITED STATES
ZIP / Postal Code*: 27599-1350
Phone Number*: (919) 962-3950 Fax Number: Email: grants@unc.edu

6. EMPLOYER IDENTIFICATION NUMBER *(EIN) or (TIN)** 1-566001393-A1

7. TYPE OF APPLICANT* H: Public/State Controlled Institution of Higher Education

Other (Specify):
Small Business Organization Type ❍ Women Owned ❍ Socially and Economically Disadvantaged

8. TYPE OF APPLICATION*
● New ❍ Resubmission
❍ Renewal ❍ Continuation ❍ Revision

If Revision, mark appropriate box(es).
❍ A. Increase Award ❍ B. Decrease Award ❍ C. Increase Duration
❍ D. Decrease Duration ❍ E. Other *(specify)* :

Is this application being submitted to other agencies?* ❍Yes ●No What other Agencies?

9. NAME OF FEDERAL AGENCY*
National Institutes of Health

10. CATALOG OF FEDERAL DOMESTIC ASSISTANCE NUMBER
TITLE:

11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT*
Novel advances in stroke and arterial thrombus lysis through image guided cavitation with clot penetrating nanodroplets

12. PROPOSED PROJECT
Start Date* Ending Date*
07/01/2021 06/30/2025

13. CONGRESSIONAL DISTRICTS OF APPLICANT
NC-004

SF 424 (R&R) APPLICATION FOR FEDERAL ASSISTANCE

14. PROJECT DIRECTOR/PRINCIPAL INVESTIGATOR CONTACT INFORMATION

Prefix: Dr. First Name*: Paul Middle Name: A Last Name*: Dayton Suffix: PhD

Position/Title: William R. Kenan, Jr. Distinguished Professor
Organization Name*: University of North Carolina at Chapel Hill
Department: Biomedical Engineering
Division:
Street1*: 116 Manning Drive, 9018A Mary Ellen Jones Bldg
Street2: Campus Box 7575
City*: Chapel Hill
County: Orange
State*: NC: North Carolina
Province:
Country*: USA: UNITED STATES
ZIP / Postal Code*: 27599-7575
Phone Number*: (919) 843-9521 Fax Number: Email*: padayton@email.unc.edu

15. ESTIMATED PROJECT FUNDING

a. Total Federal Funds Requested*	$3,022,187.00
b. Total Non-Federal Funds*	$0.00
c. Total Federal & Non-Federal Funds*	$3,022,187.00
d. Estimated Program Income*	$0.00

16.IS APPLICATION SUBJECT TO REVIEW BY STATE EXECUTIVE ORDER 12372 PROCESS?*

a. YES ❍ THIS PREAPPLICATION/APPLICATION WAS MADE AVAILABLE TO THE STATE EXECUTIVE ORDER 12372 PROCESS FOR REVIEW ON:

DATE:

b. NO ● PROGRAM IS NOT COVERED BY E.O. 12372; OR

❍ PROGRAM HAS NOT BEEN SELECTED BY STATE FOR REVIEW

17. By signing this application, I certify (1) to the statements contained in the list of certifications* and (2) that the statements herein are true, complete and accurate to the best of my knowledge. I also provide the required assurances * and agree to comply with any resulting terms if I accept an award. I am aware that any false, fictitious, or fraudulent statements or claims may subject me to criminal, civil, or administrative penalties. (U.S. Code, Title 18, Section 1001)

● I agree*

** The list of certifications and assurances, or an Internet site where you may obtain this list, is contained in the announcement or agency specific instructions.*

18. SFLLL or OTHER EXPLANATORY DOCUMENTATION File Name:

19. AUTHORIZED REPRESENTATIVE

Prefix: Dr. First Name*: Terry Middle Name: R Last Name*: Magnuson Suffix: PhD

Position/Title*: Vice Chancellor for Research
Organization Name*: University of North Carolina at Chapel Hill
Department: Office of Sponsored Research
Division:
Street1*: 104 Airport Dr. Ste. 2200
Street2: Campus Box 1350
City*: Chapel Hill
County: Orange
State*: NC: North Carolina
Province:
Country*: USA: UNITED STATES
ZIP / Postal Code*: 27599-1350
Phone Number*: (919) 966-3411 Fax Number: (919) 962-5011 Email*: resadminosr@unc.edu

Signature of Authorized Representative*	Date Signed*
Terry R Magnuson	10/13/2020

20. PRE-APPLICATION File Name:

21. COVER LETTER ATTACHMENT File Name:Cover_Letter1046094907.pdf

424 R&R and PHS-398 Specific
Table Of Contents

SF 424 R&R Cover Page 1
Table of Contents 3
Performance Sites 4
Research & Related Other Project Information 5
Project Summary/Abstract(Description) 6
Project Narrative 7
Facilities & Other Resources 8
Equipment 12
Research & Related Senior/Key Person 15
Research & Related Budget Year - 1 48
Research & Related Budget Year - 2 51
Research & Related Budget Year - 3 54
Research & Related Budget Year - 4 57
Budget Justification 60
Research & Related Cumulative Budget 63
PHS398 Cover Page Supplement 64
PHS 398 Research Plan 66
Specific Aims 67
Research Strategy 68
PHS Human Subjects and Clinical Trials Information 80
Study 1: Novel advances in stroke and arterial thrombus lysis through image guided cavitati on with clot penetrating nanodroplets 82
Inclusion Enrollment Reports 88
Vertebrate Animals 95
Multiple PD/PI Leadership Plan 97
Bibliography & References Cited 98
Letters of Support 102
Authentication of Key Biological and/or Chemical Resources 104

OMB Number: 4040-0010
Expiration Date: 12/31/2022

Project/Performance Site Location(s)

Project/Performance Site Primary Location

❍ I am submitting an application as an individual, and not on behalf of a company, state, local or tribal government, academia, or other type of organization.

Organization Name:	University of North Carolina at Chapel Hill
Duns Number:	608195277
Street1*:	116 Manning Drive, 9018A Mary Ellen Jones Bldg
Street2:	Campus Box 7575
City*:	Chapel Hill
County:	Orange
State*:	NC: North Carolina
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	27599-7575
Project/Performance Site Congressional District*:	NC-004

Project/Performance Site Location 1

❍ I am submitting an application as an individual, and not on behalf of a company, state, local or tribal government, academia, or other type of organization.

Organization Name:	North Carolina State University
DUNS Number:	042092122
Street1*:	1840 Entrepreneur Dr; EBIII 3282
Street2:	Campus Box 7910
City*:	Raleigh
County:	Wake
State*:	NC: North Carolina
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	27695-7910
Project/Performance Site Congressional District*:	NC-004

Additional Location(s) File Name:

Tracking Number: GRANT13224045
Funding Opportunity Number: PAR-19-158. Received Date: 2020-10-13T11:12:56.000-04:00

OMB Number: 4040-0010
Expiration Date: 12/31/2022

RESEARCH & RELATED Other Project Information

1. Are Human Subjects Involved?* ● **Yes** ❍ **No**

1.a. If YES to Human Subjects

Is the Project Exempt from Federal regulations? ❍ Yes ● No

If YES, check appropriate exemption number: _ 1 _ 2 _ 3 _ 4 _ 5 _ 6 _ 7 _ 8

If NO, is the IRB review Pending? ● **Yes** ❍ **No**

IRB Approval Date:

Human Subject Assurance Number 00004801

2. Are Vertebrate Animals Used?* ● **Yes** ❍ **No**

2.a. If YES to Vertebrate Animals

Is the IACUC review Pending? ● Yes ❍ No

IACUC Approval Date:

Animal Welfare Assurance Number A3410-01

3. Is proprietary/privileged information included in the application?* ❍ **Yes** ● **No**

4.a. Does this project have an actual or potential impact - positive or negative - on the environment?* ❍ **Yes** ● **No**

4.b. If yes, please explain:

4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ❍ Yes ❍ No

4.d. If yes, please explain:

5. Is the research performance site designated, or eligible to be designated, as a historic place?* ❍ **Yes** ● **No**

5.a. If yes, please explain:

6. Does this project involve activities outside the United States or partnership with international collaborators?* ❍ **Yes** ● **No**

6.a. If yes, identify countries:

6.b. Optional Explanation:

	Filename
7. Project Summary/Abstract*	Abstract1046094848.pdf
8. Project Narrative*	Narrative1046094867.pdf
9. Bibliography & References Cited	References1046185396.pdf
10.Facilities & Other Resources	Facilities1046094859.pdf
11.Equipment	Equipment1046185737.pdf

Tracking Number: GRANT13224045

Funding Opportunity Number: PAR-19-158. Received Date: 2020-10-13T11:12:56.000-04:00

PROJECT SUMMARY

Arterial thrombosis, including ischemic stroke and myocardial infarction, remains a leading cause of death in the United States and the world. In particular, ischemic stroke is the fifth leading cause of death in the United States, and a major cause of disability in non-fatal cases. The only FDA approved non-surgical treatment for ischemic stroke is the administration of recombinant tissue-plasminogen activator (rt-PA). However, timeliness of rt-PA treatment is crucial: effectiveness of this therapy declines rapidly within hours, while prolonged ischemia worsens tissue damage.

The application of ultrasound for thrombus disruption, termed sonothrombolysis, has been studied for decades. Although this approach effectively breaks up thrombi under certain conditions such as fresh thrombus, and has demonstrated clinical efficacy in a small subset of cases, it has failed to be fully realized as an impactful therapy during several large clinical trials. One of the hypotheses for the failure of sonothrombolysis is that aged thrombi undergo platelet-mediated contraction (consolidation), making them dense and resistant to fibrinolysis, and performance of ultrasound with rt-PA alone has been shown to be very limited in these denser clots. The amplification of cavitation through the addition of microbubble contrast agents mechanically enhances clot disruption, but has still not demonstrated efficacy in treating older, contracted and denser clots, and therefore has not demonstrated the improvements needed for clinical impact.

Our team proposes a novel approach to sonothrombolysis for arterial thrombosis, which replaces microbubbles with novel low-boiling point phase change nanodroplets, invented by MPI Dayton. Phase-change nanodroplets are hundred-nanometer, liquid perfluorocarbon filled nanoparticles which convert to gas-filled microbubbles upon excitation at low-acoustic energies. Our preliminary studies have shown a drastic improvement in treatment of clots with nanodroplets – our nanoparticles diffuse into areas of the clot inaccessible to microbubbles and cavitate in the acoustic field; and preliminary data demonstrates fast and efficient clot lysis even in aged and contracted clots. Ultrasound feedback can provide image guidance with overlay confirming active cavitation at the clot site. Our studies show a novel mechanism of action – clot breakup initiates internally, which results in dissolution in dense clots unachievable with ultrasound alone or microbubbles even with rt-PA. Due to this novel ‘permeation and internal disruption’ mechanism, we hypothesize that this may be a ‘game changer’ with stroke and other arterial thrombus treatments. We have assembled a world-class team to test this hypothesis. Our approach will consist of studying and optimizing nanodroplet formulation, including modification for improved clot permeation and retention, and optimizing acoustic parameters. We will test studies first in bovine blood, and then in human blood, as well as in animal models. Successful results will prepare the way for large animal studies and early clinical trials.

RELEVANCE TO PUBLIC HEALTH

Arterial thrombosis, including ischemic stroke and myocardial infarction, remains a leading cause of death in the United States and the world. Our team proposes a novel approach to sonothrombolysis for arterial thrombosis, which utilizes novel low-boiling point phase change nanodroplets. Our preliminary studies have shown a drastic improvement in treatment of aged clots with nanodroplets compared to standard of care, and successful results will prepare the way for large animal studies and early clinical trials.

FACILITIES AND OTHER RESOURCES

The UNC-NC State Joint Department of Biomedical Engineering is a single department that encompasses labs and faculty across The University of North Carolina at Chapel Hill and North Carolina State University at Raleigh, which are located very close to each other as part of the Research Triangle in North Carolina. There are no subcontracts as these two campuses work with BME as one single entity.

Facilities at The University of North Carolina at Chapel Hill

Dr. Dayton is a member of the Department of Biomedical Engineering, Eshelman School of Pharmacy, and Lineberger Comprehensive Cancer Center. His laboratories occupy ~2000 ft^2 in the newly renovated Mary Ellen Jones building on the UNC Chapel Hill Campus, and contain equipment and facilities for small animal ultrasound imaging and therapy, contrast agent manufacture, ultrasonic testing, and image and data processing.

The **Small Animal Ultrasound Core**, part of the Dayton Lab, includes four isoflurane veterinary anesthesia systems, a SonoVol Vega system, a Visualsonics Vevo 770 ultrasound system, a Visualsonics Vevo 2100 ultrasound system, two Acuson Sequoia 512 ultrasound systems with Cadence contrast imaging software, a Boston Scientific Galaxy Ultrasound System, a Volcano IVUS system, two Verasonics 256 channel programmable ultrasound systems, a Philips TIPS therapeutic ultrasound system, and computers for data processing and analysis.

Shared laboratory space in 9013 Mary Ellen Jones for tissue culture includes class IIA laminar clean hoods, incubators, an autoclave, and a tissue culture microscope.

Dr. Dayton's Contrast Agent Formulation Lab contains a chemical fume hood, laminar clean hood, centrifuge, sonicators, vortexer, freezers and refrigerators, two inverted and one upright fluorescence microscopes with z-stacking capability, two Photron high-speed camera systems, particle sizers, adhesion assay systems, and other general equipment for contrast agent formulation and analysis.

The Ultrasonic Testing Lab contains ultrasonic pulsers and receivers, transducers, digital oscilloscopes, waveform generators, a 3-axis motion system for beam field characterization, calibrated hydrophones, 14-bit 100 MHz A-D systems, and other technology for imaging system development.

Dr. Dayton's laboratory has three Assistant Research Professors, three post-docs, eight PhD students, two technicians and several undergraduate research assistants. Each lab member has a desk and a computer available for word and image processing.

SonoVol Vega and Strata small animal ultrasound system have been loaned by SonoVol to UNC, and are available. They are located in the Small Animal Ultrasound Core, part of the Dayton lab space.

Biostatistics Shared Resource: The principal objective of the Biostatistics Shared Resource (BIOS Core) is to provide high quality statistical consultation services to UNC's Lineberger Comprehensive Cancer Center (LCCC) members. The BIOS Core provides a full collaborative scientific resource focused on providing LCCC members support for the power calculations, design and analyses, and generation of manuscripts for their research.

UNC Lineberger Comprehensive Cancer Center: The University of North Carolina (UNC) and Lineberger Comprehensive Cancer Center (LCCC) have a long-standing dedication to advancing clinical care through scientific research. The UNC LCCC, founded in 1975, is one of 41 National Cancer Institute (NCI) designated comprehensive cancer centers in the country. Rated as "exceptional" at its last CCSG review in 2015, UNC Lineberger members hold more than $69 million in NCI funding (7th nationally in 2014). A significant infrastructure for animal models, genomics, proteomics, bioinformatics, small animal imaging, immune system assessment and translational research makes LCCC a Cancer Center that fosters innovation and high impact. The equipment and facilities relevant to this application and available to Dr. Dayton include the Division of Laboratory Animal Medicine, Animal Studies Facility, Confocal and Multiphoton Imaging Facility, Small Animal Imaging, Tissue Culture and Molecular Biology Support, Flow Cytometry Facility, and Animal Histopathology Core among others. A wide variety of equipment and instrumentation is also readily available for the applicant through the Centralized Lab Services Core. Importantly, LCCC provides services by the Biostatistics Shared Resource (BIOS Core) and Bioinformatics Core, which are full collaborative scientific resources focused on providing LCCC members support for the design, conduct and analysis of genomics research projects.

The UNC Flow Cytometry Core Facility: This facility provides state-of-the-art flow cytometry analysis instrumentation and cell sorting services and offers analytic flow cytometry utilizing Cytek-modified 5-color

FACSCalibur, 8-color Stratedigm S1000Ex, 18-color BD LSR II and three 9-color Dako CyAns. BSL1 sorting is provided by a Dako MoFlo and a Dako MoFlo XDP. BSL-2 sorting is available on an iCyt Reflection and a BD FACS Aria II. Skilled staff provides help with instrument setup, data analysis, and consultation for experiment design.

Animal Histopathology Core Facility: This facility is run by a board-certified veterinary pathologist and offers high-quality preparation and processing of frozen and fixed tissue sections, along with education regarding appropriate harvesting, fixation, immunology, and selection of histological procedures.

Small Animal Imaging: The Biomedical Research Imaging Center (BRIC) across the street from the Dayton labs at UNC includes ~ 5000 square feet of small animal imaging, procedure, and storage facilities, available on a recharge basis. The Small Animal Imaging Core, part of the BRIC (separate from Dr. Dayton's ultrasound core), has a Bruker 9.4T 30cm bore MRI, GE eXplore Vista PET/CT, GE eXplore speCZT SPECT/CT, several bioluminescence/fluorescence IVIS Kinetic and Lumina systems and an additional Visualsonics 2100 ultrasound system available for use at core recharge rates

Large Animal Imaging and HIFU: The Biomedical Research Imaging Center has world class large-bore human and large animal research systems, dedicated for research use. These include a Siemens clinical 3T Magnetom Prisma MRI, Siemens Biograph mMR PET/MR, Siemens Biograph mCT PET/CT, and a Siemens Magnetom 7T MRI system. A Focused Ultrasound Instruments LP100 MR-guided HIFU system can be utilized with the Magnetom 7T or 3T for large animal use, available on a recharge basis.

Tissue Culture Facility: The Tissue Culture Facility (TCF) is the UNC Lineberger's most comprehensive and widely used core facility. The TCF is a full-service cell culture shared resource and provides Cancer Center members and UNC colleagues with a wide variety of services and products. The TCF stocks media, media supplements, serum, salt solutions, antibiotics, cell culture flasks and dishes and many other in vitro cell culture-related items. The TCF offers a wide range of cell culture services from frozen vials and starter cultures to large-scale suspension systems of over 75 liters, specialized multiple-layer units for anchorage cells, and hollow-fiber bioreactors for monoclonal antibody or other cell metabolite production. Anchorage and suspension-type cells can be grown in most any size or volume to meet the researcher's needs. The TCF offers individual consultation, training sessions, and training materials in cell and tissue culture techniques, including media and systems for in vitro propagation, testing, and cell cryopreservation. The staff is available for help in developmental work as well as cell culture problem solving.

Animals: Dr. Dayton's animal work is conducted in fully accredited animal facilities in Genetic Medicine Building (GMB). Animal facility in GMB includes a transgenic/knockout core; as well as two Cesium and two X-ray irradiators and air filtered cubicle to house animals for the transplantation studies described in this application. The facility also has instrumentation for bioluminescence imaging (BLI) and ultrasound imaging of tumors. The facility is an excellent NIH approved facility for the care and housing of small animals. The institution follows all guidelines established by the NIH for the care of animals and is committed to following the standards established by the Animal Welfare Acts and Principals of Use of Animals and Guide for Care and Use of Laboratory Animals. All protocols are reviewed annually by an internal animal use and welfare committee. The **Animal Studies Core** facility will help facilitate the majority of these experiments. They are a full service core that provides the following services at a cost of $35/hour: Orthotopic Injections, Laboratory Coordinator Services, Tumor Passage, Tissue/Blood collection, Injections, Colony Management, Aseptic Surgery, Drug Dosing/Oral Gavage, Intra-cardiac Injections, IVIS Imaging, Animal Identification, and Protocol Assistance. This core is directed by Dr. Bernard Weissman and Charlene Santos.

Clinical Resources: The University of North Carolina Hospitals is the principal venue for clinical/translational research activities at UNC-CH. The University of North Carolina Hospitals at Chapel Hill now includes: North Carolina Memorial Hospital, North Carolina Clinical Cancer Center, North Carolina Neurosciences Hospital, North Carolina Women's Hospital, North Carolina Children's Hospital and North Carolina Cancer Hospital. As North Carolina's only state-owned acute care facility, UNC Hospitals continues to serve as the primary referral center for each of the 100 counties in the state. The 771 attending physicians plus a house staff comprising of 672 residents and postdoctoral fellows provide clinical care in the UNC Health Care System.

The North Carolina Cancer Hospital opened its doors in August 2009. The N.C. Cancer Hospital is 315,000 ft^2, which is more than 3 times larger than previous space in the UNC Gravely clinic. The total project costs were $207,000,000 - $178M for the N.C. Cancer Hospital and $29M allocated for the Physician's Office Building component. The State of North Carolina provided $180M for the project. Multi-disciplinary and other clinic space

includes 101 examination, treatment, consultation, and procedure rooms. To facilitate the multidisciplinary care of patients and for tumor board conferences for patients seen throughout NC, there is a dedicated Conference Center on the first floor of the Cancer Hospital with high definition video-conferencing capabilities. To facilitate and expand clinical trial capabilities, there is a dedicated Clinical Trials Unit (CTU) with specially trained personnel. The inpatient facility includes 50 inpatient beds for medical oncology and bone marrow transplant (an increase of 15 beds over preexisting). The outpatient facility includes 72 infusion stations for adults (48), children (14), and clinical trials (10) (more than double the preexisting pediatric stations and triple the preexisting adult stations).

Facilities at North Carolina State University

Dr. Jiang's Micro/Nano Engineering Laboratory (MNEL) occupies ~ 800 ft^2 in Engineering Building III on the North Carolina State University at Raleigh, and contains equipment and facilities for electromechanical device design and modeling; precision lapping and polishing; sensors, actuators and transducers assembly; piezoelectric materials and devices characterization; ultrasonic testing; vibration measurements; actuator and sensor testing; and precision charge measurements. Dr. Jiang's group currently has 3 postdocs, 8 PhD students and 3 undergraduate students.

Other available facilities (NCSU): The PI has access to the equipment and facilities in the following NCSU facilities: (1) *Nanofab facility at Centennial Campus at North Carolina State University*. The Nanofab has a complete list of equipment for micro/nanofabrication including Lithography, Chemical Vapor Deposition, Physical Vapor Deposition, Furnace Processes, Rapid Thermal Annealing, Plasma Etching, Chemical Hoods, and Metrology; (2) North Carolina Supercomputing Center (NCSC), Research Triangle Park, NC. The Center provides unlimited computing access to faculties and student for research and teaching; (3) The NCSU *Analytical Instrumentation Facility (AIF)*: a North Carolina State University resource for materials characterization. AIF operates a number of major analytical instruments and numerous other instruments including VPSEM, FESEM, FIB, TEM, SPM, SIMS, TOFSIMS, XPS, Microtomy, Metallography, and Prolifometry. (4) The Department of Mechanical and Aerospace Engineering has a full-service and wellequipped machine shop in which can manufacture small and large-scale steel fabrication, specialized equipment, precision models, etc.

Dr. Brown has a primary appointment in the Joint Department of Biomedical Engineering at North Carolina State University (NSCU) and the University of North Carolina at Chapel Hill (UNC). This department brings together the exceptional strengths and national reputations of the College of Arts & Sciences and the School of Medicine at UNC-Chapel and the College of Engineering and the Veterinary School at NCSU. The joining of these top-rated institutions provides a fertile environment for biomedical engineers to develop innovative and outstanding opportunities in the pursuit of important biomedical problems facing our society. Faculty members have complete access to facilities and opportunities at both universities. Importantly for this project, as faculty member at UNC, Dr. Brown regularly interacts with faculty members at the McAllister Heart Institute at UNC, including the director of the Small Animal Imaging Core at UNC, Dr. Brian Cooley. The McAllister Heart Institute provides a world class environment for basic, preclinical, and applied cardiovascular research. Additionally, at UNC, a number of clinically based core facilities are available to BME facility members, including the Small Animal Imaging Core, the Clinical and Translational Research Center, the Cytokine Analysis Facility, the Tissue Procurement Facility, the Translational Pathology Laboratory, and the UNC CFAR Clinical Pharmacology and Analytical Chemistry Core.

At NCSU, Dr. Brown has access to all BME core facilities, including the Microbiology Suite, Cell Biology Lab, Wet Lab, and Histology Room. Additional university administered facilities available to researchers at NCSU include the Additive Manufacturing Facility, the Genomic Science Laboratory, and the Cellular and Molecular Imaging Facility. Of particular interest to development of translation materials, NCSU has a nationally ranked Veterinary School, with GMP facilities. Collaboration with the Veterinary School provides biomedical engineers rich opportunities to translate materials into viable products. Core facilities at the Veterinary School include a Biomedical Imaging Resource Facility, a BSL3 Biocontainment Facility, a Clinical Trials Center, Media and Design, a Flow Cytometry and Cell Sorting Laboratory, Laboratory Animal Resources, and a Laboratory for Advanced Electron and Light Optical Methods. Additional core facilities available to researchers at NCSU include Sequencing and Library Preparation, Bioinformatics and Statistical Consulting, High-Performance Computing, NMR, X-Ray Crystallography, Mass Spectrometry, Scanning Microscopy, Dynamic Light Scattering, Surface Science, X-Ray Diffraction, Nanofabrication, wide-field and confocal imaging, laser imaging and vibrational spectroscopy, electron microscopy, small and large animal imaging, and flow cytometery.

Laboratory: Dr. Brown's Laboratory occupies approximately 700 sq. ft. on the second floor of the recently opened Biomedical Partnership Building at the Veterinary School Campus at NCSU and consists of a main wet lab with a fume hood and a separate tissue culture room.

Animal: Animal work at NCSU will be conducted at the nationally ranked NC State College of Veterinary Medicine (CVM) campus. The CMV campus includes operating and procedure rooms, portable isoflurane units, a full-time animal husbandry staff and attending veterinarian, the Biomedical Imaging Resource Facility, a BSL3 Biocontainment Facility, a Clinical Trials Center, Media and Design, a Flow Cytometry and Cell Sorting Laboratory, Laboratory Animal Resources, and a Laboratory for Advanced Electron and Light Optical Methods. Dr. Brown's laboratory is located on the CVM campus; this proximity to world class veterinary resources primes this project for success.

Computer: Dr. Brown has one personal MacBook computer for word processing, presentation construction, literature searching, and data analysis. Four additional computers are available in the lab for equipment control and bulk data processing and analysis.

Office: Student and Post-Doc office space (approximately 400 sq. ft.) is available immediately adjacent to Dr. Brown's office. Dr. Brown's office (approximately 200 sq. ft.) is located on the second floor of the Biomedical Partnership Building.

MAJOR EQUIPMENT

Dr. Paul Dayton's laboratory (available 100%)

Ultrasonics: SonoVol Vega small animal 3-D robotic ultrasound imaging system with image-guided HIFU upgrade, Verasonics 256 channel programmable ultrasound systems, low and high frequency range systems,Verasonics multi-system synchronization box, Verasonics 1024 channel programmable ultrasound systems (custom build), Acuson Sequoia S512 Ultrasound System, Acuson Sequoia C512 Ultrasound Systems, Visualsonics Vevo 2100 ultrasound system, Visualsonics Vevo 770 ultrasound system (2), Boston Scientific Galaxy intravascular ultrasound system, Volcano IVG3 phased array intravascular ultrasound system, Philips TIPS ultrasound system, Ultrasound Instruments LP-100 computer controlled HIFU system, 3-D computer controlled water tank hydrophone system, Onda 400 micron and 200 micron calibrated needle hydrophones, Assorted Siemens, Philips, Visualsonics, Vermon, and Verasonics linear and phased array transducers, Newport high-precision 3-axis computer controlled positioning systems (2)

High-speed and video microscopy: Photron APX-RS ultra-high speed camera, Photron S1 ultra-high speed camera, Olympus IX-71 inverted microscope with custom microbubble positioning systems, Olympus BX-51 upright fluorescence microscope, various water immersion and long-working distance objectives, Narishige Micromanipulators (4)

Biological: Thermo Class IIa Cell culture hood, Thermo water jacket CO2 incubators (2), Assorted centrifuges

Particle Analysis: Accusizer 780 particle sizing system, Accusizer Nano-FX sub-micron particle sizing system, Thermo Nanodrop spectrophotometer, Tecan Plate Reader, Coulter Multizer

Pre-clinical: Vetmed IsoFlurane Anesthesia System (4), Visualsonics respiration and pulse monitoring systems (4)

Dr. Wolberg's laboratory (available 100%)

Major equipment includes two UV-visible automated kinetic microplate readers (Molecular Devices Spectramax Plus 384), two Fluoroskan Ascent® fluorometers and software for calibrated automated thrombography (Thrombinoscope), three TEG® 5000 Hemostasis Analyzers (Haemonetics), a chromatography system (BioRad NGC™ Quest 10 system with BioFrac fraction collector), a BioRad gel electrophoresis and western blotting system, a Pharmacia Phastgel gel electrophoresis and Western blotting system, Nanodrop spectrophotometer, refrigerated centrifuges (Beckman Coulter, Allegra 6R), two CO_2 tissue culture incubators (Sanyo), laminar flow hood (Baker), 3 -20°C freezers, two 70 ft^3 ultra-low freezers (Revco), refrigerators, and other general laboratory equipment. Dr. Wolberg also has an Eagle Eye Model 150 anesthesia machine and dedicated bench space for mouse surgical procedures. A gel and western blot imaging system (Li-Cor Biosciences Odyssey Fc Imaging System) is shared equipment within the Department. An STP 120 tissue processor (Thermo Scientific), a Histostar embedding station (Thermo Scientific), an AccuCut SRM 300LT microtome (Sakura), a ChemiDoc MP gel/blot imaging system (Biorad), a Verti 96 well thermal cycler (Applied Biosystems), and a StepOne Plus Real-time PCR system (Applied Biosystems) are shared with Dr. Matt Flick, who shares research space with Dr. Wolberg in Mary Ellen Jones Building.

Dr. Jiang's laboratory (available 50%)

Key equipment which has been installed or purchased include:

Design and modeling: Eight desktop and two laptop computers equipped with design, modeling and data acquisition software including AUTOCAD, ANSYS, MATLAB, SIMULINK, Field II, Labview and office software.

Processing and assembly: a Logitech PM5 precision lapper and polisher, a Heidenhain CT6001 length gauge (sub-micron resolution), Olympus ST-6 measurement microscope (sub-micron resolution), 2 celestron stereomicroscopes, precision irons for wire attachments, Al AccuTemp vacuum drying oven.

Characterization: Brucker Nano AFM/PFM imaging system (Dimension Icon), Polytec OFV-534 Laser Vibrometer, Agilent HP 4294 impedance analyzer, Agilent HP 4284 impedance analyzer, Agilent D7104B 1GHz digital oscilloscope, Olympus 5900PR pulser/receiver, Olympus 5077PR pulser/receiver, SRS PS350 DC power supply (5000V, 25 W), ENI 3100L RF power amplifier, AR 75A250A RF power amplifier, JZ-4B d33 meter, Tektronix AFG3101 waveform generator, Ohmic UPM-DT-1AV acoustic powermeter (10 mW resolution), SRS 830 lock-in amplifier, Onda HGL-0085 hydrophone, B&K 2706 power amplifier, KH 7602M wideband amplifier,

KCF V201 vibration shaker, MTI GSL1100X high temperature furnace, a Newport RS2000 Table Top W/Tuned Damping (4 ft x 8 ft x 8 in) and a Newport ATS overhead table shelf system, and one 18 in x 26 in x 22 in water tank with X-Y-Z-Theta scanning stages.

Dr. Brown's laboratory (available 50%)

Dr. Brown's laboratory is equipped with an Asylum MFP3D-Bio AFM, a Malvern Nanosight NS300 and an EVOS FL Auto Cell Imaging System with Onstage Incubator. Additionally, Dr. Brown, as a part of the Dept. of Biomedical Engineering, has full access to BME Culture Core equipment within Engineering Building III. The Microbiology suite (700+ square feet) contains a chemical fume hood, two flammable storage cabinets, a water deionization system, a Nuaire model 425-600 6ft class II type A2 biological safety cabinet, a Nuaire model NU-5510 incubator with CO2 option and sanitize cycle, AccuSpin Micro17 centrifuge, Denver Instrument XP-300 top loading and TP-64 analytical balances, New Brunswick Scientific I Series 24 rotary shaker/incubator, Sorvall St 40R benchtop refrigerated centrifuge, and an Isotemp -80°C freezer. The Cell Biology Core (236 square feet) contains a Leica DMIL inverted fluorescence microscope with heated stage and QImaging Micropublisher 3.3 cooled RTV CCD color camera, a Nuaire model 4750D CO2 incubator, a Nuaire model NU-425-600 class II type A2 6 ft. biological safety cabinet, Tecan GENios microplate reader, and a liquid nitrogen storage unit. The Histology core contains a Fisher Histo Center paraffin tissue processor, a Leitz 1512 rotary microtome, a Thermo Scientific HN 525NX Cryostat, and a Leica EZ4D stereo zoom microscope. Additionally, available at no cost to BME researchers in the Engineering III Building are a 4°C walk-in cold room, an autoclave, an ethylene oxide sterilizer, and ice machines. As a member of the Comparative Medicine Institute, Dr. Brown also has access to facilities at the nationally ranked NC State College of Veterinary Medicine (CVM) campus. The CMV campus includes a Biomedical Imaging Resource Facility, a BSL3 Biocontainment Facility, a Clinical Trials Center, Media and Design, a Flow Cytometry and Cell Sorting Laboratory, Laboratory Animal Resources, and a Laboratory for Advanced Electron and Light Optical Methods.

Other Relevant Campus Resources

In the **UNC Microscopy Services Lab**, a 5-minute walk from Dr. Wolberg and Dayton's laboratories: 3 confocal laser scanning microscopes: a Zeiss Pascal with multiline argon (458, 488, 514 nm) and green HeNe lasers; an Olympus FV500 with multiline argon, green HeNe and red HeNe lasers; and a Zeiss LSM 710 multispectral system with 405 nm laser, multiline argon, green HeNe, orange HeNe, and red HeNe lasers. These systems are all capable of simultaneous and sequential acquisition of multi-channel data. The Zeiss LSM 710 is equipped for spectral deconvolution. All of these systems are also equipped for bright field, DIC and standard epi-fluorescence imaging. The research level transmission electron microscope is a 120 kV Zeiss EM 910 capable of 0.4 nm resolution. It is equipped with a eucentric goniometer stage computer controlled in x, y, z, and tilt, and a Gatan SC1000 digital camera. The Microscopy Services Lab also offers light and scanning electron microscopy services.

In the **UNC Animal Surgery Core Lab**, a 5-minute walk from Dr. Wolberg and Dayton's laboratories: 6 operating microscopes, 2 light microscopes (including a Leica SPE TCS scanning confocal microscope), an extensive variety of microsurgical instrumentation, a fume hood, thermal cyclers, electrophoresis equipment, water baths, autoclave and refrigerator/freezers. The existing intravital imaging system includes C-mounts for digital camera attachment a DVC 1412 low-light, infrared digital video camera, several lasers of various wavelengths (405, 473, 532, 594, 650, 670, and 780 nm wavelengths), a shutter with shutter driver, filter wheel and 3 beam expanders. This current in vivo fluorescence imaging system is used for live imaging of thrombosis models using laser excitation for fluorescence image capture. This imaging system is housed and run in a dedicated 200 ft^2 room under Dr. Cooley's direction.

In the **UNC Animal Histopathology & Laboratory Medicine (AHLM) Core**, a 5-minute walk from Dr. Wolberg and Dayton's laboratories: 2 Leica tissue processors, a Leica autostainer, a Leica coverslipper, a Leica Bond Max immunostainer, a Ventana Discovery Ultra immunostainer, 2 Leica histoembedders, a Thermo Scientific Cryostat, 5 Leica microtomes, A Milestone BoneSTATION decalcification system, Alfa Wasserman Vet Axcel® Chemistry System, IDEXX ProCyte DxTM Hematology Analyzer, Luminex MAGPIX system using magnetic bead-based multi-analytes, BioTek automated microplate washer, and numerous incubators, water baths, and microscopes. This equipment allows the AHLM to provide investigators high-quality, customizable pathology at a convenient on-campus location with competitive pricing and a short turnaround time. Currently, the AHLM employs three full-time and one part-time histotechnologists experienced in routine histopathologic techniques.

Through The Biomedical Research Imaging Center

Bruker 30cm bore 9.4T small animal MRI, Visualsonics Vevo 2100 Ultrasound imaging systems, Xenogen IVIS 100, GE eXplore Vista micro PET/CT, GE eXplore speCZT system micro SPECT/CT, SCANCO µCT 40, iCyt Reflection, Siemens Biograph mCT clinical bore PET/CT, Siemens Biograph mMR clinical bore PET/MR, Siemens Prisma 3T MRI, Siemens Magnetom 7T MRI, GE PETrace 880 cyclotron and radiochemistry facilities, ABT Biomarker Generator cyclotron

OMB Number: 4040-0010
Expiration Date: 12/31/2022

RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator

Prefix: Dr. First Name*: Paul Middle Name A Last Name*: Dayton Suffix: PhD

Position/Title*: William R. Kenan, Jr. Distinguished Professor
Organization Name*: University of North Carolina at Chapel Hill
Department: Biomedical Engineering
Division:
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Street2: Campus Box 7575
City*: Chapel Hill
County: Orange
State*: NC: North Carolina
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Zip / Postal Code*: 27599-7575

Phone Number*: (919) 843-9521 Fax Number:

E-Mail*: padayton@email.unc.edu

Credential, e.g., agency login: PADAYTONNCSU

Project Role*: PD/PI Other Project Role Category:

Degree Type: PhD Degree Year: 2001

Attach Biographical Sketch*: File Name: Biosketch_Dayton1046185242.pdf

Attach Current & Pending Support: File Name:

Tracking Number: GRANT13224045
Funding Opportunity Number: PAR-19-158 . Received Date: 2020-10-13T11:12:56.000-04:00

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Alisa Middle Name S Last Name*: Wolberg Suffix: PhD

Position/Title*: Professor
Organization Name*: University of North Carolina at Chapel Hill
Department: Pathology & Lab Medicine
Division:
Street1*: 116 Manning Drive , 8018A Mary Ellen Jones Bldg
Street2: Campus Box 7035
City*: Chapel Hill
County: Orange
State*: NC: North Carolina
Province:
Country*: USA: UNITED STATES
Zip / Postal Code*: 27599-7035

Phone Number*: (919)962-8943 Fax Number:

E-Mail*: alisa_wolberg@med.unc.edu

Credential, e.g., agency login: ALISA_WOLBERG

Project Role*: PD/PI Other Project Role Category:

Degree Type: PhD Degree Year: 1996

Attach Biographical Sketch*: File Name: Biosketch_Wolberg1046094993.pdf

Attach Current & Pending Support: File Name:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Xiaoning Middle Name Last Name*: Jiang Suffix: PhD

Position/Title*: 4261
Organization Name*: North Carolina State University
Department: Mechanical & Aerospace Engin.
Division:
Street1*: 1840 Entrepreneur Dr; EBIII 3282
Street2: Campus Box 7910
City*: Raleigh
County: Wake
State*: NC: North Carolina
Province:
Country*: USA: UNITED STATES
Zip / Postal Code*: 27695-7910

Phone Number*: (919) 515-5240 Fax Number:

E-Mail*: xjiang5@ncsu.edu

Credential, e.g., agency login: XIAONING

Project Role*: Co-Investigator Other Project Role Category:

Degree Type: PhD Degree Year: 1996

Attach Biographical Sketch*: File Name: Biosketch_Jiang1046094990.pdf

Attach Current & Pending Support: File Name:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: George Middle Name A. Last Name*: Stouffer Suffix: MD

Position/Title*: Ernest and Hazel Craige Dist Professor
Organization Name*: University of North Carolina at Chapel Hill
Department: Medicine - Cardiology
Division: School of Medicine
Street1*: 7075 Cath Lab 2227 Old Infirmary
Street2: Campus Box 7075
City*: Chapel Hill
County: Orange
State*: NC: North Carolina
Province:
Country*: USA: UNITED STATES
Zip / Postal Code*: 27599-7075

Phone Number*: (919) 966-5141 Fax Number:

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Credential, e.g., agency login: GEORGE_STOUFFER

Project Role*: Co-Investigator Other Project Role Category:

Degree Type: MD Degree Year: 1987

Attach Biographical Sketch*: File Name: Biosketch_Stouffer1046185736.pdf

Attach Current & Pending Support: File Name:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: Ashley Middle Name C. Last Name*: Brown Suffix: PhD

Position/Title*: Assistant Professor
Organization Name*: North Carolina State University
Department: Biomedical Engineering
Division: School of Medicine
Street1*: 1840 Entrepreneur Dr; EBIII 4204B
Street2: Campus Box 7115
City*: Raleigh
County: Wake
State*: NC: North Carolina
Province:
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Zip / Postal Code*: 27695-7115

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Credential, e.g., agency login: ACARSON

Project Role*: Co-Investigator Other Project Role Category:

Degree Type: PhD Degree Year: 2011

Attach Biographical Sketch*: File Name: Biosketch_Brown1046094988.pdf

Attach Current & Pending Support: File Name:

PROFILE - Senior/Key Person

Prefix: Dr. First Name*: James Middle Name K Last Name*: Tsuruta Suffix: PhD

Position/Title*: Research Assistant Professor
Organization Name*: University of North Carolina at Chapel Hill
Department: Biomedical Engineering
Division: School of Medicine
Street1*: 116 Manning Drive, 9018A Mary Ellen Jones Bldg
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City*: Chapel Hill
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Province:
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Phone Number*: (919) 966-4436 Fax Number:

E-Mail*: james_tsuruta@med.unc.edu

Credential, e.g., agency login: JTSURUTA

Project Role*: Co-Investigator Other Project Role Category:

Degree Type: PhD Degree Year: 1991

Attach Biographical Sketch*: File Name: Biosketch_Tsuruta1046094992.pdf

Attach Current & Pending Support: File Name:

OMB No. 0925-0001 and 0925-0002 (Rev. 03/2020 Approved Through 02/28/2023)

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: **Dayton, Paul Alexander**

eRA COMMONS USER NAME (credential, e.g., agency login): **PADAYTONNCSU**

POSITION TITLE: **William R. Kenan Jr. Distinguished Professor**

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Villanova University, Villanova, PA	B.S.	1995	Physics
Villanova University, Villanova, PA	B.S.	1995	Comprehensive Science
University of Virginia, Charlottesville, VA	M.E.	1998	Electrical Engineering
University of Virginia, Charlottesville, VA	Ph.D.	2001	Biomedical Engineering
University of California, Davis, CA	Post-doc	2001	Biomedical Engineering

A. Personal Statement

Dr. Dayton has been working in the field of biomedical ultrasonics technologies for 25 years, with a focus on microbubbles and nanodroplets, and their applications in biomedical ultrasonics. He has over 200 publications in the field, several book chapters, and has presented numerous national and international seminars. Dr. Dayton's laboratory has launched two companies, SonoVol, Inc., and Triangle Biotechnology, Inc. which are commercializing novel ultrasound tools and contrast agents.

Specific to this project, Dr. Dayton is inventor of the low-boiling point phase change contrast agent technology, and he has been working on applications of these nanodroplets over the past decade. Over the past two years, he has been working with collaborators Dr. Jiang and more recently Co-PI (MPI) Dr. Wolberg and Brown on nanodroplet-enhanced sonothrombolysis, with exciting preliminary results. This project will greatly expand this project and collaborative team to test the hypothesize that low-boiling point PCCAs might be an enabling technology to push the success of sonothrombolysis for treating arterial clots, where ultrasound alone or ultrasound with microbubbles could not achieve the desired success.

B. Selected Positions and Honors

IEEE Ultrasonics Technical Program Committee 2008-
Director of the Graduate Program, UNC-NCSU Joint Department of Biomedical Engineering, 2009-2012
Associate Director – UNC Biomedical Research Imaging Center, 2011 – current
Associate Department Chair – UNC-NCSU Joint Department of Biomedical Engineering, 2012 – 2019
Professor – UNC Eshelman School of Pharmacy, 2013-
Professor – UNC-NCSU Joint Department of Biomedical Engineering, 2013-
Co-Chair – Certificate In Biomedical Imaging Science, 2015-
Co-Chair – 2015 Southeast Regional Biomedical Engineering Society meeting
Imaging Chair – 2015 National Biomedical Engineering Society meeting
Recipient of the 2016 SPIE Pioneer Award in Biomedical Wellness
American Institute of Ultrasound in Medicine Technical Standards Committee, 2016-2019
Fellow, American Institute for Medical and Biological Engineering, 2018-
William J. Kenan Jr. Distinguished Professor, 2019-
Interim Chair, UNC-NCSU Joint Department of Biomedical Engineering, 2019-present
Technical Program Chair, IEEE International Ultrasonics Symposium, 2020
Recipient, IEEE UFFC Carl Hellmuth Hertz Award, 2020.

NIH study sections: ZRG1 DTCS-A (81), ZRG1 SBIB-Z (03), CSRAC, ZMH1 ERB-C (07), ZRG1 DTCS-A (81), ZRG1 SBIB-Z (03), ZRG1 SBIB-D (57), ZRG1 SBIB-Z (03), ZMH1 ERB-C (09), ZRG1 DTCS-A (81), MEDI (Charter Member 2009-2013), ZRG1 SBIB-Y (31) (Chairperson), ZRG1 SBIB-X (30), ZRG1 SBIB-W (55), CMIP, ZRG1 MEDI-S (09), ZRG1 SBIB-S (91), ZRG1 BST-R (02), ZRG1 MEDI-A (08), ZRG1 MEDI-A (09), ZRG1 MEDI-S (09)

C. Contribution to Science

1) Applications of cavitation enhancing agents in sonothrombolysis

Recently, Dr. Dayton has been working with collaborators to evaluate new technologies for sonothrombolysis, and has discovered the potential of low-boiling point phase change nanodroplets as a potential new tool to treat aged and contracted clots.

a) A Comparison of Sonothrombolysis in Aged Clots Between Low Boiling Point Phase Change Nanodroplets and Microbubbles of the Same Composition, J. Kim, R. DeRuiter, L. Goel, Z. Xu, X. Jiang, and P. A. Dayton, Ultrasound in Medicine & Biology, 2020. (doi.org/10.1016/ j.ultrasmedbio.2020.07.008)
b) Intravascular forward-looking ultrasound transducers for microbubble-mediated sonothrombolysis, Jinwook Kim, Brooks D. Lindsey, Wei-Yi Chang, Xuming Dai, Joseph M. Stavas, Paul A. Dayton, Xiaoning Jiang, Sci Rep. 2017; 7: 3454. Published online 2017 Jun 14. doi: 10.1038/s41598-017-03492-4
c) Microbubble mediated dual-frequency high intensity focused ultrasound thrombolysis: an In vitro study, D. Suo, Z. Jin, X. Jiang, P. Dayton, and Y. Jing, Applied Physics Letters, 110, 2, 023703 (2017).
d) Examining the influence of low dose Tissue Plasminogen Activator on Microbubble Mediated Forward-Viewing Intravascular Sonothrombolysis, , L. Goel, H. Wu, H. Kim, B. Zhang, J. Kim, P. Dayton, Z. Xu, and X. Jiang, Ultrasound in Medicine & Biology (2020) (doi.org/10.1016/j.ultrasmedbio.2020.03.012).

2) The development, study, and application of low-boiling point phase change contrast agents

One of Dr. Dayton's recent contributions to the field of ultrasonic is the development and application of low-boiling point phase change contrast agents. Dr. Dayton is the inventor of a new technique to formulate ~100-400 nanometer sized droplets which can be activated by ultrasound. He has extensively studied their physical response and performance under various conditions, and demonstrated their unique utility and advantages in enhancing high intensity focused ultrasound therapy as well as various imaging applications.

a) Contrast-Enhanced Ultrasound Imaging and in Vivo Circulatory Kinetics with Low-Boiling-Point Nanoscale Phase-Change Perfluorocarbon Agents. Sheeran PS, Rojas JD, Puett C, Hjelmquist J, Arena CB, Dayton PA. Ultrasound Med Biol. 2015 Mar;41(3):814-31. doi: 10.1016/j.ultrasmedbio.2014.10.020. Epub 2015 Jan 22.
b) Phase change events of volatile liquid perfluorocarbon contrast agents produce unique acoustic signatures. Sheeran PS, Matsunaga TO, Dayton PA. Phys Med Biol. 2014 Jan 20;59(2):379-401. doi: 10.1088/0031-9155/59/2/379. Epub 2013 Dec 19.
c) Phase-shift perfluorocarbon agents enhance high intensity focused ultrasound thermal delivery with reduced near-field heating. Phillips LC, Puett C, Sheeran PS, Wilson Miller G, Matsunaga TO, Dayton PA. J Acoust Soc Am. 2013 Aug;134(2):1473-82. doi: 10.1121/1.4812866. Erratum in: J Acoust Soc Am. 2013 Dec;134(6):4575.
d) Phase-Change Contrast Agents for Imaging and Therapy, Paul S. Sheeran, Paul A. Dayton, Curr Pharm Des. 2012; 18(15): 2152–2165.

3) The development and implementation of acoustic angiography

Dr. Dayton is the inventor of acoustic angiography, a contrast enhanced microvascular imaging technique. Although superharmonic imaging has been previously described by various groups, it was not until Dr. Dayton's work with collaborators to develop the necessary multi-frequency transducers, define optimal imaging parameters, and interface the prototype transducers with an imaging system that acoustic angiography (based on superharmonic imaging) has been a feasible imaging technique. Dr. Dayton has furthermore led the additional testing and development of this technique, and has led its implementation in both pre-clinical and clinical imaging studies. Acoustic angiography is a new paradigm in high-frequency contrast imaging, enabling the detailed visualization of microvascular morphology without tissue background. This technique has already demonstrated substantial utility in cancer imaging applications.

a) Acoustic angiography: a new imaging modality for assessing microvasculature architecture. Gessner RC, Frederick CB, Foster FS, Dayton PA. Int J Biomed Imaging. 2013;2013:936593. doi: 10.1155/2013/936593. Epub 2013 Jul 17.
b) Mapping microvasculature with acoustic angiography yields quantifiable differences between healthy and tumor-bearing tissue volumes in a rodent model. Gessner RC, Aylward SR, Dayton PA. Radiology. 2012 Sep;264(3):733-40. doi: 10.1148/radiol.12112000. Epub 2012 Jul 6.
c) Quantification of microvascular tortuosity during tumor evolution using acoustic angiography. Shelton SE, Lee YZ, Lee M, Cherin E, Foster FS, Aylward SR, Dayton PA. Ultrasound Med Biol. 2015 Jul;41(7):1896-904. doi: 10.1016/j.ultrasmedbio.2015.02.012.
d) Super harmonic ultrasound for motion-independent localization microscopy: applications to microvascular imaging from low to high flow rates. Kierski TM, Espindola D, Newsome IG, Cherin E, Yin J, Foster FS, Demore CEM, Pinton GF, Dayton PA. IEEE Trans Ultrason Ferroelectr Freq Control. 2020 Jan 10. doi: 10.1109/TUFFC.2020.296576.

4) **The demonstration, study, and application of acoustic radiation force to enhance targeted therapy and ultrasound molecular imaging**
Dr. Dayton was one of the first investigators to describe, model, and evaluate the effects of acoustic radiation forces on ultrasound contrast agents. He then demonstrated the utility of applying radiation force to enhance the targeting of microbubbles for both imaging and therapeutics applications.
a) An in vivo validation of the application of acoustic radiation force to enhance the diagnostic utility of molecular imaging using 3-d ultrasound. Gessner RC, Streeter JE, Kothadia R, Feingold S, Dayton PA. Ultrasound Med Biol. 2012 Apr;38(4):651-60. doi: 10.1016/j.ultrasmedbio.2011.12.005. Epub 2012 Feb 15.
b) Ultrasound radiation force modulates ligand availability on targeted contrast agents. Borden MA, Sarantos MR, Stieger SM, Simon SI, Ferrara KW, Dayton PA. Mol Imaging. 2006 Jul;5(3):139-47.
c) Radiation-force assisted targeting facilitates ultrasonic molecular imaging. Zhao S, Borden M, Bloch SH, Kruse D, Ferrara KW, Dayton PA. Mol Imaging. 2004 Jul;3(3):135-48.
d) The magnitude of radiation force on ultrasound contrast agents. Dayton PA, Allen JS, Ferrara KW. J Acoust Soc Am. 2002 Nov;112(5 Pt 1):2183-92.

5) **The development and application of ultra-high speed photomicroscopy systems for understanding the physical behaviors of ultrasound contrast agents.**
Dr. Dayton worked with labmates to develop the first ultra-high-speed microscopy imaging system for physical analysis of ultrasound contrast agents exposed to acoustic fields. The system he developed was used in some of the first manuscripts which reported the radial oscillations and fragmentation behavior of contrast agents. Ultra-high speed microscopy is now the gold standard for microbubble analysis in an ultrasonic field.
a) Optical and acoustical dynamics of microbubble contrast agents inside neutrophils. Dayton PA, Chomas JE, Lum AF, Allen JS, Lindner JR, Simon SI, Ferrara KW. Biophys J. 2001 Mar;80(3):1547-56.
b) Threshold of fragmentation for ultrasonic contrast agents. Chomas JE, Dayton P, May D, Ferrara K. J Biomed Opt. 2001 Apr;6(2):141-50.
c) Nondestructive subharmonic imaging. Chomas J, Dayton P, May D, Ferrara K. IEEE Trans Ultrason Ferroelectr Freq Control. 2002 Jul;49(7):883-92.
d) Optical and acoustical observations of the effects of ultrasound on contrast agents. Dayton P, Morgan K,. Klibanov A, Brandenburger G, Ferrara K, IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control., vol. 46, No. 1, pp. 220-232, 1999

Lists of Published Work in Digital Bibliographies: **h-index: 65 (Google Scholar 9/18/20)**

http://www.ncbi.nlm.nih.gov/pubmed/?term=Dayton+PA
http://scholar.google.com/citations?user=bgfWy5kAAAAJ&hl=en

D. Research Support

Ongoing Research Support (several projects in NCE or that are unrelated are not listed due to space constraints)

R21CA246550 (MPIs: Dayton, Pylayeva-Gupta) 04/01/2020-03/31/2022
NIH
Parametric optimization of ultrasound-mediated immuno-modulation for pancreatic cancer therapy
In this proposal we will perform a mechanistic study across varying acoustic parameters aimed at informing both biological mechanism as well as optimal acoustic parameters to achieve immunostimulation. This effort will inform the community on key parameters for focused ultrasound (FUS)-directed immunotherapy, as we facilitate the translation of this novel approach towards treating PDAC in clinic.

R21CA246414 (MPIs: Brudno, Dayton) 02/15/2020-01/31/2022
NIH / NCI
Image-guided, ultrasound-enhanced long-term intracranial drug delivery
In order to deliver drugs to brain, one must pass through the blood brain barrier, something which is possible with BBB disruption, but allows for a short window of time (~1 hour) to administer drugs. We propose combining ultrasound-mediated BBB disruption with refillable drug depots, which allow for local and sustained drug release following bolus administration of systemic doses of inert prodrugs to present therapeutic agents to the brain over months. This approach could have a significant impact in the treatment of brain diseases.

R01CA220681 (MPIs: Dayton, Pinton) 08/10/2017-07/31/2022
NIH
High Frame Rate 3-D Super Resolution Ultrasound Microvascular Imaging
Our motivation is to develop super-resolution imaging as a novel new approach for imaging angiogenesis – one of the hallmarks of cancer, as a new biomarker target for both diagnostics and assessment of response to therapy.

R01NS113285 (MPIs: Pinton, Oralkan) 09/01/2019-05/31/2022
NIH
A Fully Ultrasonic Approach for Combined Functional Imaging and Neuromodulation in Behaving Animals
For this project, we will develop hardware and software for a combined simultaneous neurostimulation and super-resolution functional imaging approach that will visualize functional changes in microvessel flow during acoustically induced neurostimulation in the macaque somatosensory brain circuit using a high channel-count high frame-rate ultrasound device.
Role: Co-Investigator

R01CA232148 (PI: Dayton) 06/01/2018-05/31/2022
NIH
Treating Tumoral Hypoxia via Ultrasound-Guided Oxygen Release for Improving Radiation Therapy
We hypothesize that we can guide specially formulated OMBs via ultrasound imaging, and preferentially release oxygen, in the tumor for radiosensitization to significantly improve the radiotherapy therapeutic ratio. In order to test our hypothesis, will optimize microbubble formulations and administration parameters, evaluate biological mechanisms and kinetics, and validate our hypothesis both in a rodent model and then in a translational large animal model across two of the leading veterinary schools in the country. To achieve these goals, we approach this project with a collaborative team of leading experts in the fields of microbubbles, oxygen transport, radiation oncology, and tumor biology.
Role: Principal Investigator

R01EB026897 (MPIs: Oralkan, Dayton) 07/01/2018-06/30/2022
NIH
Acoustic Angiography Using Dual-Frequency and Ultrawideband CMUT Arrays
We hypothesize that CMUT transducers will be an enabling technology for acoustic angiography and we have assembled a team of experts to develop and test this hypothesis. Our specific aims are thus as follows: Aim 1: Design and characterize new dual-frequency and ultrawideband transducers based on CMUT technology. Aim 2: Optimize imaging approaches using the CMUT arrays. Aim 3: In-vivo contrast imaging characterization and validation.

R01HL141967-01A1 (PI: Jiang) 08/15/2018-06/30/2022
NIH
Forward Viewing Catheter-Delivered Microbubble Enhanced Sonothrombolysis (FV-CAMUS)
We propose to develop a new ultrasound imaging monitored intravascular sonothrombolysis technique using Forward Viewing CAtheter-directed MicrobUbble-enhanced Sonothrombolysis (FV-CAMUS), which can be effective to treat both partially occluded and completely occluded clots.
Role: Co-Investigator

R21EB025306 (PI: Johnson) 09/01/2018-06/30/2021
NIH
Novel Targeted Iodinated Microbubbles for Molecular Imaging With Computed Tomography
The long-term goal of this research is to develop a MI paradigm for CT for use in clinical practice. The objective of this application is to generate a MI probe for CT comprised of targeted microbubbles doped with radiopaque polyesters.
Role: Co-Investigator

No Grant Number (PI: Johnson) 11/01/2019-10/31/2020
North Carolina Diabetes Research Collaborative
Development and validation of ultrasound molecular imaging agents for diabetic kidney disease by measuring bio-molecular interaction with a novel, sensitive, homogeneous proximity assay (AlphaScreen™)
The goal of this project is to develop an in vitro assay based on amplified luminescent proximity homogeneous assay screen to produce sensitive and specific targeted microbubbles for diabetic kidney disease (DKD). The assay will assess and optimize microbubble ligand-receptor interactions necessary to make the targeted contrast agent. Once the targeted microbubbles are developed, we will compare the optimized to non-optimized targeted microbubbles in preclinical models of DKD to assess improvement. The result will be an assay to quickly generate new targeted microbubbles for ultrasound molecular imaging. Ultimately, ultrasound molecular imaging will provide a screening tool for clinicians to detect which diabetic patients will progress to DKD and offer an opportunity for timely intervention.
Role: Co-Investigator

OMB No. 0925-0001 and 0925-0002 (Rev. 03/2020 Approved Through 02/28/2023)

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Alisa S. Wolberg, Ph.D., FAHA

eRA COMMONS USER NAME (credential, e.g., agency login): ALISA_WOLBERG

POSITION TITLE: Professor of Pathology and Laboratory Medicine

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
University of North Carolina at Chapel Hill, NC	B.S.	05/1991	Biology
University of North Carolina at Chapel Hill, NC	Ph.D.	10/1996	Biology
Duke University Medical Center, Durham, NC	Postdoctoral	10/2001	Hematology

A. Personal Statement

In this project, I look forward to expanding my collaboration with Dr. Dayton and the team to develop a novel strategy for in-situ arterial thrombus treatment, mediated by ultrasound activated clot-penetrating nanodroplets. My experience in this area is extensive and I am well qualified to be a co-lead on this project. For over 25 years, I have conducted research in physiologic and pathophysiologic mechanisms in hemostasis and thrombosis. We employ in vivo mouse models of thrombosis and bleeding, ex vivo and in vitro assays using human tissue and blood samples, and novel recombinant proteins to investigate biological, biochemical, and biophysical mechanisms contributing to coagulation and diseases associated with abnormal blood clotting. We have uncovered mechanisms regulating multiple aspects of clot formation, including mechanisms by which cellular and plasma activities promote thrombosis and thrombus stability, pathologic pathways that differentiate arterial and venous thrombosis, the mechanism of action of hemostatic therapies for hemophilia, and regulation of tissue factor activity on cells and cell-derived microparticles. Most recently, we identified a newly-recognized role of transglutaminase factor XIII in venous thrombosis in which fibrin crosslinking promotes retention of red blood cells in venous thrombi. Through these investigations, I have developed a vibrant research training program for junior investigators at the undergraduate, graduate, and post-graduate levels. The strength of their experience in my laboratory is evident in their publication record and matriculation into subsequent scientific endeavors. The publications listed underscore my recognized expertise in arterial and venous thrombosis/thromboembolism and mouse models to study these diseases.

a. Machlus KR, Cardenas JC, Church FC, **Wolberg AS**. 2011. Causal relationship between hyperfibrinogenemia, thrombosis, and resistance to thrombolysis in mice. *Blood*, 117(18):4953-63. PMID: 21355090, PMC3100702 [Commentary by Ariens RAS. 2011. Elevated fibrinogen causes thrombosis. *Blood,* 117(18):4687-8.]
b. Aleman MM, Walton BL, Byrnes JR, Wang JG, Heisler MJ, Machlus KR, Cooley BC, **Wolberg AS**. 2013. Elevated prothrombin promotes venous, but not arterial, thrombosis in mice. *Arterioscl, Thromb Vasc Biol*, 33: 1829-36. PMID: 23723374, PMC3779620
c. **Wolberg AS**, Rosendaal FR, Weitz JI, Jaffer IH, Agnelli G, Baglin T, Mackman N. 2015. Venous thrombosis. *Nature Rev Dis Pri*, http://www.nature.com/articles/nrdp20156, 1:15006. PMID: 27189130
d. Byrnes JR, **Wolberg AS**. 2017. Red blood cells in thrombosis. *Blood*, 130(16):1795-9. PMID: 28811305, PMC5649548

B. Positions and Honors

Positions and Employment

1990 – 1991 Internship, Burroughs Wellcome, Research Triangle Park, NC
2001 – 2001 Assistant Research Professor, Dept of Pathology, Duke Univ Medical Center

2001 – 2007	Res Asst Professor, Pathology & Laboratory Medicine, Univ of North Carolina at Chapel Hill
2007 – 2011	Assistant Professor, Pathology & Laboratory Medicine, Univ of North Carolina at Chapel Hill
2011 – 2016	Associate Professor (tenured), Pathology & Laboratory Medicine, Univ of North Carolina at Chapel Hill
2016 – pres	Professor, Pathology & Laboratory Medicine, Univ of North Carolina at Chapel Hill
2019 – pres	Director, UNC Program in Translational Medicine

Other Experience and Professional Memberships

Study Sections: American Heart Association (AHA) Natl Thrombosis (2006-2010, 2015); NIH/NHLBI (HT standing member 2018-2024)

Editorial Positions: Associate Editor, *Res Practice Thromb Haemost* (2017-pres) and *Semin Thromb Hemost* (2015-pres); Editorial Board, *Arterio Thromb Vasc Biol* (2010-pres), *J Thromb Haem* (2011-2013, 2016-pres), *Blood Advances* (2016-2019), and *Blood* (2018-pres)

Societies (Current Committees): International Society on Thrombosis and Haemostasis (ISTH) (Council, Governance, Membership); AHA/ATVB (Programming, Leadership); American Society of Hematology (ASH) (Scientific Affairs, Dynamic Research, PhD Task Force, HT Working Group, Women in Hematology Working Group, Media Relations); Hemostasis Thrombosis Research Society (Board of Directors); Intl Fibrinogen Research Society (Board of Councillors)

Honors and Awards

UNC Graduate School Travel Award (1993); National Research Service Award, NIH Postdoctoral Fellowship (1998-2001); ASH Travel Award (1999); Durham VA Medical Center Institute for Medical Research Intramural Grant Award (2000); AHA Beginning Grant-In-Aid (2001-2003); UNC Bayer Corporate Fellow (2001-2002); UNC Building Interdisciplinary Research Careers in Women's Health (BIRCWH) Scholar (2003-2004); Early Career Development Award, Bayer Hemophilia Awards Program (2003-2005); Novo Nordisk Award in recognition of outstanding research of an original nature leading to further understanding of bleeding disorders and their treatment (2005); UNC Joe W. Grisham Award for Excellence in Graduate Student Teaching (2008); Finalist, AHA/ATVB Kenneth M. Brinkhous Young Investigator Prize in Thrombosis (2008); Associate Editor, *Arterioscl Thromb Vasc Biol* Special Issue on DVT (2011); Elected Fellow of the AHA (FAHA, 2011); Co-Chair, 6th Symposium on Hemostasis with Special Focus on TF, FVIIa, and TFPI: Hemostasis and Beyond (2012); Guest Editor, *Thromb Res* Suppl Issue (2012); AHA/ATVB Special Recognition Award in Thrombosis (2012); Top Ten Reviewer, *Arterioscl Thromb Vasc Biol* (2014); 17th Biennial Award for Contributions to Hemostasis (BACH) Investigator Recognition Award, ISTH (2015); Vice-Chair, Gordon Research Conference on Hemostasis (2016); EHRA Merit Bonus for teaching excellence, UNC School of Medicine (2016); Elected Council Member, ISTH (2016-2022); Chair, Gordon Research Conference on Hemostasis (2018); 19th Annual Mosesson Lecture/Versiti (2020); Scientific Co-Chair, ASH Annual Meeting (2020)

C. Contribution to Science

1. **Factor XIII in venous thrombosis (VT).** In contrast to platelet-rich arterial "white" clots, venous thrombi are so-called "red thrombi" because of their high red blood cell (RBC) content. For decades, the prevailing paradigm presumed RBCs are only passively entangled within the fibrin network during thrombogenesis. I led research to investigate pathophysiologic mechanisms promoting VT and discovered that coagulation factor XIII (FXIII) activity mediates RBC retention in clots and consequently, determines venous thrombus size. We extended this finding to show this effect is mediated specifically by formation of high molecular weight crosslinked fibrin species. We also identified fibrin(ogen) residues that comprise the long-elusive binding site for FXIII and showed that these residues are required for normal FXIII activation and fibrin crosslinking, and consequently, for RBC retention in thrombi. We identified plasmin as the endogenous inactivator of FXIIIa. We showed that inhibiting FXIIIa activity or blocking FXIII-fibrin(ogen) interactions reduces thrombus RBC content and thrombus size. These findings are paradigm-shifting for understanding pathophysiologic mechanisms that mediate VT, and suggest FXIII(a) is a novel therapeutic target for reducing VT.
   a. Aleman MM, Byrnes JR, Wang J-G, Tran R, Lam WA, Di Paola J, Mackman N, Degen JL, Flick MJ, **Wolberg AS**. 2014. Factor XIII activity mediates red blood cell retention in venous thrombi. *J Clin Invest*, 124(8):3590-600. PMID: 24983320, PMC4109540 [Cover]
   b. Byrnes JR, Duval C, Wang Y, Hansen C, Ahn B, Mooberry MJ, Clark MA, Johnsen JM, Lord ST, Lam W, Meijers JCM, Ni H, Ariëns RAS, **Wolberg AS**. 2015. Factor XIIIa-dependent retention of red blood

cells in clots is mediated by fibrin α-chain crosslinking. *Blood*, 126(16):1940-8. PMID: 26324704, PMC4608241

c. Byrnes JR, Wilson C, Boutelle AM, Brandner CB, Flick MJ, Philippou H, **Wolberg AS**. 2016. The interaction between fibrinogen and zymogen FXIII-A_2B_2 is mediated by fibrinogen residues γ390-396 and the FXIII-B subunits. *Blood*, 128(15):1969-78. PMID: 27561317, PMC5064719

d. Kattula S, Byrnes JR, Martin SM, Holle LA, Cooley BC, Flick MJ, and **Wolberg AS**. Factor XIII in plasma, but not in platelets, mediates red blood cell retention in clots and venous thrombus size in mice. *Blood Adv*, 2(1):25-35. PMID: 29344582, PMC5761627

2. **Hypercoagulability and thrombosis.** Blood hypercoagulability is an established risk factor for thrombosis. Beginning with my postdoctoral research, I developed models to measure thrombin generation and fibrin formation, structure, and stability, and identify contributions of hypercoagulability to arterial and venous thrombosis. We showed that elevated plasma fibrinogen, factor VIII (FVIII), or prothrombin promote vessel occlusion via unique mechanisms. Whereas hyperfibrinogenemia directly augments thrombus fibrin(ogen) content and stability, elevated FVIII and prothrombin increase thrombin generation and alter platelet accumulation and clot permeability, elastic modulus, and stability. We have also shown that elevated hematocrit promotes arterial thrombus formation by enhancing platelet accumulation in thrombi. These findings have critical implications for predicting thrombosis risk from global coagulation assays and understanding the operant mechanisms. These findings also provide rationale for the use of anticoagulants and thrombolytics in stroke, myocardial infarction, and deep vein thrombosis.

   a. **Wolberg AS**, Monroe DM, Roberts HR, Hoffman M. 2003. Elevated prothrombin results in clots with an altered fiber structure: a possible mechanism of the increased thrombotic risk, *Blood*, 101(8):3008-13. PMID: 12506014, PMC3779620

   b. Machlus KR, Lin FC, **Wolberg AS**. 2011. Procoagulant activity induced by vascular injury determines contribution of elevated factor VIII to thrombosis and thrombus stability in mice. *Blood*, 118(14):3960-8. PMID: 21828144, PMC3193271

   c. Walton BL, Getz TM, Bergmeier W, Lin FC, Uitte de Willige S, **Wolberg AS**. 2014. The fibrinogen γA/γ' isoform does not promote acute arterial thrombosis in mice. *J Thromb Haemost*, 12(5): 680-89. PMID: 24916154, PMC4098759

   d. Walton BL, Lehmann M, Skorczewski T, Holle LA, Beckman JD, Cribb JA, Mooberry MJ, Wufsus AR, Cooley BC, Homeister JW, Pawlinski R, Falvo MR, Key NS, Fogelson AL, Neeves KB, **Wolberg AS**. 2017. Elevated hematocrit enhances platelet accumulation following vascular injury. *Blood*, 129(18):2537-46. PMID: 28251913, PMC5418635 [Commentary by Machlus KR, Battinelli EM. 2017. RBCs pin platelets against the (thrombus) wall. *Blood*, 129(18): 2460-1]

3. **Bleeding and development of hemostatic therapies.** Both traumatic injury and congenial and acquired bleeding disorders, including hemophilia A, B, and C are challenging clinical presentations. Beginning with my graduate work, I have investigated mechanisms contributing to bleeding. We showed that fibrin network structure and stability are sensitive and robust biomarkers for detecting bleeding risk, even in situations in which abnormal thrombin generation cannot detect significant abnormalities. These findings have led to the use of clot structure and stability assays in the pre-clinical development of novel hemostatic agents in both academic and industry settings. The publications listed describe our pre- and post-clinical studies to characterize clot quality in congenital and acquired bleeding disorders, improve novel hemostatic therapies for both on-demand and prophylactic administration, and develop assays to identify patients with increased risk of bleeding.

   a. Allen GA, Persson E, Campbell RA, Ezban M, Hedner U, **Wolberg AS**. 2007. A variant of recombinant factor VIIa with enhanced procoagulant and antifibrinolytic activities in a model of hemophilia. *Arterio Thromb Vasc Biol*, 27(3):683-9. PMID: 17204663

   b. Zucker M, Seligsohn U, Salomon O, **Wolberg AS**. 2014. Abnormal plasma clot structure and stability distinguish bleeding risk in patients with severe factor XI deficiency. *J Thromb Haemost*, 12(7): 1121-30. PMID: 24815347, PMC4107079

   c. Gidley GN, Holle LA, Bolton-Maggs PHB, Lin F-C, **Wolberg AS**. 2018. Abnormal plasma clot formation and fibrinolysis reveal bleeding tendency in patients with partial factor XI deficiency. *Blood Adv*, 2(10):1076-88. PMID: 29760205, PMC5965046

   d. Macrae FL, Duval C, Papareddy P, Baker S, Yuldasheva N, Kearney KJ, McPherson HR, Asquith N, Konings J, Casini A, Degen JL, Connell SD, Philippou H, **Wolberg AS**, Herwald H, Ariëns RAS. 2018. A fibrin biofilm covers the blood clot and protects from microbial invasion. *J Clin Invest*, PMID: 29723163

[Commentary by SX Gu, SR Lentz. 2018. Fibrin films: overlooked hemostatic barriers against microbial infiltration. *J Clin Invest*, 128(8): 3243-5]

4. **Procoagulant and fibrinolytic mechanisms in disease.** Thrombosis is a complication in many diseases, including antiphospholipid syndrome, diabetic nephropathy, environmental pollution, obesity, and liver injury. Beginning with my postdoctoral work, I have both led and collaborated on studies to determine operant mechanisms and elucidate new potential therapeutic targets for these pathologies. These studies have demonstrated that tissue factor (TF) exposure and activation of coagulation is a cross-cutting mechanism in disease etiology. The papers listed summarize our findings on mechanisms that regulate TF activity in quiescent and activated cells, contributions of cellular microvesicles to disease, and dysregulation of fibrinolytic activity in obesity.
   a. Machlus KR, Colby EA, Wu JR, Koch GG, Key NS, **Wolberg AS**. 2009. Effects of tissue factor, thrombomodulin, and elevated clotting factor levels on thrombin generation in the calibrated automated thrombogram. *Thromb Haemost*, 102(5):936-44. PMID: 19888532, PMC2904819
   b. Campbell RA, Overmyer KA, Sheridan BC, Selzman CH, **Wolberg AS**. 2009. Contributions of extravascular and intravascular cells to fibrin network formation, structure, and stability. *Blood*, 114(23):4886-96. PMID: 19797520, PMC2786294
   c. Aleman MM, Gardiner C, Harrison P, **Wolberg AS**. 2011. Differential contributions of monocyte- and platelet-derived microparticles towards thrombin generation and fibrin formation and stability. *J Thromb Haem*, 9(11):2251-61. PMID: 21883880, PMC3206146
   d. Miszta A, Kopec A, Pant,A, Holle LA, Byrnes JR, Lawrence DA, Hansen KC, Flick MJ, Luyendyk JP, de Laat B, **Wolberg AS**. 2020. A high fat diet delays plasmin generation in a thrombomodulin-dependent manner in mice. *Blood*, 135(19):1704-17. PMID: 32315384

5. **Cancer-associated thrombosis**. Thrombosis and bleeding are frequent complications of cancer and cancer-directed therapy, and contribute substantively to morbidity and mortality associated with cancer. Throughout my career, I have collaborated on studies of cancer-associated thrombosis and provided important expertise in coagulation and fibrinolytic pathways promoted by tumor cells, as well as the use of mouse models of thrombosis for evaluating these pathways in disease progression. These studies have revealed surprising roles for TF, FXIII, and the plasminogen activation system in cancer and cancer-associated coagulopathy.
   a. Carter JC, Campbell RA, Gibbons JA, Gramling MW, **Wolberg AS**, Church FC. 2010. Enhanced cell-associated plasminogen activator pathway but not coagulation pathway activity contributes to motility in metastatic breast cancer cells. *J Thromb Haemost,* 8:1323-32. PMID: 20180817
   b. Unruh D, Ünlu B, Lewis CS, Qi X, Chu Z, Sturm R, Keil R, Ahmad SA, Sovershaev T, Adam M, Van Dreden P, Woodhams BJ, Ramchandani D, Weber GF, Rak JW, **Wolberg AS**, Mackman N, Versteeg HH, Bogdanov VY. 2016. Antibody-based targeting of alternatively spliced tissue factor: a new approach to impede the primary growth and aggressiveness of pancreatic ductal adenocarcinoma. *Oncotarget*, 7(18):25264-75. PMID: 26967388, PMC5041902
   c. Porrello A, Leslie PL, Gorentla BK, Harrison EB, Azam S, Kattula S, Ghosh SK, Holtzhausen A, Hayward MC, Waugh TA, Bae S, Godfrey V, Randell SH, Oderup C, Makowski L, Wilkerson MD, Weiss J, Hayes DN, Earp HS, Baldwin AS, **Wolberg AS**, Pecot CV. 2018. Factor XIIIA-expressing inflammatory monocytes promote lung squamous cancer through fibrin cross-linking. *Nat Comm*, 9:1988. PMID: 29777108, PMC5959879
   d. McCann JV, Xiao L, Khan O, Kowalski P, Anderson D, Pecot P, Azam S, Parker J, Tsai YS, **Wolberg AS**, Tatsumi K, Mackman N, Dudley AC. 2019. Endothelial cell subtypes subvert a TGFb/miR-30c-driven wound healing response that supports tumor growth. *J Clin Invest*, 129(4):1654-70. PMID: 30855280, PMC6436861 [Cover]

Complete List of Published Work in MyBibliography:
https://www.ncbi.nlm.nih.gov/sites/myncbi/alisa.wolberg.1/collections/43600532/public/

D. Research Support

Ongoing Research Support

R33HL141791 (MPIs: Lowenstein, Smith, Wolberg) 04/01/2020-03/31/2023
NIH/NHLBI
Model System of Oral Contraceptive-Induced VTE: Integrating Genomic, Transcriptomic, and Proteomic Discovery with Functional Biology.
Major goals are to develop a model to assess prothrombotic effects of oral contraceptives in premenopausal women.
Role: MPI

U01HL143403 (MPIs: Flick, Fishel, Han, Wolberg) 08/01/2018-07/31/2023
NIH/NHLBI
Targeting the Plasminogen Activation System to Limit Pancreatic Cancer Progression and Associated Thrombosis
Major goals are to investigate the contribution of PAR-1/uPA/uPAR to pancreatic ductal adenocarcinoma (PDAC), illuminate key mechanisms coupling the PA system to PDAC-associated thrombophilia, and provide essential proof-of-principle data in experimental animals and with patient-derived material to facilitate translation of findings into new treatments for PDAC and cancer-associated thrombosis.
Role: MPI

R01HL141291 (MPIs: Morrison, Wolberg) 02/01/2019-01/31/2023
NIH/NHLBI
Using Genomics and Functional Biology to Understand Fibrinogen and its Effect on Thrombotic and Atherosclerotic Outcomes
Major goal is to generate new biological knowledge about genomic regulation of fibrinogen and its importance in disease.
Role: MPI

T32GM122741-01A1 (PI: Wolberg) 07/01/2018-06/30/2023
NIH/NHGMS
Enhancement training for the next generation of translational Ph.D. scientists
The goal of this program is to train students in translational science, implementing integrative approaches through multidisciplinary collaborations.

R01HL147149 (PI: Mackman) 04/01/2019-03/31/2024
NIH/NHLBI
Mechanism of Venous Thrombosis in Pancreatic Cancer
Major goals are to determine the role of the intrinsic coagulation pathway in mouse models of pancreatic CAT, determine the role of neutrophils and NETs in mouse models of pancreatic CAT, and determine the role of PAI-1 in mouse models of pancreatic CAT.
Role: Co-Investigator

R01HL146226 (PI: Key) 01/01/2019-12/31/2021
NIH/NHLBI
Fibrinolysis evaluation in A-TREAT ('FEAT' Study)
Major goals are to investigate the role of fibrinolysis in chemotherapy-associated bleeding.
Role: Co-Investigator

R01HL126974-05A1 (PI: Wolberg) 07/01/2020-06/30/2025
NIH/NHLBI
Fibrinogen and factor XIII in venous thromboembolism
Major goals are to determine mechanisms by which factor XIII and fibrin crosslinking affect venous thrombosis and pulmonary embolism risk.

Ongoing Research Support (last three years)

None

OMB No. 0925-0001 and 0925-0002 (Rev. 03/2020 Approved Through 02/28/2023)

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Jiang, Xiaoning

eRA COMMONS USER NAME (credential, e.g., agency login): XIAONING

POSITION TITLE: Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Shanghai Jiaotong University, Shanghai, China	B.S.	1990	Mechanical Engineering
Tianjin University, Tianjin, China	M.S.	1992	Mechanical Engineering
Tsinghua University, Beijing, China	Ph.D.	1996	Precision Instruments
Nanyang Technological University, Singapore	Postdoc	1996-1997	MEMS
The Pennsylvania State University, USA	Postdoc	1997-1999	MEMS

A. Personal Statement

The goal of this project is to develop a stroke treatment technique that utilizes image guidance with phase change nanodroplets for enhanced clot disruption. Stroke is the fifth leading cause of death in the United States, and remains a major cause of disability even in those cases that are not fatal. Every four minutes, someone dies of stroke in the United States, and disproportionate death rates are nearly twice that for African Americans as whites. About 87% of all strokes are ischemic strokes, in which a clot impedes blood flow to the brain. There exists a vital need for innovative noninvasive treatment of ischemic stroke, especially for patients that are excluded from rt-PA treatment or those that are not able to receive treatment within the short window open to rt-PA treatment (>3 hours of thrombosis). In this project, we hypothesize that a novel stroke treatment will combine image guidance with phase change nanodroplet enhanced cavitation for clot disruption. The nanodroplets diameters and concentration, acoustic conditions and in-vivo nanodroplets mediated sonothrombolysis tests will be studied to demonstrate significantly enhanced clot lysis rate for effective clinical treatment of stroke.

I have the expertise, leadership and motivation necessary to successfully carry out the proposed work. I have over 19 years project and R&D management experience, and over 20 years of experiences on researching and developing advanced electromechanical sensors, actuators and transducers using novel micro/nanofabrications incorporated with a broad range of smart materials and structures for biomedical applications including imaging, therapy and drug delivery. In addition, I successfully administered the projects (e.g. staffing, research protections, budget), collaborated with other researchers, and produced several peer-reviewed publications from each project. As a result of these previous experiences, I am aware of the importance of frequent communication among project members and of constructing a realistic research plan, timeline, and budget. In summary, the current application builds logically on my prior work, and my expertise and experience have prepared me to conduct this challenging and novel research on nanodroplets mediated sonothrombolysis technology. Also of note, I have been collaborating with PI Dr. Dayton and Dr. Stouffer on several NIH and other funded projects for almost a decade, including intravascular ultrasound imaging, intravascular sonothrombolysis for DVT treatment, etc. Hence, our team has experience in all aspects of the proposed work and prior success and is well poised to achieve the project goals.

B. Positions and Honors

Positions and Employment

1999-2001	Research Associate, Penn State University, University Park, PA
2001	Research Engineer, Standard MEMS, Inc., Hauppauge, NY
2001-2005	Research Scientist, TRS Technologies, Inc., State College, PA
2005-2007	Senior Research Scientist, TRS Technologies, Inc., State College, PA
2007-2009	Chief Scientist, TRS Technologies, Inc., State College, PA
2008-2009	VP Technology, TRS Technologies, Inc., State College, PA
2009-2015	Assoc. Professor of Mechanical and Aerospace Engineering, NC State University, Raleigh, NC
2010-	Adjunct Professor of Biomedical Engineering, NC State University, Raleigh, NC
2015-2018	Professor of Mechanical and Aerospace Engineering, NC State University, Raleigh, NC
2016-2017	Visiting Professor of Mechanical and Biomedical Engineering, City University of Hong Kong, HK
2018-2020	Distinguished Professor of Mechanical and Aerospace Engineering, NC State University
2020-	Dean F. Duncan Distinguished Professor of Mechanical and Aerospace Engineering, NC State

Other Experience and Professional Memberships

2000-	Member of Program Committee of International Berkeley MEMS Conference'2001
	Member of Program Committee of IEEE Sensors (2007, 2008, 2009)
	Member of Program Committee of SPIE Smart Materials, Structures and NDT
	Member of Technical Program Committee of IEEE Ultrasonics Symposium (TPC 5, 2014-)
	Member of Technical Program Committee of ASME Annual Congress (IMECE) (Track 3, 2015-)
	Member of Program Committee of IEEE Nano and IEEE NMDC conferences
2001-	Member of ASME; Senior Member of SPIE; Senior Member of IEEE
2001-	Reviewer for more than 40 Journals
2009-	NIH proposal review study sections, NSF panels, NASA and DOE proposal review
	IEEE UFFC Representative to IEEE Nanotechnology Council (2014-)
	Chair, ASME IMECE BED PhD Student paper competition committee (2014, 2015)
	Member of Editorial Board and Guest Editor, Sensors, Special Issue on Ultrasonic Sensors and Transducers for Applications in Biology, Medicine and NDT (2014-).
2017-	NanoAcoustics Technical Committee Chair, IEEE Nanotechnology Council
2020-	Co-EIC of IEEE Nanotechnology Magazine

Honors

1986-1990	Six times, the People Scholarship of Shanghai Jiaotong University
1991	First Order Scholarship of Tianjin University
1993	First Order of Guanghua Scholarship of Tsinghua University
1994	Gaotian Scholarship of Tsinghua University
1996	First Prize of China Instrument Society Scholarship
2008-2009	5 Certificates of Recognition for Invention, NASA Inventions and Contributions Board
2010	NC State faculty research development program award
2011	Winner, Best Technology Development of Energy Harvesting, Presented by Energy Harvesting & Storage USA, Wireless Sensor Networks and RTLS USA, IDTechEX (2011)
2011	IEEE Ultrasonics Symposium Student Paper Competition Award (2011 (Advisor)
2011	Best Research Publication, National Institute of Aerospace
2008-	9 US Patent Awards
2013	Faculty Grant Award recipient of ASNT
2013	DOD DURIP Award recipient
2014	Recipient of 2014 MAE Outstanding Research Award
2015	Recipient of the 2015 NC State Chancellor's Innovation Fund Award
2015	Recipient of 2015 MAE Research Excellence Award
2016	Recipient of University Faculty Scholar
2016	1st Prize Keynote Speaker (Mr. Jinwook Kim), 2016 MAE Graduate Symposium (Advisor)
2016	1st Prize Student Paper Award (Mr. Jinwook Kim), 2016 IEEE Ultrasonics Symposium (Advisor)
2016	R&D100 Award Finalist—Special Recognition
2017-	University Faculty Scholar, NC State University, Raleigh, NC.
2018, 2019	Distinguished Lecturer, IEEE Nanotechnology Council
2019-	ASME Fellow

C. Contribution to Science

1. **Sonothrombolysis**: We have extensive work in the area of developing ultrasound hardware and technologies for sonothromboysis, largely in collaboration with Dr. Dayton [a-d].
    a. Kim, J., B. Lindsey, W. Chang, X. Dai, J. Starvas, P. Dayton and **X. Jiang**, "Intravascular forward-looking ultrasound transducers for microbubble-mediated sonothrombolysis", *Scientific Reports* 7, 3454, (2017). DOI: 10.1038/s41598-017-03492-4.
    b. D. Suo, Z. Jin, X. Jiang, P. Dayton, and Y. Jing, "Microbubble mediated dual-frequency high intensity focused ultrasound thrombolysis: an In vitro study", Applied Physics Letters, 110, 2, 023703 (2017).
    c. Goel, L., H. Wu, H. Kim, B. Zhang, J. Kim, P. Dayton, Z. Xu, and **X. Jiang**, "Examining the influence of low dose Tissue Plasminogen Activator on Microbubble Mediated Forward-Viewing Intravascular Sonothrombolysis", Ultrasound in Medicine & Biology (2020) (doi.org/10.1016/j.ultrasmedbio.2020.03.012).
    d. Kim, J., R. DeRuiter, L. Goel, Z. Xu, **X. Jiang**, and P. Dayton, "A Comparison of Sonothrombolysis in Aged Clots Between Low Boiling Point Phase Change Nanodroplets and Microbubbles of the Same Composition", Ultrasound in Medicine & Biology, 2020. (doi.org/10.1016/j.ultrasmedbio.2020.07.008)

2. **Piezoelectric-composite micromachined ultrasound transducer technology:** conventional high frequency ultrasound transducers use piezo plates or thick/thin piezo films, which are known with relatively low electromechanical coupling coefficients, leading to poor HF transducer performances. I Invented piezoelectric composites based micromachined ultrasound transducers (PC-MUT) technology, which has presented the highest electromechanical coupling coefficient for piezoelectric transducer structures at frequency > 40 MHz. PC-MUT 40-60 MHz transducers have been successfully demonstrated in swine model IVUS imaging with unprecedented high resolution. PC-MUT will also significantly enhance ultrasound for other disease diagnosis, new drug discovery, as well as high resolution NDE [a-b].
    a. **Jiang, X.**, W.S. Hackenberger, and K. Snook, "Micromachined Piezoelectric Ultrasound Transducer Arrays", *US Patent* # 8008842 (2011).
    b. Li, S., J. Tian, P. Han, **X. Jiang**, "20 MHz/40 MHz Dual-layer Micromachined Composite Transducer for Tissue Harmonic Imaging", *IEEE Ultrasonics Symposium*, Chicago, IL, Sept. 2014.
    c. Li, S., J. Tian, and **X. Jiang**, "A Micromachined PMN-PT Single Crystal Composite Circular Array for Intravascular Ultrasound Imaging", ASME J of Medical Diagnostics 2(2), 021001 (Jan 18, 2019).

    L. Goel and X. Jiang, "Advances in Sonothrombolysis Techniques Using Piezoelectric Transducers", Sensors 20.5 (2020): 1288.

3. **Multifrequency ultrasound imaging:** conventional medical ultrasound transducers transmit or/and receive ultrasound waves at one center frequency with limited bandwidth (e.g. -6 dB fractional bandwidth < 100%). We have been investigating dual frequency ultrasound for super-harmonic imaging (or acoustic angiography) in collaboration with Dr. Dayton for contrast enhanced-IVUS imaging [a, b]. We also demonstrated dual frequency 3 MHz/18 MHz transducer and 7.5 MHz/15 MHz co-linear array for superharmonic imaging and multimode imaging, respectively [c, d]. Multifrequency ultrasound technology will play a significant role in the next generation ultrasound in medicine and biology.
    a. Dayton, P. and **X. Jiang**, "Catheter Device Implementing High Frequency, Contrast Imaging Ultrasound Transducer, and Associated Method", *US Patent Application* # 61/654,335, June 2012; PCT/US2013/043002, May 2013.
    b. Ma, J., K. Martin, P. Dayton, **X. Jiang**, "A Preliminary Engineering Design of Intravascular Dual-Frequency Transducers for Contrast Enhanced Acoustic Angiography and Molecular Imaging", *Ultrasonics, Ferroelectrics and Frequency Control, IEEE Transactions* on , 61(5), 870-80 (2014).
    c. Wang, Z., S. Li, T. Czernuszewicz, C. Gallippi, R. Liu, X. Geng, **X. Jiang**, "Design, fabrication and characterization of a bi-frequency co-linear array", *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control,* 63(2), 266-274 (2016).
    d. Li, S., J. Kim, Z. Wang, S. Kasoji, B. Lindsey, P. Dayton, **X. Jiang**, "A Dual-frequency Co-linear Array for Acoustic Angiography in Prostate Cancer Evaluation", *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, Volume: 65, Issue: 12, pp. 2418-2428, 2018.

My Bibliography Link: https://www.ncbi.nlm.nih.gov/myncbi/xiaoning.jiang.1/bibliography/public/

D. Research Support

Ongoing Research Support

No Grant Number (PI: Jiang) 08/01/2020-08/01/2021
Lineberger Comprehensive Cancer Center
Development of a Miniaturized Multidirectional Ultrasound Ablation Array (MiDUSa) for Endobronchoscopic Lung Nodule Ablation
The goal of this project is to develop a high intensity focused ultrasound catheter for endobronchoscopic lung nodule ablation.

R41HL154735 (PI: Jiang) 07/01/2020-06/30/2021 (award in negotiation)
NIH / SonoVascular
Dual Excitation Catheter-delivered Laser Ultrasound Thrombolysis (DECLUT) for Deep Vein Thrombosis (DVT) Treatment
The goal of this project is to develop a dual mode ultrasound catheter for intravascular sonothrombolysis.

DMR-2011978 (PIs: Liu, Jiang, Kumah) 07/01/2020-06/30/2024
NSF
Domain-Engineering Enabled Thermal Switching in Ferroelectric Materials
The goal of this project is to study the domain structures in ferroelectric materials (including relaxor-PT piezoelectric single crystals) and the associated thermal conductivity.
Role: Co-PI

R01HL152023 (PI: Kim) 06/15/2020-05/31/2024
University of Pittsburgh / NIH
Prevent Unnecessary Carotid Intervention and Stroke using Noninvasive Transcutaneous Ultrasound Thermal Strain Imaging (US-TSI)
The goal of this project is to develop a dual mode ultrasound array for thermal strain imaging.
Role: Co-Investigator / Subaward Lead

R21EB027304 (PI: Jiang) 06/01/2019-03/31/2021
NIH
Photoacoustic-imaging-guided intravascular sonothrombolysis
The goal of this project is to develop photoacoustic-imaging-guided intravascular sonothrombolysis.

R01HL141967 (PI: Jiang) 08/15/2018-06/30/2022
NIH
Forward viewing catheter-delivered microbubble enhanced sonothrombolysis (FV-CAMUS)
The goal of this project is to develop forward viewing catheter delivered microbubble/nano-droplet enhanced sonothrombolysis technique for effective treatment of venous thrombosis disease.

OPP1191684 (PI: Jiang) 05/01/2020-04/30/2021 (award in negotiation)
UNC-Chapel Hill / Bill and Melinda Gates Foundation
Fetal Age and Machine Learning Initiative (ULTRA – Ultrasound for Resource-limited Areas)
The goal of this project is to investigate a position and orientation sensing system for ultrasound probe tracking for intelligent ultrasound scanners.

N00014-18-1-2538 (PI: Jiang) 07/01/2018-06/30/2021
ONR
Effect of AC Poling on Dielectric and Piezoelectric Properties of Relaxor Single Crystals for Acoustic Sensors
The goal of this project is to investigate the AC poling technology for relaxor-PT piezoelectric single crystals so that sensors with enhanced performance can be resulted.

DE-NE0008708 (PI: Jiang) 10/01/2017-09/30/20201
DOE
High Temperature Embedded/Integrated Sensors (HiTEIS) for Remote Monitoring of Reactor and Fuel Cycle System
The goal of this project is to develop high temperature and radiation resistant sensors including laser ultrasound transducers, acoustic emission sensors, etc. for structural health monitoring of reactor and fuel cycle systems in nuclear power plants.

Completed Research Support (last three years)

2017-2869 (PI: Jiang) 02/20/2018-05/15/2020
Goodix Technology
Novel Biomedical Ultrasound Sensors
The goal of this project is to develop ultrasound sensors for fingerprint display.

R03EB022311 (PI: Muller) 09/30/2016-07/31/2019
NIH
Ultrasonic Characterization of Atherosclerotic Plaque using Multiple Scattering
The goal of this project is to characterize atherosclerotic plaque using dual frequency transducers and multiple scattering from microbubbles.
Role: Co-Investigator

N00014-15-1-2418 (PI: Jiang) 06/01/2015-08/31/2018
ONR
Surface and Domain Nano-Engineering of Relaxor Single Crystals for Acoustic Sensors
The goal of this project is to investigate the structure-property relationship of relaxor-PT piezoelectric single crystals via nano-domain engineering.

R01EB015508 (PI: Jiang) 08/01/2012-05/31/2018
NIH
Dual-frequency Intravascular Arrays for Functional Imaging of Atherosclerosis
The goal of this project is to develop dual frequency ultrasound transducers and arrays for contrast enhanced IVUS imaging.

OMB No. 0925-0001 and 0925-0002 (Rev. 03/2020 Approved Through 02/28/2023)

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Stouffer, George A.

eRA COMMONS USER NAME (credential, e.g., agency login): GEORGE_STOUFFER

POSITION TITLE: Ernest and Hazel Craige Distinguished Professor of Medicine, Chief, Division of Cardiology and Co-Director of McAllister Heart Institute

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Bucknell University, Lewisburg, PA	BSME, BA	1980	Mechanical Engineering, Mathematics
University of Maryland, Baltimore, MD	MD	1987	Medicine

A. Personal Statement

My interests as both an interventional cardiologist and as a basic science and clinical researcher include identifying factors that contribute to the development and progression of atherosclerosis, understanding the relationship between atherosclerosis and thrombosis, and determining responses to treatments for acute coronary syndromes such as percutaneous coronary interventions, anti-platelet drugs, anticoagulants and thrombolytic agents. My basic science laboratory has a long-standing interest in understanding the factors that regulate thrombosis. I have >25 years of experience in interpreting intravascular ultrasound (IVUS) images and look forward to working on this project to develop sonothrombolytic therapies.

B. Positions and Honors

Positions and Employment

University of Virginia, Charlottesville, VA

1987-1990 Intern and Resident, Department of Internal Medicine
1990-1992 NIH Cardiovascular Research Fellow, Department of Molecular Physiology and Biological Physics
1992-1994 Clinical Fellowship, Division of Cardiology
1994-1995 Interventional Fellowship, Division of Cardiology and Instructor in Medicine

University of Texas Medical Branch, Galveston, TX

1995-1999 Assistant Professor of Medicine and Co-Director of Clinical Research, Division of Cardiology
1999-2000 Associate Professor of Medicine, Co-Director of Clinical Research and Associate Director of the Cardiac Catheterization Laboratory, Division of Cardiology
1995-2000 Principal Investigator, Sealy Center for Molecular Cardiology

University of North Carolina, Chapel Hill, NC

2000-2004 Associate Professor of Medicine
2000- Member of the McAllister Heart Institute and the Lineberger Comprehensive Cancer Center
2001-2015 Director of the Cardiac Catheterization Laboratories and Director of Interventional Cardiology
2004- Professor of Medicine
2005-2018 Henry A. Foscue Distinguished Professor of Medicine and Cardiology
2005-2007 Vice Chief of the Division of Cardiology for Clinical Affairs
2007-2013 Chief of Clinical Cardiology
2013- Chief of Cardiology
2015- Physician in Chief, UNC Medical Center Heart and Vascular Service Line
2016- Co-Director of McAllister Heart Institute
2018- Ernest and Hazel Craige Distinguished Professor of Medicine

Honors

Magna Cum Laude graduate, Bucknell University (1980)
Alpha Omega Alpha Honor Medical Society (1986)
Phi Beta Kappa National Honor Society (1979)
Tau Beta Pi National Engineering Honor Society (1979)
Pi Mu Epsilon National Mathematics Honor Society (1980)
Omicron Delta Kappa National Leadership Honor Society (1980)
World Bank Design Award for Photovoltaic Powered Water Pumping System (1982)
First Prize, American College of Physicians Residents Competition (1989)
Finalist, American College of Cardiology Young Investigator Award in Cellular and Molecular Cardiology (1997)
First Prize, Astra-Merck Young Investigators Forum (1998)
American Federation for Medical Research Junior Faculty Award (1999)
Advisory Assistant Editor for Journal of Thrombosis and Haemostasis (2001 – 2005)
Editorial Board member of the Journal of the American College of Cardiology – Cardiovascular Interventions and the American Journal of Cardiology
Member of the Guidelines Committee for Invasive Diagnostic Procedures, Society of Cardiac Angiography and Interventions (2012 – 2014)
Ad hoc reviewer for the Atherosclerosis and Inflammation Cardiovascular Sciences Study Section (AICS), NIH (2003 – 2005)
Elected to membership in The American Clinical and Climatological Association (2009)
Medical Student Research Mentor Award given by the Academy of Educators (2014)
Councilor of the NC Chapter of the American College of Cardiology (2013 – 2018)

C. Contribution to Science

1. A focus of recent clinical research is on patients who have a ST elevation myocardial infarction (STEMI) while hospitalized for a non-cardiac condition. There has been a large national effort to improve care of patients who have a STEMI as an outpatient. In contrast, little is known about STEMI occurring in hospitalized patients ("inpatient STEMI"). My group has published the largest study in this area and also developed a quality improvement program to expedite treatment of these patients. We are now coordinating a multi-center study to look at interventions to improve outcomes in these patients.
    a. Dai X, Bumgarner J, Spangler A, Meredith D, Smith SC, Stouffer GA. Acute ST Elevation Myocardial Infarction in Patients Hospitalized for Non-cardiac Conditions. Journal of the American Heart Association 2013;2:e000004. PMCID: PMC3647284
    b. Kaul P, Federspiel JJ, Dai X, Stearns SC; Smith Jr, SC; Yeung M; Beyhaghi H, Zhou L, Stouffer GA. Association of inpatient-onset vs outpatient-onset of ST elevation myocardial infarction with treatment and clinical outcomes. JAMA 2014; 312:1999-2007. PMCID: PMC4266685
    c. Dai X, Kaul P, Smith SC, Stouffer GA. Predictors, Treatment and Outcomes of Inpatient ST Elevation Myocardial Infarction. Nature Reviews Cardiology 2016;13:148-154. PMID: 26525542
    d. Dai X, Meredith D, Sawey E, Kaul P, Smith Jr, SC; Stouffer GA. A Quality Improvement Program Improved Recognition and Treatment Times For Inpatient ST-Elevation Myocardial Infarctions JAMA Cardiol 2016;1:1077-1079. Doi:10.1001

2. Another focus of my clinical research is on using CYP2C19 genotyping to identify the best anti-platelet agent for patients with acute coronary syndromes and/or undergoing percutaneous coronary intervention. Bioactivation of clopidogrel to its active thiol metabolite is dependent on CYP2C19. Loss-of-function polymorphisms in CYP2C19 are common and confer a reduced capacity for clopidogrel bioactivation and platelet inhibition compared with wild-type. Although the relationship between CYP2C19 genotype, clopidogrel use and clinical outcomes remains controversial, multiple studies have shown higher rates of stent thrombosis and myocardial infarction in CYP2C19 non-responders.
    a. Rossi JS, Cammarata M, Dharmavaram J, Weck K, Walko C, Gabriel D, Kuritzky J, Muldrew K, Stouffer GA. (2014). Clopidogrel Dose Adjustment after Outpatient Screening for CYP2C19 Variant Alleles: A Pilot Study. Pharmacogenomics,15:915-23. PMID: 24956245
    b. Lee JA, Lee CR, Reed BN, Plitt DC, Polasek MJ, Howell LA, Cicci JD, Tasca KE, Weck KE, Rossi JS, Stouffer GA (2015). Implementation and evaluation of a CYP2C19 genotype-guided antiplatelet therapy algorithm in high-risk coronary artery disease patients. Pharmacogenomics, 16:303-313. PMID: 25823779

c. Cavallari LH, Lee CR, Beitelshees AL, Cooper-DeHoff, RM, Duarte JD, Voora D, Kimmel SE, McDonough CW, Gong Y, Dave CV, Pratt VM, Alestock, TD, Anderson RD, Alsip J, Ardati AK, Brott BC, Brown L, Chumnumwat S, Clare-Salzler MJ, Coons JC, Denny JC, Dillon C, Elsey AR, Hamadeh I, Harada S, Hillegass WB, Hines L, Horenstein RB, Howell LA, Jeng LJB, Kelemen MD, Lee YM, Magvanjav O, Montasser M, Nelson DR, Nutescu EA, Nwaba DC, Pakyz RE, Palmer K, Peterson JF, Pollin TI, Quinn AH, Robinson SW, Schub J, Skaar TC, Smith DM, Sriramoju VB, Starostik P, Stys TP, Stevenson JM, Varunok N, Vesely MR, Wake D, Weck KE, Weitzel KW, Wilke RA, Willig J, Zhao RY, Kreutz RP, Stouffer GA, Empey PE, Limdi NA, Shuldiner AR, Winterstein AG, Johnson JA on behalf of the IGNITE Network Pharmacogenetics Working Group (2018). Multi-site Investigation of Outcomes with Implementation of CYP2C19 Genotype-Guided Antiplatelet Therapy after Percutaneous Coronary Intervention. JACC-Cardiovascular Interventions,11:181-91. PMCID: PMC5775044.

d. Lee CR, Sriramoju VB, Cervantes A, Howell LA, Varunok N, Madan S, Hamrick K, Polasek MJ, Lee JA, Clarke M, Cicci JD, Weck KE, **Stouffer GA**. Clinical outcomes and sustainability of using CYP2C19 genotype-guided antiplatelet therapy after percutaneous coronary intervention. Circulation-Genomics and Precision Medicine 2018;11:e002069. PMCID: PMC5889089

3. My laboratory has a long-standing interest in understanding the direct effects of thrombin on smooth muscle cells (SMC). Thrombin is present at high concentrations at sites of vascular injury and elicits many effects on SMC including proliferation, migration, matrix production and cell adhesion which contribute to atherosclerosis and the development of restenosis after percutaneous coronary intervention (PCI). More recently, my laboratory has focused on identifying thrombin-induced signaling pathways in SMC.
   a. Stouffer GA, Sarembock IJ, McNamara CA, Gimple LW, Owens GK. Thrombin-induced mitogenesis of vascular SMC is partially mediated by autocrine production of PDGF-AA. American Journal of Physiology 1993;265:C806-C811. PMID: 8214037
   b. Stouffer GA, Runge MS: Thrombin and Restenosis - Has the question been answered?. Journal of American College of Cardiology, 1996;28:1856-1857. PMID: 8962576
   c. Stouffer GA and Smyth SS: Effects of thrombin on interactions between $\beta 3$ integrins and extracellular matrix in platelets and vascular cells. Arteriosclerosclerosis, Thrombosis, Vascular Biology 2003;23:1971-1978. PMID: 12947018
   d. Sajid M, Zhao R, Pathak A, Smyth SS, Stouffer GA: $\alpha_v\beta_3$ integrin antagonists inhibit thrombin-induced proliferation and focal adhesion formation in smooth muscle cells. American Journal Physiology- Cell Physiology 2003;285:C1330-C1338. PMID: 12878490

4. My laboratory has a long-standing interest in the role that β_3 integrins play in mediating SMC responses to thrombin. In the normal artery, vascular expression of β_3 integrins is generally limited to the luminal endothelial monolayer. Increased β_3 integrin expression is found in human atherosclerotic lesions and following vascular injury in numerous animal models, including baboons, rats, rabbits, mice and pigs. β_3 integrins antagonists reduce neointimal formation following vascular injury in animal models. Our recent studies have focused on identifying mechanisms whereby β_3 integrins mediate responses to thrombin and other SMC agonists.
   a. Stouffer GA, Hu Z, Sajid M, Jin G, Nakada MT, Hanson SR, Runge MS: β_3 integrins are upregulated following vascular injury and mediate proliferation of cultured smooth muscle cells. Circulation 1998;97:907-915, doi.org/10.1161/01
   b. Lele M, Sajid M, Wajih N, Stouffer GA: Eptifibatide and 7E3, but not tirofiban, inhibit $\alpha_v\beta_3$ integrin-mediated binding of smooth muscle cells to thrombospondin and prothrombin. Circulation 2001;104:582-587. PMID: 11839641
   c. Stouffer GA, Pathak A, Zhao R, Huang J. Down but not out: New insights into the role of $\alpha_v\beta_3$ integrins in vascular healing. Arteriosclerrosis, Thrombosis, Vascular Biology 2005;25:1309-1310. PMID: 15980268
   d. Pathak A, Zhao R, Huang J, and Stouffer GA. Eptifibatide and abciximab inhibit insulin-induced focal adhesion formation and proliferative responses in human aortic smooth muscle cells. Cardiovascular Diabetology 2008;7:36. PMCID: PMC2628888

Complete List of Published Work in MyBibliography:
http://www.ncbi.nlm.nih.gov/sites/myncbi/george.stouffer.1/bibliography/42302248/public/?sort=date&direction=descending

D. Research Support

Ongoing Research Support

None

Completed Research Support (last three years)

None

OMB No. 0925-0001 and 0925-0002 (Rev. 03/2020 Approved Through 02/28/2023)

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Brown, Ashley Carson

eRA COMMONS USER NAME (credential, e.g., agency login): ACARSON

POSITION TITLE: Assistant Professor of Biomedical Engineering

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Clemson University, Clemson, SC, USA	BS	08/2002	Biosystems Engineering
Georgia Institute of Technology, Atlanta, GA, USA	PhD	08/2011	Bioengineering
Georgia Institute of Technology, Atlanta, GA, USA	Postdoc	02/2014	Chem and Biochem/ Biomedical Engineering

A. Personal Statement

My long-term research goals are to understand mechanisms involved in coagulation and develop novel therapies that augment the body's native clotting and subsequent healing processes. My doctoral and postdoctoral training provided me with experimental and theoretical knowledge in the areas of coagulation, biomaterial design, fibrin mechanics, and cellular mechanotransduction mechanisms in fibrosis and wound healing, thus providing a firm foundation for the pursuit of my long-term research goals. My current research interests include developing novel biomaterial strategies to direct cellular behavior and promote pro-regenerative responses. A primary research focus in my group in this area is the development of fibrin-targeting materials for treatment of bleeding and targeted delivery of fibrinolytics, including tPA. I have published 46 peer reviewed articles, 5 book chapters, and 1 editorial in these various areas, with 26 of those as corresponding author. More recently, I have been working with Dr. Dayton and Dr. Wolberg to explore technologies to optimize delivery of nanodroplets to thrombus, and I look forward to working with the team to develop novel arterial clot disruption therapies in this project. Specifically, I will extend my experience in these areas to the development of thrombus targeting nanodroplets and their combination reduced-dose tPA delivery. My group will work with Dr. Dayton's team to characterize clot targeting behavior of functionalized nanodroplets. The relevant publications are as follows:

a) Brown AC[+] and Levy J. Maintaining hemostatic balance in treating disseminated intravascular coagulopathy. *Anesthesiology*. 2019, doi: 10.1097/ALN.0000000000002862.
b) Mihako, E., Huang, K., Sproul, E., Cheng, K., Brown, A.C[+]. Targeted Treatment of Ischemic and Fibrotic Complications of Myocardial Infarction Using a Dual-Delivery Microgel Therapeutic. *ACS Nano.* 2018, doi: 10.1021/acsnano.8b01977.
c) Nandi, S., Sproul, E., Nellenbach, K., Erb, M., Gaffney, L., Freytes, D., Brown, A.C.[+], Platelet-like particles dynamically stiffen fibrin matrices and improve wound healing outcomes. *Biomaterials Science,* 2018, doi: 10.1039/C8BM01201F.
d) Brown, A.C.[*], Stabenfeldt, S.E.[*], Ahn, B., Hannan, R., Dhada, K., Herman, E., Stefanelli, V., Guzzetta, N., Alexeev, A., Lam, W.A., Lyon, L.A., Barker, T.H. Ultrasoft microgels displaying emergent platelet-like behaviours. *Nature Materials,* 2014, **12**, 1108-14. doi:10.1038/nmat4066.

B. Positions and Honors

Positions and Employment

2014-2015 Research Scientist II, School of Chemistry and Biochemistry, Atlanta, GA
2015-present Assistant Professor, Joint Department of Biomedical Engineering, North Carolina State University, Raleigh, NC and University of North Carolina, Chapel Hill, NC

2016-present Affiliated Assistant Professor, Department of Materials Science and Engineering, North Carolina State University, Raleigh, NC

Honors and Awards

1. Tissue Engineering and Regenerative Medicine Society (TERMIS) AM Young Investigator Award (2020)
2. *Cellular and Molecular Bioengineering* Young Innovator (2020)
3. American Society for Matrix Biology Junior Investigator Award (2020)
4. North Carolina State University Faculty Scholar (2020)
5. National Science Foundation CAREER Award (2019)
6. American Heart Association Kenneth M. Brinkhous Young Investigator Prize in Thrombosis Finalist (2019)
7. *Biomaterials Science* Emerging Investigator (2019)
8. National Academy of Engineering Frontiers of Engineering Invitee (2018)
9. Biomaterials Gordon Research Conference Young Investigator Poster Award (2017)
10. American Heart Association Scientist Development Award Recipient (2016)
11. American Heart Association Postdoctoral Fellow (2012-2014)
12. American Society of Hematology Abstract Achievement Award (2012)
13. Society for Biomaterials Student Travel Achievement Recognition (STAR) Award (2011)
14. TERMIS-EU 50 Best Abstracts Award (2010)
15. 13th Annual Regenerative Medicine Conference Travel Award (2009)
16. Georgia Tech Cell and Tissue Engineering Training Grant Awardee (2008-2010)
17. 12th Annual Regenerative Medicine Conference Travel Award (2008)
18. Georgia Institute of Technology Presidential Fellowship (2006-2010)

Recent Professional Activities and Memberships

2020-present	Biomedical Engineering Society Student Affairs Subcommittee Co-chair
2020	Ad hoc reviewer, NIH Oral, Dental, and Craniofacial Sciences (ODCS) Study Section
2019-present	Society for Biomaterials Council Member
2019-present	Society for Biomaterials Engineering Cells and Their Microenvironment Special Interest Group Chair
2018-present	American Society for Matrix Biology Council Member
2016-present	Biomedical Engineering Society Student Affairs Subcommittee Member
2008-present	Member, American Society for Matrix Biology
2008-present	Member, Society for Biomaterials
2010-present	Member, Tissue Engineering and Regenerative Medicine International Society
2012-present	Member, American Heart Association
2012-present	Member, Biomedical Engineering Society
2012-present	Member, American Society for Hematology
2014-present	Member, International Fibrinogen Research Society
2018-present	Member, International Society on Thrombosis and Haemostasis

C. Contribution to Science

1. **Designing bioresponsive platelet-like particles**: A major focus of my research is the development of fibrin-targeting platelet-like particles (PLPs) that interact specifically with nascent fibrin fibers at wound sites to augment clot formation and stop bleeding following traumatic injury. To maximize interactions with fibrin networks, we utilized highly deformable, ultra-low crosslinked (ULC) pNIPAm microgels with multiple sites for chemoligation as the base material for our hemostatic material. To impart fibrin specificity to the microgels, humanized synthetic single domain variable fragment (sdFv) antibodies with high affinity for fibrin were identified through molecular evolutionary techniques and then coupled to the microgels. Like natural platelets, these PLPs specifically target fibrin rather than fibrinogen, interactive extensively with fibrin networks, augment clotting *in vitro* and decrease bleeding times in rodent models of traumatic injury. Importantly, PLPs were found to actively collapse fibrin networks, an emergent behavior that mimics *in vivo* clot contraction. These studies demonstrate that this clot collapse feature was intimately related to 1) the high level of deformability of ULC particles and 2) the high affinity interactions of PLPs with the fibrin networks. I have shown the utility of these particles in treating bleeding following injury and in promoting healing. I have served as a primary investigator and PI in these studies.

a) Brown, A.C.*, Stabenfeldt, S.E.*, Ahn, B., Hannan, R., Dhada, K., Herman, E., Stefanelli, V., Guzzetta, N., Alexeev, A., Lam, W.A., Lyon, L.A., Barker, T.H. Ultrasoft microgels displaying emergent platelet-like behaviours. *Nature Materials,* 2014, **12**, 1108-14. doi:10.1038/nmat4066.
b) Myers, D., Qiu, Y., Chester, D., Fay, M., Sakurai, Y., Tran, R., Ciciliano, J., Ahn, B., Mannino, R., Bunting, S., Bennett, C., Briones, M., Smith, M., Brown, A.C., Sulchek, T., Lam, W. Single-platelet nanomechanics measured by high-throughput cytometry. *Nature Materials,* 2016, doi: 10.1038/nmat4772.
c) Sproul, E., Nandi, S., Roosa, C., Schreck, L., Brown, A.C.+ Biomimetic microgels with controllable deformability improve healing outcomes. *Advanced Biosystems*, 2018, doi: 10.1002/adbi.201800042.
d) Nandi, S., Sproul, E., Gaffney, L., Freytes, D., Brown, A.C.+, Platelet-like particles dynamically stiffen fibrin matrices and improve wound healing outcomes. *Biomaterials Science,* 2018, doi: 10.1039/C8BM01201F. *****Featured in the 2019 Biomaterials Science Emerging Investigators Collection.***

2. **Engineering cellular microenvironments to promote wound healing:** In a number of projects, where I have contributed as a primary investigator and corresponding author, I have utilized ECM biology principles to design environments for eliciting specific cellular responses to promote healing. This has included 1) designing fibrin-based colloids and fibrin-targeting colloids (i.e. PLPs) to enhance clot properties to promote wound healing and 2) utilizing integrin-specific recombinant fibronectin fragments coupled with biophysical cues to direct EMT responses for wound healing.
   a) Nandi, S., Mohanty, K., Nellenbach, K., Erb, M., Muller, M., Brown, A.C.+, Ultrasound enhanced synthetic platelet therapy for augmenting wound repair. *ACS Biomaterials Science and Engineering,* 2020. Doi: 10.1021/acsbiomaterials.9b01976.
   b) Chee, E., Nandi, S., Nellenbach, K., Mihalko, E., Snider, D., Morrill, L., Bond, A., Sproul, E., Sollinger, J., Cruise, G., Hoffman, M. Brown, A.C.+, Nanosilver composite pNIPAm microgels for the development of antimicrobial platelet-like-particles. *Journal of Biomedical Materials Research Part B.* 2020. Doi: 10.1002/jbm.b.34592.
   c) Muhamed, I., Sproul, E., Ligler, F., Brown, A.C.+ Fibrin nanoparticles coupled with keratinocyte growth factor enhance dermal wound healing rate. *ACS Applied Materials & Interfaces*, 2019, doi: 10.1021/acsami.8b21056.
   d) Brown, A.C., Dysart, M.M., Clarke, K.C., Stabenfeldt, S.E., Barker, T.H. Integrin $\alpha 3\beta 1$ binding to fibronectin is dependent on the 9th type III repeat. *Journal of Biological Chemistry*, 2015, Oct 16;290(42):25534-47. doi: 10.1074/jbc.M115.656702.

3. **Understanding the role of ECM mechanics and protein composition in tissue repair responses:** ECM mechanics and composition are dynamic during wound healing. During my doctoral studies, I demonstrated that increased tissue stiffness is a key regulator of disease associated epithelial to mesenchymal transitions (EMT) in pulmonary fibrosis. These studies identified matrix stiffness and/or cell contractility as critical targets for novel therapeutics for fibrotic diseases. These studies also demonstrated that the responses are dependent on integrin-specific interactions, which can be quite dynamic in the wound microenvironment. I was a primary investigator in these studies. In my independent group, I have focused on developing novel therapeutic strategies to combat fibrotic responses, including targeted delivery of cell contractility inhibitors to sites of fibrosis. I have also focused on developing novel colloidal based material platforms to further study the role of viscoelasticity in modulating cell contractility and downstream effects on fibrotic responses.
   a) Brown, A.C., Fiore, V.F., Sulchek, T.A., Barker, T.H. Physical and chemical microenvironmental cues orthogonally control the degree and duration of fibrosis associated epithelial to mesenchymal transitions. *J Pathol*. 229, 25-35, 2013.
   b) Brown, A.C., Dysart, M.M., Clarke, K.C., Stabenfeldt, S.E., Barker, T.H. Integrin $\alpha 3\beta 1$ binding to fibronectin is dependent on the 9th type III repeat. *Journal of Biological Chemistry,* 2015, Oct 16;290(42):25534-47. doi: 10.1074/jbc.M115.656702.
   c) Mihako, E., Huang, K., Sproul, E., Cheng, K., Brown, A.C+. Targeted Treatment of Ischemic and Fibrotic Complications of Myocardial Infarction Using a Dual-Delivery Microgel Therapeutic. *ACS Nano,* 2018, doi: 10.1021/acsnano.8b01977.
   d) Chester, D., Kathard, R., Nortey, J., Nellenbach, K. Brown, A.C.+, Viscoelastic properties of microgel thin films control fibroblast modes of migration and pro-fibrotic responses. *Biomaterials*, 2018, doi: 10.1016/j.biomaterials.2018.09.012.

4. **Understanding coagulopathy in neonates following cardiopulmonary bypass:** Neonates have documented quantitative (concentration) and qualitative (functional) deficiencies of many coagulation proteins including both thrombin and fibrinogen. The consequences of fetal fibrinogen on clot structure in neonates, particularly in the context of surgery-associated bleeding, were not well characterized at the time of this investigation. In these studies, we examined the sequential changes in clotting components and resultant clot structure in a small sample of neonates undergoing cardiac surgery and cardiopulmonary bypass (CPB). We found that clots formed from neonatal plasma collected after CPB was more porous than baseline clots, and that the transfusion of adult fibrinogen component (current standard of care) did not result in a return to baseline structure. Further analysis with purified neonatal and adult fibrinogen demonstrated that the structure of neonatal and adult clots differed sustainably, and neonatal clots degraded significantly faster than adult clots. The two fibrinogen components did not integrate well when mixed clots (simulating transfusion conditions) were formed. Furthermore, mixed clots were found to have degradation properties similar to pure adult clots. Such properties could lead to adverse thrombotic events and these findings suggest that differential treatment strategies for neonates should be pursued to reduce the demonstrated morbidity of blood product transfusion. I was a primary investigator in these studies.
    a) Brown, A.C., Hannan, R., Timmins, L., Fernandez, J., Barker, T.H., Guzzetta, N. Fibrin network changes in neonates after cardiopulmonary bypass. *Anesthesiology,* 2016, 5, 124, pp 1021-1031. doi: 10.1097/ALN. 0000000000001058
    b) Nellenbach, K., Guzzetta, N., and Brown, A.C.+ Analysis of the structural and mechanical effects of procoagulant agents on neonatal fibrin networks following cardiopulmonary bypass. *Journal of Thrombosis and Haemostasis,* 2018, doi: 10.1111/jth.14280.
    c) Nellenbach, K., Nandi, S., Kyu, A., Sivadanam, S., Guzzetta, N., and Brown, A.C.+ Comparison of neonatal and adult fibrin clot properties between porcine and human plasma. *Anesthesiology,* 2020, doi: 10.1097/ALN.0000000000003165.
    d) Nellenbach, K., Nandi, S., Peeler, C., Kyu, A., Brown, A.C.+ Neonatal fibrin scaffolds promote enhanced cell adhesion, migration, and wound healing *in vivo* compared to adult fibrin scaffolds. *Cellular and Molecular Bioengineering.* 2020, doi: 10.1007/s12195-020-00620-5. *****Featured in the 2020 Young Innovators Collection.***

Complete List of Published Work in MyBibliography:
http://www.ncbi.nlm.nih.gov/sites/myncbi/1Pq0z-9lmtoQc/bibliography/47877502/public/?sort=date&direction=ascending.

D. Research Support

Ongoing Research Support

R01HL146701 (PI: Brown) 08/15/2019-07/31/2024
NIH NHLBI
Targeted treatment of thrombotic occlusions using a dual-microgel therapeutic
The goal of this project is to develop fibrin-targeting core-shell microgels for delivery of tissue plasminogen activator and a ROCK inhibitor to treat thrombotic occlusions and ischemia reperfusion injury.
Role: PI

DMR-1847488 (PI: Brown) 04/01/2019-03/31/2024
NSF
CAREER: Dynamic microgels that mimic platelet behavior to promote healing
The goal of this project is to develop a platelet-mimetic particle that changes shape in response to wound triggers.
Role: PI

CMMI-1825398 (PI: Brown) 09/01/2018-08/31/2021
NSF
A multiscale material approach to understanding the effects of viscoelasticity on cell adhesion, migration, and TGFβ activation/signaling
The goal of this project is to utilize microgel based thin films in conjunction with DNA origami to understand the role of material viscoelasticity on directly cell adhesion, migration, and cytokine responses.
Role: PI

No Grant Number (PI: Brown) 07/01/2018-12/30/2020 (NCE)
NCSU Chancellor's Innovation Fund
Antimicrobial platelet-like-particles to stop hemorrhage and promote healing
The goal of this project is to evaluate the translational potential of antimicrobial PLPs.
Role: PI

R01HL130918-01A1 (MPIs: Barker, Lam) 07/01/2016-06/30/2021 (NCE)
University of Virginia Sub-award; NIH NHLBI
Platelet-like-particles for augmenting hemostasis
The goal of this project is to investigate the efficacy of hemostatic PLPs in various models of platelet deficiency.
Role: Co-I

W81XWH-15-1-0485 (PI: Barker) 09/15/2015-11/29/2020 (NCE)
University of Virginia Sub-award; DOD
Development of platelet-like particles for augmentation of hemostasis in congenital heart defect patients at high risk for bleeding during cardiac surgery
The goal of this study is to characterize the safety of synthetic hemostatic platelet-like particles in various animal models of bleeding.
Role: Co-I

R01AR071985-01A1 (PI: Fisher) 04/01/2018-03/31/2023
NIH NIAMS
Sex- and Age-dependent ACL Function in the Growing Knee Joint
The goal of this proposal is to determine how age and sex impact ACL maturation during skeletal growth and joint function after ACL injury and to assess if this knowledge can be applied to improve treatment.
Role: Co-I

Completed Research Support (last three years)

R21AR071017 (PI: Brown) 07/01/2017-06/30/2020 (NCE)
NIH NIAMS
Ultrasound enhanced platelet-like particle therapy for accelerated wound repair
The goal of this project is to utilize ultrasound stimulation to enhance PLP deformation of fibrin networks to promote cell migration.
Role: PI

16SDG29870005 (PI: Brown) 07/01/2016-06/30/2020 (NCE)
AHA Scientist Development Grant (Mid-Atlantic Affiliate)
Development of anti-microbial platelet-like-particles for augmentation of hemostasis and advanced wound repair
The goal of this project is to develop anti-microbial platelet-like particles to promote hemostasis and wound repair following trauma
Role: PI

DMR-1726294 (PI: LeBeau) 08/15/2017-07/31/2019
NSF
MRI: Acquisition of a Transmission Electron Microscope for In-situ Studies of Soft and Hard Matter
This equipment grant seeks to acquire a Cryogenic TEM to facilitate in-situ studies of soft and hard materials.
Role: Co-PI

UL1TR001117 (PIs: Brown, Hoffman) 07/01/2017-06/30/2018
NIH: Duke/NCSU Translational Research Grant
Platelet-mimetic nanogels for augmentation of wound repair
The goal of this project is to investigate utility of platelet-mimetic nanogels in augmentation of impaired wound healing in hemophilia.
Role: PI

OMB No. 0925-0001 and 0925-0002 (Rev. 03/2020 Approved Through 02/28/2023)

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Tsuruta, James K

eRA COMMONS USER NAME (credential, e.g., agency login): JTSURUTA

POSITION TITLE: Asst Professor of Biomedical Engineering

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Whitman College, Walla Walla, WA	BA	05/1984	Chemistry
Washington State University, Pullman, WA	MS	01/1990	Biochemistry
Washington State University, Pullman, WA	PhD	11/1991	Biochemistry
University of North Carolina, Chapel Hill, NC	NIH training grant	11/1994	

A. Personal Statement

The foundation of my scientific training is built on a solid knowledge of and experience with the chemistry of proteins. My first training in biochemistry emphasized protein purification by two-dimensional electrophoresis and sequencing of proteins using Edman degradation. My graduate training shifted to using molecular biology to understand the role of Sertoli cells during spermatogenesis. At the University of North Carolina at Chapel Hill, I was a member of the Laboratories for Reproductive Biology (LRB) where I studied the molecular biology of the male gamete with Deborah O'Brien. The interaction between Sertoli cells and the developing male gamete provided a fascinating system to study the paracrine interactions between these cells utilizing techniques spanning biochemistry, cell biology, immunology, and molecular biology. While I was a member of the LRB, I collaborated with Paul Dayton (Biomedical Engineering, UNC) and Michael O'Rand (Cell Biology, UNC) to explore the use of therapeutic ultrasound as a reversible male contraceptive.

Currently, I am the lead formulation chemist for Paul Dayton's diagnostic ultrasound laboratory where I've been able to contribute my training in biochemistry, cellular, and molecular biology. My experiences synthesizing targeted-contrast agents and my involvement in molecular imaging studies have introduced me to an entirely different perspective on biology where the physics of acoustics is paramount. I have formulated microbubble contrast for drug delivery, for monitoring αvβ3 integrin expression, and for molecular imaging of Secreted Frizzled Related Protein-2 expression in angiosarcoma. Our studies on the accelerated blood clearance (ABC) of PEGylated ultrasound contrast agents identified immune clearance as the mechanism responsible for the decreased signal intensity of contrast wash-in curves after repeated imaging sessions. Our current application proposes strategies to control the dose of contrast agent perfusing an ROI so that results from molecular imaging studies can be compared between animals, throughout the course of longitudinal studies. Most importantly, the combination of my training in chemistry, biochemistry, extensive experiences formulating novel targeted and phase-change nanodroplet ultrasound contrast agents, and investigation of the immune clearance of PEGylated ultrasound contrast agents makes me ideally suited to contribute to this project.

a. Fix SM, Nyankima AG, McSweeney MD, Tsuruta JK, Lai SK, Dayton PA. Accelerated Clearance of Ultrasound Contrast Agents Containing Polyethylene Glycol is Associated with the Generation of Anti-Polyethylene Glycol Antibodies. Ultrasound Med Biol. 2018 Jun;44(6):1266-1280. PubMed PMID: 29602540; PubMed Central PMCID: PMC6171506.
b. Tsuruta JK, Schaub NP, Rojas JD, Streeter J, Klauber-DeMore N, Dayton P. Optimizing ultrasound molecular imaging of secreted frizzled related protein 2 expression in angiosarcoma. PLoS One. 2017;12(3):e0174281. PubMed PMID: 28333964; PubMed Central PMCID: PMC5363853.
c. Kasoji SK, Pattenden SG, Malc EP, Jayakody CN, Tsuruta JK, Mieczkowski PA, Janzen WP, Dayton PA.

Cavitation Enhancing Nanodroplets Mediate Efficient DNA Fragmentation in a Bench Top Ultrasonic Water Bath. PLoS One. 2015;10(7):e0133014. PubMed PMID: 26186461; PubMed Central PMCID: PMC4505845.

d. Tsuruta JK, Dayton PA, Gallippi CM, O'Rand MG, Streicker MA, Gessner RC, Gregory TS, Silva EJ, Hamil KG, Moser GJ, Sokal DC. Therapeutic ultrasound as a potential male contraceptive: power, frequency and temperature required to deplete rat testes of meiotic cells and epididymides of sperm determined using a commercially available system. Reprod Biol Endocrinol. 2012 Jan 30;10:7. PubMed PMID: 22289508; PubMed Central PMCID: PMC3340307.

B. Positions and Honors

Positions and Employment

1981-1982 Student Academic Advisor, Whitman College, Walla Walla, WA
1983-1984 Laboratory Assistant, Whitman College, Walla Walla, WA
1985-1986 Teaching Assistant, Washington State University, Pullman, WA
1986-1991 Research Assistant, Washington State University, Pullman, WA
1991-1994 Postdoctoral Trainee, University of North Carolina, Chapel Hill, NC
1994-2000 Postdoctoral Research Associate, University of North Carolina, Chapel Hill, NC
2000-2008 Assistant Professor of Pediatrics, University of North Carolina, Chapel Hill, NC
2008- Assistant Professor of Biomedical Engineering, University of North Carolina, Chapel Hill, NC

Honors

1979 National Merit Scholar, Whitman College, Walla Walla, WA
1979 Honors at Entrance, Whitman College, Walla Walla, WA
1985 Departmental Award, Washington State University, Department of Chemistry, Pullman, WA
1991 New Investigator Award, The Society for the Study of Reproduction, Annual Meeting, Vancouver, BC, Canada
1995 Invited Speaker, North American Testis Workshop, Raleigh, NC
1999 Junior Investigator Award, Andrew W. Mellon Foundation
2008 Invited Speaker, First International Workshop on Male Reproductive Biology, Botucatu, Brazil
2010 Grand Challenges Explorations Grant, Ultrasound as a long-term, reversible male contraceptive, Bill and Melinda Gates Foundation

C. Contribution to Science

1. My work in the Dayton laboratory as Lead Formulation Chemist for ultrasound contrast agents has provided a solid foundation in the design, and formulation of targeted molecular imaging reagents. I have formulated microbubble contrast for drug delivery, for nanodroplet formation, for monitoring αvβ3 integrin expression, and for molecular imaging of Secreted Frizzled Related Protein-2 expression in angiosarcoma. I successfully formulated dexamethasone-loaded ultrasound contrast to carry this hydrophobic drug in the lipid shell of microbubbles and determined the amount of drug loading using a specific enzyme-linked immunosorbent assay (ELISA). As molecular imaging relies upon the dose-responsive binding between targeted contrast agents and cancer biomarkers, our characterization of the accelerated blood clearance of PEGylated contrast agents after repeated exposures made it clear that tumors were being under-dosed during longitudinal studies when the injected contrast dose was kept constant throughout the study. Our current focus is to formulate strategies to prevent the under-dosing of ROIs during molecular imaging or therapeutic applications of ultrasound.
    a. Seiler GS, Campbell N, Nixon B, Tsuruta JK, Dayton PA, Jennings S, Redding WR, Lustgarten M. FEASIBILITY AND SAFETY OF CONTRAST-ENHANCED ULTRASOUND IN THE DISTAL LIMB OF SIX HORSES. Vet Radiol Ultrasound. 2016 May;57(3):282-9. PubMed PMID: 26765518.
    b. Shelton SE, Lindsey BD, Tsuruta JK, Foster FS, Dayton PA. Molecular Acoustic Angiography: A New Technique for High-resolution Superharmonic Ultrasound Molecular Imaging. Ultrasound Med Biol. 2016 Mar;42(3):769-81. PubMed PMID: 26678155; PubMed Central PMCID: PMC5653972.
    c. Tsuruta JK, Klauber-DeMore N, Streeter J, Samples J, Patterson C, Mumper RJ, Ketelsen D, Dayton P. Ultrasound molecular imaging of secreted frizzled related protein-2 expression in murine angiosarcoma. PLoS One. 2014;9(1):e86642. PubMed PMID: 24489757; PubMed Central PMCID: PMC3906081.

d. Streeter JE, Gessner RC, Tsuruta J, Feingold S, Dayton PA. Assessment of molecular imaging of angiogenesis with three-dimensional ultrasonography. Mol Imaging. 2011 Dec;10(6):460-8. PubMed PMID: 22201537; PubMed Central PMCID: PMC3653613.

2. The foundation of my scientific training is built on a solid knowledge of and experience with the chemistry of proteins. My graduate project, mentored by Michael Griswold (Washington State University), was to identify proteins secreted by Sertoli cells with the goal of characterizing proteins that could mediate paracrine interactions between Sertoli cells and developing male germ cells. By sequencing the amino terminus of sulfated glycoprotein-1 (SGP-1), I contributed to the molecular cloning of a major secretion product of Sertoli cells, which we found was identical to the nonenzymic sphingolipid activator protein, prosaposin. Prosaposin/SGP-1 plays an important role in the male reproductive system as well as serving a protective role in the nervous system. An early foray into bioinformatics allowed me to characterize structural features of sulfated glycoprotein-2 (SGP-2) that may be important to its function, after amino terminal sequencing verified it to be identical to Clusterin, Apo lipoprotein J, and serum protein-40,40 (SP-40,40). This is another Sertoli cell secreted protein that is proving to be an important player in many processes outside of the testes, including: Alzheimer's disease, metabolic disease, and carcinogenesis. I also contributed to identifying major sperm proteins secreted by the mouse epididymis. My graduate work culminated by identifying the cysteine protease inhibitor, cystatin-C, and characterizing its stage-dependent expression in Sertoli cells and germ cells at both the mRNA and protein levels. The unique temporal and spatial expression of cystatin-C was found to be the mirror image of another highly expressed, Sertoli cell protein, cyclic-protein-2 (CP-2), which was cloned by William Wright (Johns Hopkins), and found to be identical to procathepsin-L, the proform of a cysteine protease. Our thesis was that cystatin-C's exact opposite expression pattern during the cycle of the seminiferous epithelium further regulated the already highly regulated expression of procathepsin-L. My graduate training has provided me a strong foundation in chemistry, biochemistry, cell biology, and molecular biology, allowing me to contribute to projects that span different disciplines.
   a. Tsuruta JK, O'Brien DA, Griswold MD. Sertoli cell and germ cell cystatin C: stage-dependent expression of two distinct messenger ribonucleic acid transcripts in rat testes. Biol Reprod. 1993 Nov;49(5):1045-54. PubMed PMID: 8286570.
   b. Tsuruta JK, Wong K, Fritz IB, Griswold MD. Structural analysis of sulphated glycoprotein 2 from amino acid sequence. Relationship to clusterin and serum protein 40,40. Biochem J. 1990 Jun 15;268(3):571-8. PubMed PMID: 2363694; PubMed Central PMCID: PMC1131476.
   c. Collard MW, Sylvester SR, Tsuruta JK, Griswold MD. Biosynthesis and molecular cloning of sulfated glycoprotein 1 secreted by rat Sertoli cells: sequence similarity with the 70-kilodalton precursor to sulfatide/GM1 activator. Biochemistry. 1988 Jun 14;27(12):4557-64. PubMed PMID: 3048385.

3. Whereas my graduate work focused on the role of the Sertoli cell during spermatogenesis, my post-doctoral work with Deborah O'Brien (University of North Carolina at Chapel Hill, UNC-CH) in the Laboratories for Reproductive Biology (Frank French, Director) focused on the molecular biology of the male gamete. We showed that mannose-6-phosphate (M6P) bearing glycoproteins secreted by Sertoli cells interacted with the IGF-II/M6P-receptor and were able to change gene expression in different populations of purified spermatogenic cells. Although the IGF-II/M6P-receptor was thought of as merely a trafficking receptor for lysosomal proteins, our studies with receptor specific analogs of insulin like growth factors established its signal transduction ability during spermatogonial, and spermatogenic cell development. My early work characterizing changes in gene expression were done using Northern, and RNAse protection assays but transitioned to using quantitative, real-time polymerase chain reaction (PCR) as the instrumentation for this more sensitive technique became affordable. I also contributed to the analysis of androgen-binding protein, and germ cell nuclear factor during this time. My post-doctoral training has provided me a strong foundation in cell isolation, purification, and culture, further training in molecular biology including real-time PCR, and DNA construct design, immunological techniques including immunohistochemistry, designing peptide antibodies, and ELISA assays.
   a. Tsuruta JK, Eddy EM, O'Brien DA. Insulin-like growth factor-II/cation-independent mannose 6-phosphate receptor mediates paracrine interactions during spermatogonial development. Biol Reprod. 2000 Oct;63(4):1006-13. PubMed PMID: 10993821.
   b. Zhang YL, Akmal KM, Tsuruta JK, Shang Q, Hirose T, Jetten AM, Kim KH, O'Brien DA. Expression of germ cell nuclear factor (GCNF/RTR) during spermatogenesis. Mol Reprod Dev. 1998 May;50(1):93-102. PubMed PMID: 9547515.
   c. Joseph DR, O'Brien DA, Sullivan PM, Becchis M, Tsuruta JK, Petrusz P. Overexpression of

androgen-binding protein/sex hormone-binding globulin in male transgenic mice: tissue distribution and phenotypic disorders. Biol Reprod. 1997 Jan;56(1):21-32. PubMed PMID: 9002629.

d. Tsuruta JK, O'Brien DA. Sertoli cell-spermatogenic cell interaction: the insulin-like growth factor-II/cation-independent mannose 6-phosphate receptor mediates changes in spermatogenic cell gene expression in mice. Biol Reprod. 1995 Dec;53(6):1454-64. PubMed PMID: 8562703.

4. My work on developing a noninvasive, reversible male contraceptive began in collaboration with Paul Dayton (Biomedical Engineering, UNC-CH, and NC State University), and Michael O'Rand (Cell Biology and Physiology, UNC-CH) with funding from the Parsemus Foundation, and the Bill and Melinda Gates Foundation. We were able to adapt a commercially available therapeutic ultrasound unit to cause the loss of developing spermatogenic cells, and to deplete epididymal sperm reserves. The loss of sperm reserves was rapid, and complete within two weeks. Our studies showed that temperature alone was not responsible for the contraceptive effect but that the effect was aided by elevated temperature. During this period, I was also able to contribute to a new method of using computer-aided sperm analysis to classify sperm motility patterns and to provide my expertise with real-time PCR to contribute to the characterization of prostate cancer PC-3 cells, and region-specific expression of a protease inhibitor in the rat epididymis.
   a. Silva EJ, Patrão MT, Tsuruta JK, O'Rand MG, Avellar MC. Epididymal protease inhibitor (EPPIN) is differentially expressed in the male rat reproductive tract and immunolocalized in maturing spermatozoa. Mol Reprod Dev. 2012 Dec;79(12):832-42. PubMed PMID: 23070980.
   b. Tsuruta JK, Dayton PA, Gallippi CM, O'Rand MG, Streicker MA, Gessner RC, Gregory TS, Silva EJ, Hamil KG, Moser GJ, Sokal DC. Therapeutic ultrasound as a potential male contraceptive: power, frequency and temperature required to deplete rat testes of meiotic cells and epididymides of sperm determined using a commercially available system. Reprod Biol Endocrinol. 2012 Jan 30;10:7. PubMed PMID: 22289508; PubMed Central PMCID: PMC3340307.
   c. Goodson SG, Zhang Z, Tsuruta JK, Wang W, O'Brien DA. Classification of mouse sperm motility patterns using an automated multiclass support vector machines model. Biol Reprod. 2011 Jun;84(6):1207-15. PubMed PMID: 21349820; PubMed Central PMCID: PMC3099585.
   d. Alekseev OM, Richardson RT, Tsuruta JK, O'Rand MG. Depletion of the histone chaperone tNASP inhibits proliferation and induces apoptosis in prostate cancer PC-3 cells. Reprod Biol Endocrinol. 2011 Apr 16;9:50. PubMed PMID: 21496299; PubMed Central PMCID: PMC3100250.

Complete List of Published Work in My Bibliography:
https://www.ncbi.nlm.nih.gov/myncbi/james.tsuruta.2/bibliography/public/

D. Research Support

Ongoing Research Support

R03EB026826 (PI: Tsuruta) 09/01/2018-05/31/2021 (NCE)
NIH
Increasing sensitivity and discrimination of breast cancer diagnosis using molecularly targeted, extravascular ultrasound imaging with ultrasound-activated, phase-change nanodroplet contrast agent
Our hypothesis: targeted nanodroplets for extravascular ultrasound imaging will increase the sensitivity of detecting breast cancer while maintaining the discriminatory power of targeted molecular imaging.

R21EB025306 (PI: Johnson) 09/01/2018-06/30/2021
NIH
Novel Targeted Iodinated Microbubbles for Molecular Imaging With Computed Tomography
The long-term goal of this research is to develop a MI paradigm for CT for use in clinical practice. The objective of this application is to generate a MI probe for CT comprised of targeted microbubbles doped with radiopaque polyesters.

R01HL141967-01A1 (PI: Jiang) 08/15/2018-06/30/2022
NIH
Forward Viewing Catheter-Delivered Microbubble Enhanced Sonothrombolysis (FV-CAMUS)
We propose to develop a new ultrasound imaging monitored intravascular sonothrombolysis technique using Forward Viewing CAtheter-directed MicrobUbble-enhanced Sonothrombolysis (FV-CAMUS), which can be effective to treat both partially occluded and completely occluded clots.

R21CA246414 (MPIs: Brudno, Dayton) 02/15/2020-01/31/2022
NIH
Image-guided, ultrasound-enhanced long-term intracranial drug delivery
In order to deliver drugs to brain, one must pass through the blood brain barrier, something which is possible with BBB disruption, but allows for a short window of time (~1 hour) to administer drugs. We propose combining ultrasound-mediated BBB disruption with refillable drug depots, which allow for local and sustained drug release following bolus administration of systemic doses of inert prodrugs to present therapeutic agents to the brain over months. This approach could have a significant impact in the treatment of brain diseases.

No Grant Number (PI: Johnson) 11/01/2019-10/31/2020
North Carolina Diabetes Research Collaborative
Development and validation of ultrasound molecular imaging agents for diabetic kidney disease by measuring bio-molecular interactions with a novel, sensitive, homogeneous proximity assay (AlphaScreen)
Our overall goal is to use the contrast-enhanced ultrasound (CEUS) as a clinical tool in the early diagnosis of DKD. To take these images a step further, in this project, we will feed the CEUS images into machine learning algorithms. Integrating machine learning into our research will provide a robust and objective interpretation of imaging results. It also provides a means of detection that may not be readily apparent to the human eye. This project will lay the foundation of using these sophisticated algorithms on CEUS images.

Completed Research Support (last three years)

R33CA206939 (MPIs: Pattenden, Dayton, Davis) 07/01/2017-06/30/2021 (NCE)*
NIH *Grant still active but Dr. Tsuruta's effort has ended
The Application of Enhanced Cavitation to Enable DNA and Chromatin Extraction from Archived Tissues
We have invented a unique chemically inert reagent based on a perfluorocarbon nanodroplet suspension that can be used to enhance and simplify the extraction of chromatin from formalin-fixed paraffin embedded (FFPE) tissue. We hypothesize that this unique reagent will enable detection of clinically meaningful chromatin signatures from FFPE tumor tissue potentially enabling, for the first time, the development of chromatin-based cancer diagnostics.

R43CA236177 (PI: Kasoji) 04/09/2019-03/31/2021 (NCE)*
Triangle Biotechnology / NIH *Grant still active but Dr. Tsuruta's effort has ended
SBIR: Technology for overcoming bottlenecks in chromatin extraction from challenging biological samples
Triangle Biotechnology recently developed a unique cavitation enhancing reagent (RapidShear) that substantially improves the efficiency of acoustic fragmentation of genomic DNA. However, this reagent formulation shows little to no improvement for chromatin fragmentation from fixed cells. We are developing a new reagent, MegaShear, a more aggressive formulation for acoustic cavitation enhancement designed specifically for chromatin processing. Preliminary studies indicated that MegaShear facilitates a substantial improvement in chromatin fragmentation throughput, consistency, and yield. Validation and optimization of this technology will alleviate a critical bottleneck in the ChIP workflow. MegaShear will enable a substantial leap in high quality sample preparation throughput desired by large companies, as well as an increase in accessibility by enabling the use of low-cost sonicators, desired by small academic labs.

R01CA195051 (MPIs: Borden, Dayton) 04/13/2015-03/31/2021 (NCE)*
Colorado State University/NIH *Grant still active but Dr. Tsuruta's effort has ended
Ultrasound Molecular Imaging to Assess Therapeutic Response
This proposal assessed therapeutic response with ultrasound molecular imaging.

R43CA232902 (PI: Kasoji) 07/05/2018-06/30/2020 (NCE)
NIH
Towards commercialization of cavitation-enhancing nanodroplets for DNA sample fragmentation in NGS applications.
This project optimized formulation and use of nanodroplets in preparing sheared DNA samples for NGS sequencing.

U01CA189281 (MPIs: Dayton, DeMore) 07/17/2015-06/30/2018
NIH
Improving breast ultrasound specificity through SFRP2 targeted molecular imaging
The project developed an SFRP2-molecularly targeted microbubble contrast agent to visualize breast tumor vasculature and compared the specificity of SFRP2-targeted microbubble contrast for tumor vessels to VEGF-targeted microbubble contrast.

OMB Number: 4040-0010
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, Budget Period 1

ORGANIZATIONAL DUNS*: 608195277

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: University of North Carolina at Chapel Hill

Start Date*: 07-01-2021 **End Date*:** 06-30-2022 **Budget Period: 1**

A. Senior/Key Person

	Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1 .	Dr.	Paul	A	Dayton	PhD	PD/PI	197,300.00	2.4			39,460.00	11,583.00	51,043.00
2 .	Dr.	Alisa	S	Wolberg	PhD	PD/PI	161,013.00	1.8			24,152.00	7,278.00	31,430.00
3 .	Dr.	Xiaoning		Jiang	Ph.D.	Co-Investigator	197,300.00			1	16,442.00	4,257.00	20,699.00
4 .	Dr.	George	A	Stouffer	MD	Co-Investigator	197,300.00	0.24			3,946.00	1,158.00	5,104.00
5 .	Dr.	Ashley	C.	Brown	PhD	Co-Investigator	147,875.00		1		12,323.00	3,950.00	16,273.00
6 .	Dr.	James	K	Tsuruta	PhD	Co-Investigator	78,750.00	2.76			18,112.00	6,262.00	24,374.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name:

Total Senior/Key Person **148,923.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
2	Post Doctoral Associates	24			107,839.00	20,332.00	128,171.00
3	Graduate Students	30			81,428.00	18,589.00	100,017.00
	Undergraduate Students						
	Secretarial/Clerical						
1	Statistician	0.6			7,500.00	2,284.00	9,784.00
1	Research Assistant Professor	0.6			4,200.00	1,429.00	5,629.00
2	Technician	3.6			14,227.00	5,734.00	19,961.00
9	**Total Number Other Personnel**					**Total Other Personnel**	**263,562.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**412,485.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

Tracking Number: GRANT13224045
Funding Opportunity Number: PAR-19-158 . Received Date: 2020-10-13T11:12:56.000-04:00

RESEARCH & RELATED BUDGET - SECTION C, D, & E, Budget Period 1

ORGANIZATIONAL DUNS*: 608195277

Budget Type*: ● Project ❍ Subaward/Consortium

Organization: University of North Carolina at Chapel Hill

Start Date*: 07-01-2021 **End Date*:** 06-30-2022 **Budget Period:** 1

C. Equipment Description

List items and dollar amount for each item exceeding $5,000

Equipment Item	Funds Requested ($)*
1 . Calibrated hydrophone	8,000.00
Total funds requested for all equipment listed in the attached file	
Total Equipment	**8,000.00**

Additional Equipment: File Name:

D. Travel	Funds Requested ($)*
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	2,250.00
2. Foreign Travel Costs	4,750.00
Total Travel Cost	**7,000.00**

E. Participant/Trainee Support Costs	Funds Requested ($)*
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	
Number of Participants/Trainees **Total Participant Trainee Support Costs**	

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, Budget Period 1

ORGANIZATIONAL DUNS*: 608195277
Budget Type*: ● Project ❍ Subaward/Consortium
Organization: University of North Carolina at Chapel Hill

Start Date*: 07-01-2021 **End Date*:** 06-30-2022 **Budget Period:** 1

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	24,000.00
2. Publication Costs	4,000.00
3. Consultant Services	
4. ADP/Computer Services	
5. Subawards/Consortium/Contractual Costs	
6. Equipment or Facility Rental/User Fees	10,500.00
7. Alterations and Renovations	
8 . Tuition	30,375.00
9 . Patient Incentives	250.00
Total Other Direct Costs	**69,125.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**496,610.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1 . Organized Research_On Campus	55.5	458,235.00	254,320.00
		Total Indirect Costs	**254,320.00**

Cognizant Federal Agency DHHS; Darryl Mayes; (202) 401-2808
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**750,930.00**

J. Fee	Funds Requested ($)*

K. Total Costs and Fee	Funds Requested ($)*
	750,930.00

L. Budget Justification*	File Name: Budget_Justification1046094710.pdf (Only attach one file.)

RESEARCH & RELATED Budget {F-K} (Funds Requested)

Tracking Number: GRANT13224045
Funding Opportunity Number: PAR-19-158 . Received Date: 2020-10-13T11:12:56.000-04:00

OMB Number: 4040-0010
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, Budget Period 2

ORGANIZATIONAL DUNS*: 608195277

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: University of North Carolina at Chapel Hill

Start Date*: 07-01-2022 **End Date*:** 06-30-2023 **Budget Period: 2**

A. Senior/Key Person

	Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1 .	Dr.	Paul	A	Dayton	PhD	PD/PI	197,300.00	2.4			39,460.00	11,583.00	51,043.00
2 .	Dr.	Alisa	S	Wolberg	PhD	PD/PI	161,013.00	1.8			24,152.00	7,278.00	31,430.00
3 .	Dr.	Xiaoning		Jiang	Ph.D.	Co-Investigator	197,300.00			1	16,442.00	4,257.00	20,699.00
4 .	Dr.	George	A	Stouffer	MD	Co-Investigator	197,300.00	0.24			3,946.00	1,158.00	5,104.00
5 .	Dr.	Ashley	C.	Brown	PhD	Co-Investigator	147,875.00		1		12,323.00	3,950.00	16,273.00
6 .	Dr.	James	K	Tsuruta	PhD	Co-Investigator	78,750.00	2.76			18,112.00	6,262.00	24,374.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name:

Total Senior/Key Person **148,923.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
2	Post Doctoral Associates	24			107,839.00	20,332.00	128,171.00
3	Graduate Students	30			81,428.00	18,589.00	100,017.00
	Undergraduate Students						
	Secretarial/Clerical						
1	Statistician	0.6			7,500.00	2,284.00	9,784.00
1	Research Assistant Professor	0.6			4,200.00	1,429.00	5,629.00
2	Technician	3.6			14,227.00	5,734.00	19,961.00
9	**Total Number Other Personnel**					**Total Other Personnel**	**263,562.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**412,485.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

Tracking Number: GRANT13224045
Funding Opportunity Number: PAR-19-158 . Received Date: 2020-10-13T11:12:56.000-04:00

RESEARCH & RELATED BUDGET - SECTION C, D, & E, Budget Period 2

ORGANIZATIONAL DUNS*: 608195277

Budget Type*: ● Project ❍ Subaward/Consortium

Organization: University of North Carolina at Chapel Hill

Start Date*: 07-01-2022 **End Date*:** 06-30-2023 **Budget Period:** 2

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	
Additional Equipment: File Name:	

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	2,250.00
2. Foreign Travel Costs	7,750.00
Total Travel Cost	**10,000.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		
2. Stipends		
3. Travel		
4. Subsistence		
5. Other:		
Number of Participants/Trainees	**Total Participant Trainee Support Costs**	

RESEARCH & RELATED Budget {C-E} (Funds Requested)

Tracking Number: GRANT13224045

Funding Opportunity Number: PAR-19-158 . Received Date: 2020-10-13T11:12:56.000-04:00

RESEARCH & RELATED BUDGET - SECTIONS F-K, Budget Period 2

ORGANIZATIONAL DUNS*: 608195277
Budget Type*: ● Project ❍ Subaward/Consortium
Organization: University of North Carolina at Chapel Hill

Start Date*: 07-01-2022 **End Date*:** 06-30-2023 **Budget Period:** 2

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	24,650.00
2. Publication Costs	4,000.00
3. Consultant Services	
4. ADP/Computer Services	
5. Subawards/Consortium/Contractual Costs	
6. Equipment or Facility Rental/User Fees	15,785.00
7. Alterations and Renovations	
8 . Tuition	30,375.00
9 . Patient Incentives	250.00
10 . Animal Per Diem	1,608.00
Total Other Direct Costs	**76,668.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**499,153.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1 . Organized Research_On Campus	55.5	468,778.00	260,172.00
		Total Indirect Costs	**260,172.00**

Cognizant Federal Agency DHHS; Darryl Mayes; (202) 401-2808
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**759,325.00**

J. Fee	Funds Requested ($)*

K. Total Costs and Fee	Funds Requested ($)*
	759,325.00

L. Budget Justification*	File Name: Budget_Justification1046094710.pdf (Only attach one file.)

RESEARCH & RELATED Budget {F-K} (Funds Requested)

Tracking Number: GRANT13224045
Funding Opportunity Number: PAR-19-158 . Received Date: 2020-10-13T11:12:56.000-04:00

OMB Number: 4040-0010
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, Budget Period 3

ORGANIZATIONAL DUNS*: 608195277

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: University of North Carolina at Chapel Hill

Start Date*: 07-01-2023 **End Date*:** 06-30-2024 **Budget Period: 3**

A. Senior/Key Person

	Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1 .	Dr.	Paul	A	Dayton	PhD	PD/PI	197,300.00	2.4			39,460.00	11,583.00	51,043.00
2 .	Dr.	Alisa	S	Wolberg	PhD	PD/PI	161,013.00	1.8			24,152.00	7,278.00	31,430.00
3 .	Dr.	Xiaoning		Jiang	Ph.D.	Co-Investigator	197,300.00			1	16,442.00	4,257.00	20,699.00
4 .	Dr.	George	A	Stouffer	MD	Co-Investigator	197,300.00	0.24			3,946.00	1,158.00	5,104.00
5 .	Dr.	Ashley	C.	Brown	PhD	Co-Investigator	147,875.00		1		12,323.00	3,950.00	16,273.00
6 .	Dr.	James	K	Tsuruta	PhD	Co-Investigator	78,750.00	2.76			18,112.00	6,262.00	24,374.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name:

Total Senior/Key Person **148,923.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
2	Post Doctoral Associates	21.6			97,339.00	18,326.00	115,665.00
3	Graduate Students	30			81,428.00	18,589.00	100,017.00
	Undergraduate Students						
	Secretarial/Clerical						
1	Statistician	0.6			7,500.00	2,284.00	9,784.00
1	Research Assistant Professor	0.6			4,200.00	1,429.00	5,629.00
2	Technician	3.6			14,227.00	5,734.00	19,961.00
9	**Total Number Other Personnel**					**Total Other Personnel**	**251,056.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**399,979.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

Tracking Number: GRANT13224045
Funding Opportunity Number: PAR-19-158 . Received Date: 2020-10-13T11:12:56.000-04:00

RESEARCH & RELATED BUDGET - SECTION C, D, & E, Budget Period 3

ORGANIZATIONAL DUNS*: 608195277

Budget Type*: ● Project ❍ Subaward/Consortium

Organization: University of North Carolina at Chapel Hill

Start Date*: 07-01-2023 **End Date*:** 06-30-2024 **Budget Period:** 3

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	
Additional Equipment: File Name:	

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	2,250.00
2. Foreign Travel Costs	8,750.00
Total Travel Cost	**11,000.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		
2. Stipends		
3. Travel		
4. Subsistence		
5. Other:		
Number of Participants/Trainees	**Total Participant Trainee Support Costs**	

RESEARCH & RELATED Budget {C-E} (Funds Requested)

Tracking Number: GRANT13224045

Funding Opportunity Number: PAR-19-158 . Received Date: 2020-10-13T11:12:56.000-04:00

RESEARCH & RELATED BUDGET - SECTIONS F-K, Budget Period 3

ORGANIZATIONAL DUNS*: 608195277
Budget Type*: ● Project ❍ Subaward/Consortium
Organization: University of North Carolina at Chapel Hill

Start Date*: 07-01-2023 **End Date*:** 06-30-2024 **Budget Period:** 3

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	29,150.00
2. Publication Costs	4,000.00
3. Consultant Services	
4. ADP/Computer Services	
5. Subawards/Consortium/Contractual Costs	
6. Equipment or Facility Rental/User Fees	18,935.00
7. Alterations and Renovations	
8 . Tuition	30,375.00
9 . Patient Incentives	250.00
10 . Animal Per Diem	4,308.00
Total Other Direct Costs	**87,018.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**497,997.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1 . Organized Research_On Campus	55.5	467,622.00	259,530.00
		Total Indirect Costs	**259,530.00**

Cognizant Federal Agency DHHS; Darryl Mayes; (202) 401-2808
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**757,527.00**

J. Fee	Funds Requested ($)*

K. Total Costs and Fee	Funds Requested ($)*
	757,527.00

L. Budget Justification*	File Name: Budget_Justification1046094710.pdf (Only attach one file.)

RESEARCH & RELATED Budget {F-K} (Funds Requested)

Tracking Number: GRANT13224045
Funding Opportunity Number: PAR-19-158 . Received Date: 2020-10-13T11:12:56.000-04:00

OMB Number: 4040-0010
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, Budget Period 4

ORGANIZATIONAL DUNS*: 608195277

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: University of North Carolina at Chapel Hill

Start Date*: 07-01-2024 **End Date*:** 06-30-2025 **Budget Period: 4**

A. Senior/Key Person

	Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1 .	Dr.	Paul	A	Dayton	PhD	PD/PI	197,300.00	2.4			39,460.00	11,583.00	51,043.00
2 .	Dr.	Alisa	S	Wolberg	PhD	PD/PI	161,013.00	1.8			24,152.00	7,278.00	31,430.00
3 .	Dr.	Xiaoning		Jiang	Ph.D.	Co-Investigator	197,300.00			1	16,442.00	4,257.00	20,699.00
4 .	Dr.	George	A	Stouffer	MD	Co-Investigator	197,300.00	0.24			3,946.00	1,158.00	5,104.00
5 .	Dr.	Ashley	C.	Brown	PhD	Co-Investigator	147,875.00		1		12,323.00	3,950.00	16,273.00
6 .	Dr.	James	K	Tsuruta	PhD	Co-Investigator	78,750.00	2.76			18,112.00	6,262.00	24,374.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name:

Total Senior/Key Person **148,923.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
2	Post Doctoral Associates	21.6			97,339.00	18,326.00	115,665.00
3	Graduate Students	30			81,428.00	18,589.00	100,017.00
	Undergraduate Students						
	Secretarial/Clerical						
1	Statistician	0.6			7,500.00	2,284.00	9,784.00
1	Research Assistant Professor	0.6			4,200.00	1,429.00	5,629.00
2	Technician	3.6			14,227.00	5,734.00	19,961.00
9	**Total Number Other Personnel**					**Total Other Personnel**	**251,056.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**399,979.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

Tracking Number: GRANT13224045
Funding Opportunity Number: PAR-19-158 . Received Date: 2020-10-13T11:12:56.000-04:00

RESEARCH & RELATED BUDGET - SECTION C, D, & E, Budget Period 4

ORGANIZATIONAL DUNS*: 608195277

Budget Type*: ● Project ❍ Subaward/Consortium

Organization: University of North Carolina at Chapel Hill

Start Date*: 07-01-2024 **End Date*:** 06-30-2025 **Budget Period:** 4

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	
Additional Equipment: File Name:	

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	2,250.00
2. Foreign Travel Costs	8,750.00
Total Travel Cost	**11,000.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		
2. Stipends		
3. Travel		
4. Subsistence		
5. Other:		
Number of Participants/Trainees	**Total Participant Trainee Support Costs**	

RESEARCH & RELATED Budget {C-E} (Funds Requested)

Tracking Number: GRANT13224045

Funding Opportunity Number: PAR-19-158 . Received Date: 2020-10-13T11:12:56.000-04:00

RESEARCH & RELATED BUDGET - SECTIONS F-K, Budget Period 4

ORGANIZATIONAL DUNS*: 608195277
Budget Type*: ● Project ❍ Subaward/Consortium
Organization: University of North Carolina at Chapel Hill

Start Date*: 07-01-2024 **End Date*:** 06-30-2025 **Budget Period:** 4

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	24,750.00
2. Publication Costs	8,000.00
3. Consultant Services	
4. ADP/Computer Services	
5. Subawards/Consortium/Contractual Costs	
6. Equipment or Facility Rental/User Fees	18,935.00
7. Alterations and Renovations	
8 . Tuition	30,375.00
9 . Patient Incentives	250.00
10 . Animal Per Diem	2,700.00
Total Other Direct Costs	**85,010.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**495,989.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1 . Organized Research_On Campus	55.5	465,614.00	258,416.00
		Total Indirect Costs	**258,416.00**

Cognizant Federal Agency DHHS; Darryl Mayes; (202) 401-2808
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**754,405.00**

J. Fee	Funds Requested ($)*

K. Total Costs and Fee	Funds Requested ($)*
	754,405.00

L. Budget Justification*	File Name: Budget_Justification1046094710.pdf (Only attach one file.)

RESEARCH & RELATED Budget {F-K} (Funds Requested)

Tracking Number: GRANT13224045
Funding Opportunity Number: PAR-19-158 . Received Date: 2020-10-13T11:12:56.000-04:00

BUDGET JUSTIFICATION

PERSONNEL

Key Personnel:

Paul A. Dayton, PhD, Contact Principal Investigator (2.4 calendar months effort per year). Dr. Dayton is William R. Kenan Jr. Professor in the Joint Department of Biomedical Engineering at UNC Chapel Hill and NC State University, as well as a member of the McAllister Heart Center, the Lineberger Comprehensive Cancer Center and an Adjunct Professor of Radiology. Dr. Dayton has been working with ultrasound contrast agents and novel acoustic technologies for imaging and targeted therapeutics for 25 years. He has over 200 publications in the field, and has launched two companies which are commercializing novel biomedical ultrasound technologies. Dr. Dayton has been working with Drs. Jiang and Stouffer for more than 7 years on sonothrombolysis and intravascular ultrasound techniques, and more recently with Drs. Wolberg and Brown. In this project, he will work closely with all personnel to design experiments, analyze data, and prepare publications and presentations.

Alisa Wolberg, PhD, Principal Investigator (1.8 calendar months effort per year). Dr. Wolberg is a Professor in the Department of Pathology Laboratory Medicine at the UNC Chapel Hill School of Medicine. Dr. Wolberg will work closely with Drs. Dayton, Brown, and Jiang to conduct the aims of this project, analyze data, and prepare manuscripts. Dr. Wolberg is a recognized expert in coagulation biochemistry, mechanisms in hemostasis and thrombosis, and mouse models of arterial thrombosis. Dr. Wolberg will supervise the research specialist and the postdoctoral associate responsible for the thrombosis studies and will guide experimental design and analyze experimental findings.

Xiaoning Jiang, PhD, Co-Investigator (1.0 summer month effort per year). Dr. Jiang is Dean F. Duncan Distinguished Professor in the Department of Mechanical & Aerospace Engineering at NC State University, and an Adjunct Professor of Biomedical Engineering in the Joint Department of Biomedical Engineering at UNC Chapel Hill and NC State University. Dr. Jiang will be responsible for supervising the graduate student to work on ultrasound parameters for sonothrombolysis, particularly for in-vitro study.

Rick Stouffer, MD, Co-Investigator, (0.24 calendar month effort per year). Dr. Stouffer is Ernest and Hazel Craige Distinguished Professor of Cardiovascular Medicine, Chief of the Division of Cardiology at UNC Hospitals and Co-Director of the McAllister Heart Institute. He has decades of experience in thrombosis and related clinical interventions, and will serve as a clinical consultant during this project to insure clinical relevance.

Ashley Brown, PhD, Co-Investigator (1.0 academic month effort per year). Dr. Brown is an Assistant Professor in the Joint Department of Biomedical Engineering at UNC Chapel Hill and NC State University. Dr. Brown has been working on technologies for hemostasis for several years, and is the pioneer in several novel fibrin targeting strategies which will be utilized in the proposed project to improve thrombus targeting.

James Tsuruta, PhD, Co-Investigator (2.76 calendar months effort per year). Dr. Tsuruta is a Research Assistant Professor in the Joint Department of Biomedical Engineering and a senior member of the Dayton Lab. He is classically trained in molecular biology techniques and biochemistry, and he is the lead formulation chemist for the Dayton lab. Dr. Tsuruta will lead all efforts in nanodroplet formulation optimization and production during the project, and work closely with the Brown lab to develop the targeted droplets.

Other contributors:

Statistician TBN (0.6 calendar month effort per year). We will engage the assistance of a member of the UNC Biostatistics Core throughout the duration of this study for assistance with data analysis.

Kennita Johnson, PhD, Laboratory Animal Coordinator (0.6 calendar months effort per year). Dr. Johnson is a Research Assistant Professor in the Joint Department of Biomedical Engineering and a senior member of the Dayton Lab. She is the Dayton Laboratory's Animal Coordinator and will prepare and oversee all IACUC protocols for animal studies.

Brian Velasco, Laboratory Manager and Research Technician, Dayton Lab (3.0 calendar months effort per year). Mr. Velasco is the lab manager for Dr. Dayton. He will be responsible for day-to-day organizational details for this project, including maintaining laboratory databases, ordering supplies and materials, transporting

animals, managing waste disposal, preparing reagents, autoclaving surgical equipment, managing EH&S compliance, and other general tasks for the investigators in this project.

Lori A. Holle, PhD, Research Specialist, Wolberg Lab (0.6 calendar months effort per year). Dr. Holle will have worked with the PI for over 7 years at the time of award. Dr. Holle has substantial experience in mouse husbandry and translational assays of coagulation. Dr. Holle will coordinate the mouse experiments and perform experiments to analyze fibrin degradation products.

Postdoctoral Research Assistant, Jinwook Kim, Dayton Lab (12.0 calendar months effort per year in years 1 and 2, 9.6 calendar months years 3-4). Dr. Kim has extensive experience in transducer design sononthrombolysis. He has worked closely with Drs. Jiang and Dayton over the past several years, and is lead author on their recent publication showing the efficacy of nanodroplets augmenting sononthromboysis of aged clots. Dr. Kim will conduct all of the in-vitro sonothrombolysis experiments and mentor the Dayton lab graduate student.

Postdoctoral Research Assistant, TBN, Wolberg Lab (12.0 calendar months effort per year). This post-doc will work with other team members to conduct in vitro clot studies in years 1-2 and in vivo studies in years 3-4. The postdoctoral associate will manage all details of setup of the arterial thrombosis/thrombolysis models, analyze data, review results, and assist with manuscripts.

Graduate Research Assistant, TBN, Dayton Lab (12.0 calendar months effort per year). This doctoral candidate will be a graduate research assistant in the Dayton lab working on ultrasound techniques for therapeutic approaches. This student will work closely with Drs. Kim and Tsuruta, and other members of the team to conduct all proposed studies at the University of North Carolina, and will complete a thesis on nanodroplet enhanced sonothrombolysis. This student will contribute to the in-vitro sonothrombolysis experiments and be the lead ultrasound collaborator during the animal studies in years 2-4.

Graduate Research Assistant, TBN, Brown Lab (6.0 calendar months effort per year). This doctoral candidate will be a graduate research assistant in the Brown lab working on techniques for fibrin targeting. They will work closely with the Dayton lab team members to make and test fibrin-targeted nanodroplets

Graduate Research Assistant, TBN, Jiang Lab (12.0 calendar months effort per year). This graduate research assistant will be a team member from the laboratory of Dr. Jiang trained in biomedical acoustics and sonothrombolysis. This student will calibrate ultrasound transducers and conduct experiments optimize acoustic parameters for nanodroplet cavitation enhancement.

Fringe Benefits: Fringe benefits for Research Faculty and Staff are calculated at 25.889% plus health insurance of $6,512 x % effort. Fringe benefits for Postdoctoral Research Associate are calculated at 9.49% plus health insurance of $4,808.76 x % effort. Fringe benefits for Graduate Research Assistant are calculated as 9.49% of direct salary plus health insurance of $4,137.60 x % effort.

TRAVEL

Funds are requested for senior and junior members of the team to travel to scientific meetings annually, with more so in the later project years. At this time, the exact dates and locations of these meetings are not yet determined. Possible meetings include the IEEE Ultrasonics Symposium, the International Symposia on Contrast Ultrasound, or the International Society on Thrombosis and Haemostasis. The funds will support travel, lodging, registration, and meals per diem over the duration of the meetings. Domestic travel is estimated at $9,000 over the project period and international travel is budgeted at $30,000 over the project period. Assuming 2 senior members and three junior members each attending one conference a year, this budgets approximately $1,950 per team member for international or domestic travel.

EQUIPMENT

Calibrated hydrophone: Funds are requested for a new calibrated needle hydrophone for the lab, to be purchased from Onda, Inc. This piece of equipment is crucial to measure the output pressure levels from the ultrasound system, an important parameter in cavitation generation. Needle hydrophones deteriorate over time and need to be replaced every four years, a new one will be purchased at the start of this project to be used for this project. It will be 100% dedicated to the project, and we request that it remain at UNC at the conclusion of the project. These costs are estimated at $8,000 in Year 1.

SUPPLIES

Materials and supplies are requested based on established pricing and are consistent with the scope of the project that includes both in vivo and in vitro studies. These costs are estimated at $24,000, $24,650, $29,150, and $24,750 across the four years; and include costs across the four collaborating laboratories for this project:

- Surgical costs: blades, sutures, ointments, isoflurane, anesthetics, laser Doppler probes for flow detection, surgical tools
- General laboratory disposables: chemicals, pipettes, pipet tips, microfuge tubes, PCR tubes, multi-well plates, conical tubes, and other plastic ware
- Nanodoplet preparation: lipids, perfluorocarbon, buffers, glass vials, septa, labels
- Materials for targeting ligand design: reagents
- Histology reagents and stains, cassettes, slides
- In-vitro test materials: tubing, clamps, flow pumps
- Pre-human ex-vivo blood studies: bovine blood, reagents
- Acoustics: annual hydrophone calibration, cables, ultrasound gel
- Transducers: 2 Sonic Concepts HIFU transducers at ~ $3,000 each

Animal Purchases: $9,750 total across Years 2, 3 and 4 is requested to purchase 195 mice (45/75/75/year) at approximately $50 per mouse for the UNC in-vivo arterial clot sonothrombolysis studies. $2,800 total across years 2 and 3 is requested to purchase 35 mice, at approximately $80 per mouse for the targeting studies at NC State.

OTHER EXPENSES

Publication Costs: Publication costs are requested to cover the costs for publishing research from this project. Charges for publications range from $1000-$3000 depending on page length and open access, and we anticipate one paper during the first year, 2-3 papers per year 2-3, and 3-4 papers in the final year. These funds are estimated at $4,000 per year in years 1 through 3 and $8,000 in year 4.

Equipment Repair and Service: We request funds for equipment repair/service to maintain pipettors, ultrasound systems, and other laboratory equipment necessary for the proposed studies. Maintenance on this existing equipment is necessary for the equipment's continued operation. We request $4,500 per year to cover its maintenance during the project period. A single service visit for one of the key pieces of equipment such as the Accusizer or ultrasound system typically runs $6000-$7000 and we expect at least one service visit in four years.

Animal Core: Recharge rates for the animal services core, budgeted at $35 / hour, are requested for animal handling. These costs are estimated to be on the order of three hours per mouse for each of the setup and procedures for each animal study for the arterial clot studies, and one hour per animal for the targeting studies. This will cost $5,285 in year 2, $8,435 in year 3, and $8,435 in year 4.

Core Facilities: The UNC Histology core lab will provide support for post-study thrombus analyses for both ex-vivo and in-vivo studies. Costs are estimated at $6,000 per year.

Tuition: Tuition for the graduate research assistant is based on the current annual rate for in-state graduate students of $10,552 + $1,244.20 student fees. Tuition is inflated by 3% beginning in year one. The cost of tuition and fees for the entire project is estimated to be $121,500.

Patient Incentives: Funds are requested to cover payments to human blood donors. These costs are estimated at $250 per year, and include $10 per blood draw and 25 draws per year.

Animal Per Diem: Funds are requested to house 32 mice, 4 per cage, at $3.35 per cage for 60 days at NC State. These costs are estimated at $1,608 per year in years 2 and 3. Funds are requested to house 120 mice 4 per cage, at $2.00 per cage for 90 days at UNC. These costs are estimated at $2,700 per year in years 3 and 4.

INDIRECT COSTS

The University of North Carolina at Chapel Hill computes F&A at 55.5% of Modified Total Direct costs. Modified Total Direct costs exclude Tuition, Scholarship and Fellowship costs, Equipment greater than $5,000, and subcontract costs beyond $25,000.

RESEARCH & RELATED BUDGET - Cumulative Budget

	Totals ($)	
Section A, Senior/Key Person		595,692.00
Section B, Other Personnel		1,029,236.00
Total Number Other Personnel	36	
Total Salary, Wages and Fringe Benefits (A+B)		1,624,928.00
Section C, Equipment		8,000.00
Section D, Travel		39,000.00
1. Domestic	9,000.00	
2. Foreign	30,000.00	
Section E, Participant/Trainee Support Costs		
1. Tuition/Fees/Health Insurance		
2. Stipends		
3. Travel		
4. Subsistence		
5. Other		
6. Number of Participants/Trainees		
Section F, Other Direct Costs		317,821.00
1. Materials and Supplies	102,550.00	
2. Publication Costs	20,000.00	
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees	64,155.00	
7. Alterations and Renovations		
8. Other 1	121,500.00	
9. Other 2	1,000.00	
10. Other 3	8,616.00	
Section G, Direct Costs (A thru F)		1,989,749.00
Section H, Indirect Costs		1,032,438.00
Section I, Total Direct and Indirect Costs (G + H)		3,022,187.00
Section J, Fee		
Section K, Total Costs and Fee (I + J)		3,022,187.00

PHS 398 Cover Page Supplement

OMB Number: 0925-0001

Expiration Date: 02/28/2023

1. Vertebrate Animals Section

Are vertebrate animals euthanized? ● Yes ○ No

If "Yes" to euthanasia

Is the method consistent with American Veterinary Medical Association (AVMA) guidelines?

● Yes ○ No

If "No" to AVMA guidelines, describe method and provide scientific justification

2. *Program Income Section

*Is program income anticipated during the periods for which the grant support is requested?

○ Yes ● No

If you checked "yes" above (indicating that program income is anticipated), then use the format below to reflect the amount and source(s). Otherwise, leave this section blank.

*Budget Period	*Anticipated Amount ($)	*Source(s)

Tracking Number: GRANT13224045

Funding Opportunity Number: PAR-19-158 . Received Date: 2020-10-13T11:12:56.000-04:00

PHS 398 Cover Page Supplement

3. Human Embryonic Stem Cells Section

*Does the proposed project involve human embryonic stem cells? ❍ Yes ● No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s) from the following list: http://grants.nih.gov/stem_cells/registry/current.htm. Or, if a specific stem cell line cannot be referenced at this time, check the box indicating that one from the registry will be used:

Specific stem cell line cannot be referenced at this time. One from the registry will be used.

Cell Line(s) (Example: 0004):

4. Human Fetal Tissue Section

*Does the proposed project involve human fetal tissue obtained from elective abortions? ❍ Yes ● No

If "yes" then provide the HFT Compliance Assurance

If "yes" then provide the HFT Sample IRB Consent Form

5. Inventions and Patents Section (Renewal applications)

*Inventions and Patents: ❍ Yes ❍ No

If the answer is "Yes" then please answer the following:

*Previously Reported: ❍ Yes ❍ No

6. Change of Investigator/Change of Institution Section

❑ Change of Project Director/Principal Investigator

Name of former Project Director/Principal Investigator

Prefix:

*First Name:

Middle Name:

*Last Name:

Suffix:

❑ Change of Grantee Institution

*Name of former institution:

Tracking Number: GRANT13224045

Funding Opportunity Number: PAR-19-158 . Received Date: 2020-10-13T11:12:56.000-04:00

PHS 398 Research Plan

OMB Number: 0925-0001
Expiration Date: 02/28/2023

Introduction	
1. Introduction to Application (for Resubmission and Revision applications)	
Research Plan Section	
2. Specific Aims	Specific_Aims1046185394.pdf
3. Research Strategy*	Research_Strategy1046185875.pdf
4. Progress Report Publication List	
Other Research Plan Section	
5. Vertebrate Animals	Vertebrate_Animals1046094866.pdf
6. Select Agent Research	
7. Multiple PD/PI Leadership Plan	Multi_PI_Plan1046094849.pdf
8. Consortium/Contractual Arrangements	
9. Letters of Support	LOS1046185216.pdf
10. Resource Sharing Plan(s)	
11. Authentication of Key Biological and/or Chemical Resources	Authentication_of_Key_Resources1046094850.pdf
Appendix	
12. Appendix	

SPECIFIC AIMS

Arterial thrombosis, including ischemic stroke and myocardial infarction, remains a leading cause of death in the United States and the world. In particular, ischemic stroke is the fifth leading cause of death in the United States, and a major cause of disability in non-fatal cases. The only FDA approved non-surgical treatment for ischemic stroke is the administration of recombinant tissue-plasminogen activator (rt-PA). Timeliness of rt-PA treatment is crucial: effectiveness of this therapy declines rapidly within hours as clots stabilize, while prolonged ischemia worsens tissue damage. Further limiting the only clinical therapy is the complication risk of intracerebral hemorrhage, which remains a major barrier to widespread rt-PA use.

Thus, there exists a vital need for innovative noninvasive treatment of ischemic stroke, especially for large percentage of patients that are excluded from rt-PA treatment (e.g., patients susceptible to hemorrhage or those seen >3 hours after thrombosis or with unknown time of onset) [1–3].

We hypothesize a novel arterial thrombosis treatment that combines phase change nanodroplet enhanced cavitation with ultrasound. The application of ultrasound for clot disruption, termed sonothrombolysis, has been studied for decades [4]. Although this approach effectively breaks up thrombus under certain conditions such as fresh thrombus, and has demonstrated clinical efficacy in a small subset of cases, it has failed to be fully realized as an impactful therapy during several large clinical trials [4–10]. One of the hypotheses for the failure of sonothrombolysis is that aged clots in stroke patients are denser and more resistant to fibrinolysis, where performance of ultrasound with rt-PA alone has been shown to be very limited [11,12]. The amplification of cavitation through the addition of microbubble contrast agents mechanically enhances clot disruption, and was originally hypothesized as a solution [13–15]. However, despite showing improvements over non-enhanced ultrasound, microbubble-assisted sonothrombolysis has not demonstrated efficacy in treating older denser (contracted) clots, and therefore has not demonstrated improvements needed for clinical impact.

Our team proposes a novel approach to cavitation enhanced sonothrombolysis for arterial thrombosis, which replaces microbubbles with novel low-boiling point phase change nanodroplets. Phase-change nanodroplets are hundred-nanometer, liquid perfluorocarbon filled nanoparticles which convert to gas-filled microbubbles upon excitation at low-acoustic energies, and have been studied by the PI for more than a decade. [16–19]. We propose that these droplets can be designed to penetrate into clot matrices, and then be cavitated internally to effectively disrupt the clot while avoiding bioeffects caused by high intensity ultrasound. ***Our preliminary studies support this hypothesis – we have shown that our nanodroplets diffuse internally within the clot, in contrast to microbubbles which remain superficial*** [20]***. Within the clot, we observe that nanodroplets facilitate a novel mechanism of clot disruption – erosion starting from the interior. Our data shows that this mechanism results in faster and more efficient clot lysis in denser aged and retracted clots than microbubbles, and much faster than ultrasound without enhancement agents*** [21]***. This has led to great enthusiasm about nanodroplet enhanced sonothrombolysis as a 'game-changer' therapeutic approach for arterial thrombosis.***

Specific Aim 1: Determine and optimize the amount and extent of nanodroplet penetration into clots as a function of nanodroplet diameter, nanodroplet surface chemistry, and clot density. Design a droplet formulation optimal for arterial thrombus dissolution through modification of size and surface chemistry. Study if the incorporation of fibrin-targeting moieties onto droplets to increase their delivery to clots can further improve their delivery and improve lysis performance. Evaluate performance in in-vitro studies of bovine and then human blood. Evaluate targeting strategies in-vivo.

Specific Aim 2: Evaluate and optimize acoustic conditions to optimize lysis and degradation of the fibrin structure by nanodroplets and select conditions for maximum efficacy with minimum rt-PA dose. Study and optimize the effect of acoustic parameters on clot lysis, such as acoustic pressure, frequency, and duty cycle. Evaluate clot disruption mechanisms, lysis rate, and degraded particle sizes to ensure downstream safety. Evaluate thrombolysis with nanodroplets combined with low-dose rt-PA. Design and apply acoustic strategies to monitor cavitation for image guided treatment.

Specific Aim 3: Validate efficacy and safety of nanodroplet-assisted sonothrombolysis in well characterized *in-vivo* rodent models of arterial thrombosis. Using optimized parameters from Aims 1 and 2, conduct image-guided studies to validate thrombolysis in-vivo. Evaluate efficacy of clot lysis and perform histological analysis to determine effects on thrombus and vascular integrity as well as surrounding tissue.

Completion of our Specific Aims 1-2 will establish parameters for the efficacy of nanodroplet enhanced sonothrombolysis as a function of system parameters and rt-PA dose, and treatment efficacy versus both fresh as well as aged and/or contracted blood clots in-vitro. The third aim will evaluate efficacy and safety translation of optimal parameters in-vivo to prepare for large animal studies and clinical realization.

RESEARCH STRATEGY

A. SIGNIFICANCE

A.1 Stroke treatments administered quickly will have a significant impact in reducing morbidity and mortality. The "golden hour" refers to the first hour after symptoms of stroke. If treatment is administered during this early time period, a patient is much more likely to avoid death or long-term brain damage [1,22]. With modern compact ultrasound systems, a technology such as image-guided nanodroplet-enhanced sonothrombolysis is highly portable, and can be applied in remote or mobile emergency medicine settings such as in an ambulance, enabling rapid initiation of lifesaving treatment. ***An effective mobile treatment technology would have a significant impact in stroke.***

A.2 For cases where early treatment is not possible or rt-PA is contraindicated, a new approach for fast and safe treatment of acute ischemic stroke caused by *aged blood clots* would be significant. As clots age, they undergo contraction (consolidation), wherein they become denser, less permeable to fibrinolytic enzymes, and more resistant to dissolution. The frontline therapy of ischemic stroke is therapeutic dose (0.9 mg/kg body weight, <90 mg total)-administration of fibrin-specific thrombolytic recombinant tissue plasminogen activator (rt-PA) [22,23]. However, less than 5% of total stroke cases receive this treatment, mainly due to the narrow therapeutic window – rt-PA efficacy declines rapidly as clots age, and is clinically indicated only within the first 3 hours after onset of sympoms. In addition, concerns regarding hemorrhagic complications limit the population eligible for this treatment even further [24,25]. Numerous ultrasound-assisted techniques have been suggested [26] to overcome these limitations by increasing penetration into the thrombus and/or reduce the rt-PA dose required to achieve thrombolysis. Despite the demonstrated feasibility of sonothrombolysis, there remains a limitation for treating aged clots, since the contracted, low-porosity structure of older clots restricts rt-PA access to fibrin-bound plasminogen [27], making them more resistant to rt-PA lysis than "fresh" clots. Even the cavitation-enhancement approach using microbubble contrast agents as cavitation nuclei has been ineffective for contracted clot treatment [11,12]. ***Thus a clear need exists for an effective approach for treating aged clots.***

A.3 Low-boiling point phase change nanodroplets are a significant new technology for sonothrombolysis. Although phase-change contrast agents (PCCAs) have been studied for several decades, the formulation of *low-boiling point nanodroplet* PCCAs, developed by MPI Dayton in the past decade has revolutionized the nano-sized PCCA. Phase change contrast agents traditionally have been formulated with a liquid perfluorocarbon such as perfluoropentane (C_5F_{12}) or perfluorohexane (C_6F_{14}), and stabilized by a shell such as a lipid [28,29]. These liquid droplets convert to microbubbles as their core changes phase upon the addition of energy, such as ultrasound. One challenge with traditional PCCAs, however, is that these agents require increasingly large amounts of acoustic energy to cause the phase transition as their size decreases due to several factors (Laplace pressure, acoustic focusing, and nucleation dynamics) [17,30], with the result being that submicron droplets cannot be vaporized without very high acoustic energies which also cause substantial bioeffects. We first developed the low-boiling point PCCA via starting with gas-core microbubbles filled with perfluoropropane (C_3F_8) and/or (C_4F_{10}) and encapsulated with lipids, the same biocompatible excipients used in FDA approved ultrasound contrast agents. By exposing these microbubbles to increased pressure and decreased temperature, their gaseous cores condense into the liquid state and their volume decreases by a factor of more than two orders of magnitude. However, then a unique thing happens – these nano-PCCAs reach a metastable state, and can be re-warmed to body temperature and pressure, and they remain stable until additional energy is introduced, however, this energy is much less than with traditional PCCAs. The result is a nano-sized stabilized liquid droplet, made from an FDA approved formulation, which can be reconverted into a microbubble with acoustic energy levels which are safe to use in humans. MPI Dayton has been granted two patents on this novel technology and published more than 2 dozen papers on low-boiling point nanodroplets. These low-boiling point nanodroplet PCCAs have demonstrated several advantages over traditional microbubbles, namely increased stability [17], smaller size which enables extravasation and permeation beyond that possible microbubbles [17], and the ability to be tailored to specific activation thresholds [19], with the resultant performance after phase transition akin to traditional microbubbles. Prior highly successful applications of these nanodroplets in drug delivery [18,31] and focused ultrasound enhancement [32] have suggested their strong potential in sonothrombolysis.

A.3 Significance of the Research Team: We have assembled a world-class team of experts to insure a high-liklihood of success and clinical impact. Multi-PI Dr. Paul Dayton (UNC-NC State BME) has worked in biomedical acoustics for 25 years, focusing on microbubbles and nanodroplet PCCAs for imaging and therapy. Multi-PI Dr. Alisa Wolberg, Professor of Pathology and Laboratory Medicine in the UNC Blood Research Center, has more

than 25 years of experience in thrombosis and will provide biological expertise and extensive experience with in-vivo clot models. Both PIs have experience managing large teams. Working with PIs Dayton and Wolberg will be acoustics expert Dr. Xiaoning Jiang, who will provide additional expertise on acoustics and sonothrombolysis, and Dr. Ashley Brown, expert in hemostasis and bioconjugate approaches for clot targeting. Dr. Brown will play a key role in optimizing the nanodroplets for in-clot retention. Finally, clinical perspective will be provided by Dr. Rick Stouffer, M.D., Chief of Cardiology and Co-Director of the McAllister Heart Center, who has more than four decades of experience in clinical thrombosis. Rounding out the team will be staff with expertise in biostatistics, animal studies, and molecular biology.

B. INNOVATION

B.1 Low-boiling point PCCAs. Although phase-change contrast agents (PCCAs) have been studied for decades [28,29,33,34], the new development of the low-boiling point PCCA is substantially different in that its core is a perfluorocarbon which is a gas at room temperature and pressure; in fact, their composition can be identical in both lipid shell and perfluorocarbon core as FDA-approved microbubbles [35]. These nanodroplets are formed by starting with a microbubble precursor, and then reducing temperature and increasing pressure so that the gas inside of the microbubble condenses to a liquid, mediating a phase change associated reduction in volume of more than 100 fold [17]. The result is a liquid-filled, lipid-stabilized nanodroplet with roughly a 5-6x smaller diameter than a microbubble, although with identical composition. The exciting thing about these nanodroplets is that when condensed, they remain in a metastable liquid-core state even when warmed back up to body temperature. However, a small amount of acoustic energy causes them to reconvert to bubble state [17]. Once returned to bubbles, they cavitate identically to microbubbles in an acoustic field.

B.2 Low-boiling point phase-change nanodroplets are likely a superior cavitation agent for thrombolysis than microbubbles. Our data suggest that due to the smaller size (0.1-0.2 microns in diameter) and higher stability of their liquid core, phase-change nanodroplets penetrate clots more easily than microbubbles [32,36]: This feature results in a critical difference between microbubbles for the thrombolysis of contracted clots. Despite this, outside of our handful of limited preliminary manuscripts to date, nanodroplet sonothrombolysis and nanodroplet interactions with clots have not been studied. Although our prelimnary data are encouraging, it is highly likely that optimization of nanodroplet size and formulation, as well as the addition of clot-targeting ligands to their shell will all increase their already promising clot-busting performance even further. The current knowledge gap stands in the way of a clinical trial of nanodroplet-mediated sonothrombolysis as a way to improve the current standard therapy for ischemic stroke, and our proposal to explore this gap is highly innovative.

B.3 A key treatment mechanism for contracted clots: the concept of internal fibrinolysis. Recent studies on clot retraction suggest that once t-PA enzymes penetrate a clot, a tight fibrin structure actually facilitates efficient movement of plasmin molecules between fibers [27]. This theory is applicable to nanodroplet-mediated sonothrombolysis. We expect this internal enzymatic mechanism will be enhanced by entry of rt-PA via channels created by the mechanical force created by activated nanodroplets cavitating in the ultrasound field.

B.4 Low-boiling point perfluorocarbon nanodroplets create droplet-induced cavitation at lower peak negative pressures than traditional nanodroplets. Compared to traditional nanodroplets PCCAs such as perfluoropentane and perfluorohexane that have a relatively high boiling point (>29°C) [30,37], our nanodroplets comprised of perfluorocarbons which are a gas at room temperature and pressure, require much lower pressure for vaporization (<1 MPa vs. >5 MPa) [38]. This feature allows us to conduct sonothrombolysis at much lower pressures than investigated with prior phase change nanodroplets, and will enable efficient sonothrombolysis while avoiding bioeffects in healthy tissue.

B.5 The mechanism of nanodroplet-blood clot interaction will be studied for the first time. Although microbubbles have been studied for some time, and are known to disrupt clots through 1) radiation force-induced mechanical clot *surface* agitation and 2) cavitation-induced fibrin cleaving from the outside in [15,39–41], our preliminary data have shown drastically different mechanisms for ***nanodroplets. Our studies show that cavitation and clot erosion occurs from the inside of the clot – an effect not observed with microbubbles***. This is likely due to novel mechanisms, such as **1)** <u>nanodroplets penetrating into</u> the tight fibrin structure, **2)** ultrasound initiating the phase change from nanodroplets to microbubbles when inside the clot, and the newly formed microbubbles **3)** cavitating *inside the clot*, which we hypothesize **4)** <u>creates channels allowing rt-PA access to clot-bound plasminogen</u>. The proposed research aims to optimize these novel mechanisms.

Scientific Premise: Understanding and optimizing low-boiling point nanodroplet interaction with clots and their enhancement of sonothrombolysis in-vitro and in-vivo will provide significant outcomes including 1) rational guidelines to maximize nanodroplet induced clot disruption and minimize bioeffects to healthy tissue 2) extensive

data to support a go/no decision towards clinical treatment of contracted blood clots, and 3) data to support further innovation in other vessel recanalization applications, such as deep vein thrombosis or pulmonary embolism.

C. APPROACH

Preliminary Data: We have previously invented (US Patent 9,532,769; Applications 20160361447 20130336891) and published more than 25 journal articles on low-boiling point phase-change nanodroplets and their applications [16–19,42–44]. We have demonstrated that we can change their size [17], their activation energy threshold [19], and their shell composition, including functionalizing them for targeting [45].

Specific to sonothrombolysis, our ***preliminary data provide five novel primary findings which support the potential high degree of novelty and potential impact in this project*** [20,21,46] ***1) nanodroplets penetrate to the interior of aged clots, whereas microbubbles do not 2) nanodroplet cavitation cloud are generated inside of clots whereas microbubble cavitation clouds are generated outside of the clots 3) nanodroplet cavitation causes a unique 'from the inside-out' dissolution pattern of clots, 4) Nanodroplet enhanced sonothrombolysis produces faster dissolution of aged and contracted clots much greater than rt-PA alone or US alone, and significantly greater than microbubbles plus ultrasound alone. 5) Low-boiling point nanodroplets can mediate sonothrombolysis when used with low-dose rt-PA at mechanical indices within FDA guidelines for imaging (MI<1.9).***

Nanodroplets readily penetrate aged clots, microbubbles do not:

We prepared lipid-shelled decafluorobutane (DFB, C_4F_{10}, boiling point of -2°C) microbubbles with a diameter of 1.1 +/- 0.5 microns. Some of these microbubbles were reserved for microbubble studies, where as the rest were then condensed to produce nanodroplets [17]. The nanodroplets had a diameter range of ~100-200 nanometers, approximately 5-6x less than the microbubble precursors. Otherwise, the microbubbles and nanodroplets had identical composition and concentration per unit volume.

For cavitation monitoring studies and lysis studies without rt-PA, we used aged (>3 days) clot samples with an in-vitro flow model setup to simulate blood flow around clot partially occluding the vessel. Clots were prepared with bovine blood (Densco Marketing, Inc., Woodstock, IL, USA), mixed with 2.5-3.0% calcium chloride ($CaCl_2$) solution (Fisher Scientific, Fair Lawn, NJ, USA). The flow system used a fixed reservoir height to apply constant pressure (0.1 psi) with a flow rate of 80±15 ml/min maintained during partial occlusion. A 1 MHz focused transducer (Sonic Concepts, Inc., WA, USA) mediated sonothrombolysis, while cavitation signals were measured using a secondary co-aligned 5 MHz piston transducer (V309, Panametrics, Waltham, MA, USA). The clot breakdown process was optically monitored by high resolution video camera. For image guidance and cavitation imaging studies, we used a Vantage 256 scanner (Verasonics, Kirkland, WA, US) with a A L11-5 linear array (ATL Ultrasound, Bothell, WA, USA).

B-mode imaging showed that the US (ultrasound) +ND (nanodroplet) treatment generated a cavitation cloud inside the cross-section of a clot (**Fig. 1a**), whereas the US+MB (microbubble) case produced a large but external cavitation cloud near the surface of the clot, without the presence of internal cavitation (**Fig. 1b).** We quantified these observations by defining a metric of the US video intensity ratio of cavitation clouds that were located within a cross-sectional area of a clot (**Fig. 1c**). Analysis of nine images (3 different treatment regions across 3 clot samples) resulted in a significantly different (p=0.004) ratio of cavitation internally to externally for the US+ND case (0.38±0.30) compared to the US+MB case (0.04±0.03). This result suggests that nanodroplets penetrated the aged clots, then vaporized during the ultrasound pulses to create internalized microbubbles.

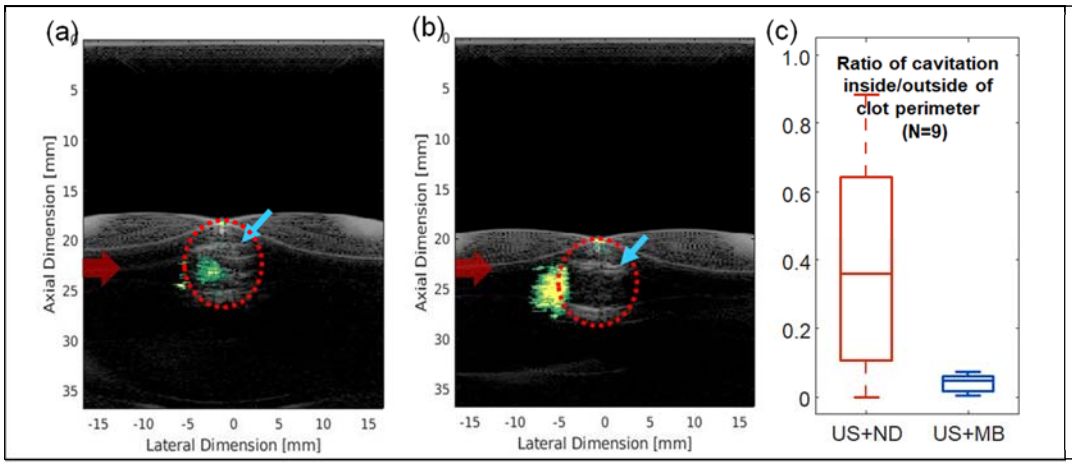

Figure 1. Cavitation imaging (green/yellow) overlaid on b-mode imaging (grayscale) of clot within phantom vessel, red arrow indicates flow. Clot mass is within red circle. Nanodroplets cause cavitation internal to clot (a) whereas bubbles cavitate mainly at upstream surface of clot (b). (c) Cavitation location ratio (N=9), where 1 is cavitation within the clot and 0 is cavitation external.

In both cases of +ND or +MB, it is the cavitation of the resultant microbubbles that cause the mechanical shearing and microstreaming that help disrupt the fibrin matrix and allow rt-PA to permeate the clot. However, in the nanodroplet case, the nanodroplets penetrate the clot first, convert to microbubbles, and then open cavities and channels within the clot. This is in comparison to the microbubbles which from the start are too large to effectively penetrate the clot, and can only affect the clot surface.

Nanodroplets erode clots both internally and externally, microbubbles erode externally: Nanodroplet-mediated cavitation exhibited a unique clot fragmentation process compared to other treatment groups. Video microscopy during treatment demonstrated that clot erosion started internally, at the center of the contracted clot (**Fig. 2**), rather than from the periphery. After multiple micro cavities appeared within 10s of the excitation, distinctive inner erosions emerged. This internal erosion was recognized as a signature of nanodroplet-mediated pulsed ultrasound treatment. The location of erosion cavities through visual observation of clots exposed to acoustic energy directed at the center of the clot samples (n=21) demonstrated this **internal erosion** mechanism **0% (0/21) for ultrasound only**, **14.3% (3/21) for US+MB**, and **76.2% (16/21) for US+ND** treatments. Thus, we interpreted this erosion pattern as evidence of internalized nanodroplets converting to microbubbles within the clot, and disrupted the clot matrix through internal cavitation, radiation forces, and microstreaming.

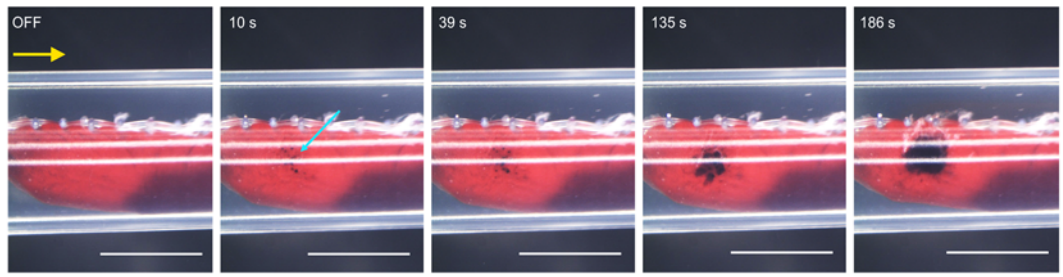

Figure 2. Optical video of nanodroplet enhanced clot dissolution, showing erosion cavities forming internal to clot surface. The clot appears red within the phantom vessel, and the erosion appears black. This mechanism was not observed with microbubbles. Scale=5mm.

Nanodroplets disrupt clots substantially faster than rt-PA alone and notably faster than microbubbles: Nanodroplets demonstrated significant and superior clot lysis rates than ultrasound alone or ultrasound plus microbubbles across more than 20 separate experiments. Without rt-PA, nanodroplets mediated effective sonothrombolysis and were superior to microbubbles at ~ 5.9 and ~ 8 MPa (**Fig. 3a, 3b**) ($p<0.05$). However, when rt-PA was added at a low-dose of 1 microgram/mL (compared to the clinical dose of 90 milligrams in a 100 kg patient, diluted in ~6 L of blood, or 15 micrograms/mL), we observed dissolution of a similar rate to the higher pressure lysis even with a greatly reduced acoustic pressure, even in aged contracted clots (**Fig. 3c**). In all cases, nanodroplet enhanced ultrasound clot lysis with rt-PA was far superior and statistically significant compared to rt-PA alone ($p<0.05$).

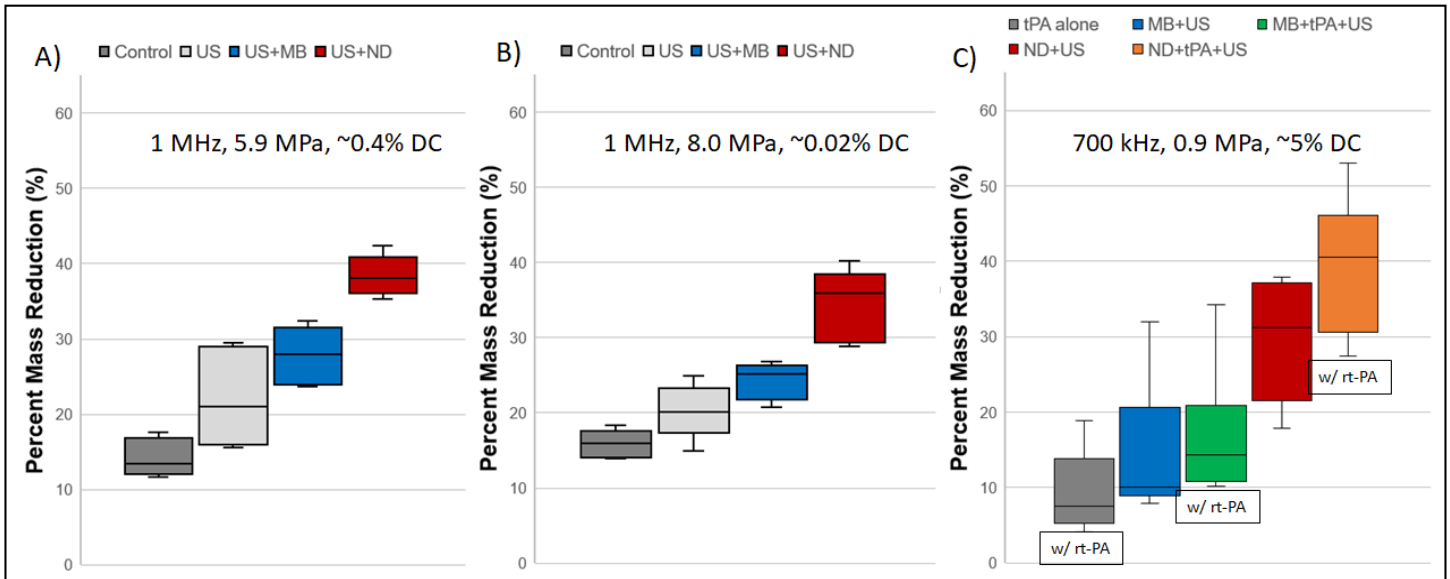

Figure 3 : Sonothrombolysis of in-vitro bovine clots with nanodroplets compared to microbubbles, as well as controls. A) and B) demonstrate clot lysis at different peak pressures, without rt-PA. In each case, ***nanodroplets + US are superior to microbubbles of the same formulation and concentration***. C) When rt-PA was added at a very low dose of 1 microgram/mL, acoustic pressure could be substantially reduced to within FDA Mechanical Index limits for biomedical imaging (MI<1.9). ***US+ND+low dose rt-PA was superior to just rt-PA in aged contracted clots (gray vs. orange bars in C).***

C.1. Research Design and Methods for Aim 1: Determine and optimize the amount and extent of nanodroplet penetration into clots as a function of nanodroplet diameter, nanodroplet surface chemistry, and clot density.

Our hypothesis: Nanodroplets will be able to penetrate into blood clots due to their sub-micron diameter. We will assay the extent of nanodroplet accumulation within fresh and contracted clots using three different methods (a) quantitatively with a high throughput, no-wash immunoassay (*ALPHA assay*) to detect and quantify tagged lipid shells of our nanodroplets trapped inside clot histologic slices, and (b) quantification of the spatial localization of vaporizing nanodroplets and resultant microbubbles by acoustically mapping vaporization and cavitation, and (c) qualitatively by visually observing clot erosion due to cavitation. These methods provide (i) *measurement of nanodroplet concentration within blood clots*, and (ii) *spatial localization of nanodroplets.*

Optical and acoustical data are acquired simultaneously during ultrasound excitation, and clot histology analysis will be performed in separate trials without acoustic energy.

Experimental in-vitro system

For in-vitro studies of nanodroplet delivery to thrombus, cavitation monitoring, and clot lysis, we will use a thrombus in a simulated flow system (**Fig. 4**), as described in preliminary data and prior publications [20]. This system uses a fixed reservoir height to apply constant pressure during each experiment. We can modify pressure and flow rate variables by adjusting the reservoir height.

Blood clot samples are prepared following the previously described method [11,12]. In our initial studies we will create clots from bovine blood (Densco Marketing, Inc., Woodstock, IL) mixed with 2.5-3.0% calcium chloride ($CaCl_2$) solution (Fisher Scientific, Fair Lawn, NJ) with a volume ratio of 10:1 (100 ml blood and 10 ml $CaCl_2$) [47,48]. The mixture is transferred to 1.8 mL-transfer pipettes (33620S, the outer diameter of 6.6 mm, Thermo Fisher Scientific, Waltham, MA). After 3 h in a 37°C water bath, the blood clot-filled pipettes are stored at 4°C for 4 days. Clot samples are on the order of A 300mg with approximate dimensions of 15 mm length and 5 mm diameter. Bovine blood will be used initially as it is readily available from commercial vendors and will allow us to optimize parameters under a range of conditions; more focused studies using human blood will then be performed.

Clots are fixed in a 8 mm silicone vessel phantom with a nylon mesh (1 mm grid size, 500 μm thick, WN1000, Industrial Netting, MN). Two ultrasound transducers are on a computer controlled 3-axis motion stage (Newport, Irvine, CA) which enables scanning of the acoustic focus over the clot volume. One ultrasound transducer is a low-frequency piston transducer which will mediate cavitation, while the second is a L11-5 linear

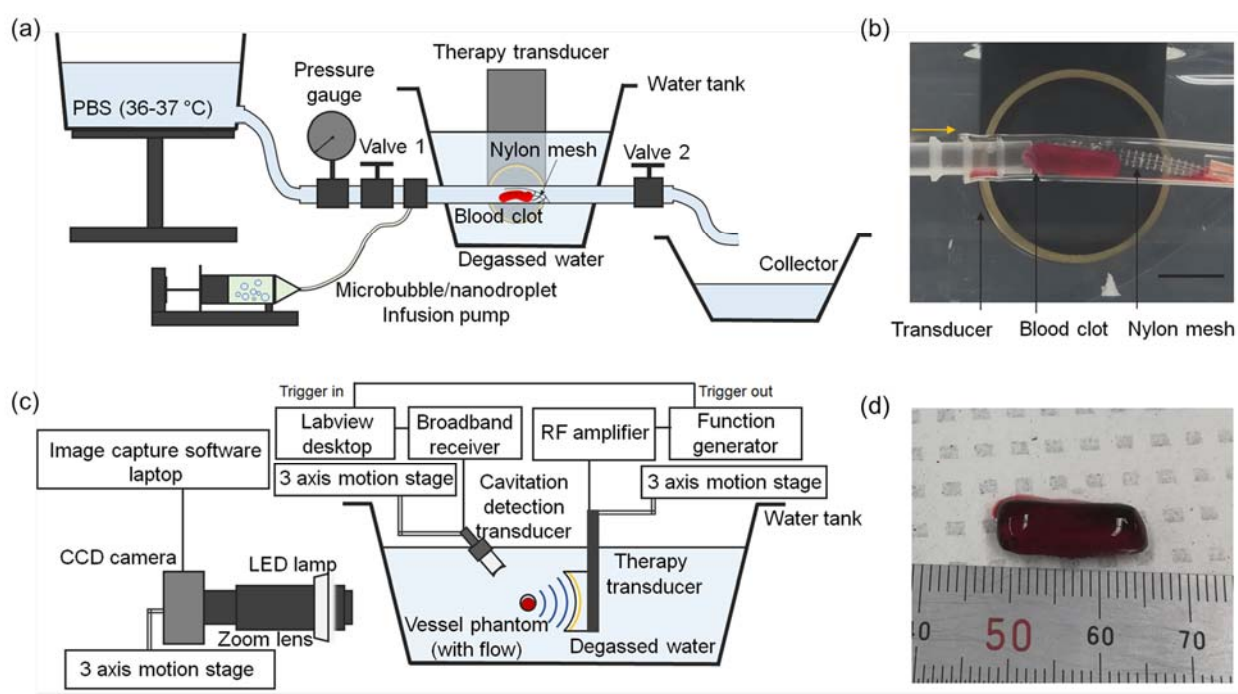

Figure 4 : Experimental in-vitro setup diagram. (a) Front view. (b) the blood clot sample positioned by the nylon mesh. A yellow arrow denotes flow direction (scale bar = 10 mm). (c) Side view of the setup. (d) photo of a contracted clot sample (288 mg). Note that for in our preliminary studies we used a piston for cavitation detection, drawn here, but we also swap this for a L11-5 linear array controlled by a Verasonics 256 programmable ultrasound system.

imaging array coupled to a Verasonics Vantage 256 (Kirkland, WA) system which will detect cavitation and provide cavitation localization imaging. A video microscopy system (Photron, San Diego, CA) with long-working distance lens is also positioned at the clot for continuous optical recording of the clot during studies.

For the studies in Aim 1 which will utilize acoustic thrombolysis, we will use the ultrasound parameters from our preliminary data, e.g. 1 MHz, 900 kPa, 5% duty cycle (**Fig. 3c**). Alternate acoustic parameters will be evaluated in Aim 2. The basic experimental workflow will be as follows: The clot will be positioned in a low-attenuation silicone vessel, secured by the nylon mesh as we have previously demonstrated [20]. Flow across the thrombus will be initiated, with an in-line pressure gauge indicating the desired pressure head. A solution of nanodroplets and/or microbubbles and/or rt-PA will be infused into the flow upstream from the clot via a Y at a controlled rate via a calibrated infusion pump (Harvard Aparattus, Holliston, MA). A manual 3-D positioner allows the long-working distance lens of the microscope to be focused on the clot. A computer controlled 3-D positioning stage (Newport, Inc.) moves both the linear array and the HIFU piston transducer, which are confocally mounted, across the clot for sonication and ultrasound cavitation imaging. The HIFU piston (Sonic Concepts, Bothell, WA) is driven by a RF power amplifier (ENI, Rochester, NY) and programmable function generator (Hewlett Packard, Palo Alto, CA). The lysis pulses, cavitation monitoring, and optical frame acquisition are synchronized via TTL for timing reference.

Human blood tests

For all first-round in-vitro studies, we will use bovine blood as previously described [20]. However, to begin translational studies that more accurately replicate the human condition, we will then examine human blood clotted in-vitro. Human blood clots will be prepared as previously described by Wolberg's team [49,50]. Briefly, blood will be collected from healthy human subjects into sodium citrate (anticoagulant). Clot formation will be triggered by re-calcification and addition of tissue factor (initiating protein). Thrombolysis experiments will then be performed on formed blood clots, as described with bovine blood.

Assessment: Nanodroplet retention within clot matrices with immuno-detection histology: We will assay nanodroplet concentration within the clot matrix with a no-wash ALPHA immuno-assay. Nanodroplets will be co-labeled with biotin and a c-myc peptide. ***Clots will not be sonicated prior to this assay.***

This study uses the amplified luminescent proximity homogeneous assay (ALPHA, Perkin Elmer, Waltham, MA) which is essentially a no-wash ELISA-like assay. Briefly, homogenate from clot slices will be mixed with assay beads and light production will be measured. The assay utilizes donor beads that release singlet oxygen upon absorption of 680 nm light. The short half-life of singlet oxygen means that they can only diffuse about 200 nm before decaying. The second half of the ALPHA assay uses acceptor beads that release 615 nm when they interact with singlet oxygen. For the purpose of detecting nanodroplets within blood clots, our assay uses commercially available streptavidin-donor beads and anti-c-myc acceptor beads. The intact nanodroplet shell will incorporate both DSPE-PEG2000-biotin (Avanti Polar Lipids) and DSPE-PEG2000-DBCO (Avanti Polar Lipids) conjugated to an azido-activated synthetic peptide carrying the c-myc epitope (GenScript) by 'click' chemistry. When acceptor and donor beads are both bound to the nanodroplet shell they are within the necessary proximity to produce a signal; acceptor or donor beads bound to free biotin or free c-myc peptide will not produce any signal. Clot sections will be analyzed in triplicate using a standard curve constructed from known dilutions of biotin/c-myc tagged nanodroplets. ***Analysis will provide quantitative data indicating the amount of nanodroplets internalized in the clot matrix.***

Assessment: Acoustic cavitation and nanodroplet vaporization monitoring: We will use the Verasonics Vantage 256 system to monitor cavitation around and within the clot volume during sonothrombolysis experiments. As shown in preliminary data (Fig. 1), we can quantify the amount of cavitation which occurs within the clot compared to that occurring outside the clot. Inertial and stable cavitation are quantified by measuring the power content in higher level acoustic harmonics as previously described [20].

Furthermore, specific nanodroplet vaporization events can be identified by their unique acoustic signatures [17,42]. When liquid core low-boiling point nanodroplets convert to gas-phase microbubbles, the newly formed microbubbles oscillate before coming to equilibrium, producing substantial characteristic 'ringing' signal that is unique to the phase-change event. This detection method is highly sensitive to nanodroplet phase change events, and is not produced by microbubbles. Data acquired with the Verasonics will be analyzed not only for cavitation content, but also for nanodroplet vaporization content. Image corresponding to the activation loci will be reconstructed according to the method of Rojas et al. [51]. ***Data will indicate the magnitude and spatial distribution of activated nanodroplets as well as bubble cavitation.***

Assessment: Optical clot dissolution monitoring
The optical clot-monitoring system is composed of 1) a long working distance microscope with a high-speed camera (>125 frames/s) and a fiber optic illuminator. During the treatment time, the clot-nanodroplet interaction will be recorded in real-time high resolution video for observation of clot erosion characteristics and completion. These ***data will provide qualitative optical visualization of clot erosion mechanisms (ie, internal vs. internal) as well as quantitative progress towards complete clot dissolution.***

Variable: Nanodroplet size: According to the Ideal Gas Law, condensing a gas core microbubble to a liquid perflurocarbon nanodroplet will result in an approximately five-times smaller diameter. Thus, we can formulate nanodroplets with different average diameters by preparing microbubble precursors of different average diameters. We can also generate fairly narrow distributions by using centrifugal sorting of precursor microbubbles. We predict that accumulation of nanodroplets will be hindered by diameters that exceed the fibrin matrix 'pores' in fresh or contracted clots. Diffusion will be facilitated by diameters smaller than the smallest pore size of our clots.

Briefly, precursor microbubbles will be manufactured in-house [52]. A (9:1) molar mixture of lipid and surfactant (lipid-polyethylene glycol) is shaken in a vial containing perfluorocarbon (PFC) gas. Microbubbles with a PFC gas core form spontaneously. Typical microbubble yield is 1E10/ml with an average diameter of ~1 micron with a range from < 0.5 micron to ~4 microns. We can *isolate different size ranges* of microbubbles by taking advantage of the differential flotation of microbubbles under centrifugation, as we have published previously [52]. This allows us to isolate populations of microbubbles with modal diameters, e.g. ~ 0.6, 1, and 2 microns, which are predicted to produce nanodroplets of ~ 120, 200, and 400 nm diameter. By lowering the temperature and increasing the pressure within vials of microbubbles, the gaseous perfluorcarbon undergoes a phase-change into liquid perfluorcarbon. We will use C_4F_{10} (decafluorobutane, DFB, freezing point ~ 2 C) and induce phase change at -13 C and 40 psi [17,19]. ***Data will identify nanodroplet size which most effectively penetrates clot matrices.***

Variables: Clot consolidation and network density: Two major determinants of clot density are the extent of platelet-driven clot contraction and the amount of fibrin contained within the clot. To test the ability of nanodroplets to penetrate clots and induce dissolution, we will modulate these determinants using methods established in the Wolberg lab [49]. First, we will compare freshly-formed (<30 minutes) clots and clots that have undergone extensive contraction (>2 hours). Second, we will compare native blood clots with clots formed from blood spiked with exogenous fibrinogen (to 4 mg/mL, final), simulating the clinical risk factor hyperfibrinogenemia that increases fibrin density and decreases the effectiveness of thrombolysis [49]. In both cases, we will determine the extent of nanodroplet penetration into the clot, by acoustical and histological methods as described above to analyze the effect of clot density on the loading of clots with nanodroplets. Since we can measure clot erosion with photomicrographs, we will determine if *non-enzymatic* thrombolytic efficacy is related to the diameter and concentration of accumulated nanodroplets. We predict that an optimal diameter, and targeting scheme (which biomarker to target, labeling index of nanodroplets) can be established to maximize the loading of clots with nanodroplets. We will evaluate penetration of nanodroplets with the aforementioned variables into clots of different density. We anticipate testing clots grouped in four different densities, from diffuse to densly contracted. ***Data will indicate nanodroplet penetration into clots of different density as a function of nanodroplet parameters***

Variable: blood pressure: Since the pressure of the fluid on the clot will play a role on flow around the clot and the penetration of the nanodroplets, we will perform a set of studies evaluating the penetration of the nanodroplet into the clot as a function of hydrostatic pressure. This will be done by adjusting the height of the flow reservoir in the in-vitro flow system. We will evaluate pressures of 40, 80, and 120 mm/Hg. ***Data will elucidate the effect of blood pressure on driving the droplet penetration into clots.***

Variable: Nanodroplet surface chemistry and targeting: Nanodroplets will accumulate within clots when the rate of diffusion into the clot is higher than the rate of diffusion out of the clot. We propose to optimize retention of nanodroplets within clots through optimizing surface chemistry specifically by creating fibrin-binding nanodroplets. Collaborator Dr. Brown has extensive experience in creating fibrin targeted nanogel droplets [53–61], and we will use her proven strategies. Specifically, we will conjugate a potent fibrin-binding peptide (AHRPYAAC), which mimics the polymerization knob sequence of native fibrinogen and binds to the fibrin B-hole, to the nanodroplet shell. We do not anticipate reactivity issues across our experiments with bovine, human, or mouse models as fibrinogen is a homologous protein that is highly conserved across species ranging from

lobster to human [62]. Targeting knob:hole interactions has the noted advantage of allowing for competition with pre-exisiting knob:hole binding within a clot to further augment clot dissolution beyond tPA action [63].

For sham studies, we will utilize a non-targeting control peptide (GPSPFPAK, GenScript). The conjugation will utilize carboxylic acid functionalized nanodroplets and standard EDC-Sulfo NHS chemistry. The success of peptide conjugation will be determined by using a CBQCA peptide quantification assay (Invitrogen). We will vary the labeling index of the targeting moiety to determine the optimal ligand density for retention in clots.

The influence of surface functionalization on fibrin targeting will be analyzed *in-vitro* using a microfluidics set-up in the presence of pre-formed clots, as we have published previously [59]. Custom PDMS fluidics chambers constructed by the Brown lab, which contains a stationary fluorescent fibrin clot, located at a T-junction and a fluidic channel that allows a solution, +/- fibrin-binding nanodroplets, to pass over the fibrin clot. An EVOS microscope will be used to take images at the T-junction every minute for 20 min. Binding of fluorescent particles will be quantified by the fluorescence intensity at the clot boundary site. Optimization of targeting will be quantified within this system, after which the best performing targeting droplet formulation will be translated to the macrofluidic system to be tested with acoustics. ***Studies will formulate a targeted nanodroplet which is preferentially retained over current non-targeted nanodroplets.***

Assessment: In-vivo nanodroplet retention assay:

After refinement of the fibrin-targeted nanodroplet formulation, we will conduct a small rodent study to confirm retention of the targeted nanodroplets within thrombus in-vivo. For this study, we will use the murine carotid artery thrombosis model, as described by Dr. Wolberg [49]. Briefly, 6- to 8-week old male C57Bl/6 mice will be anesthetized with isoflurane in oxygen. Body temperature will be maintained by a temperature-controlled heating pad. The right common carotid artery will be surgically exposed after midline cervical incision. A Doppler flow probe (Transonic Systems) will be applied and connected to a flow meter (model T206; Transonic Systems). The Doppler probe provides a gold standard of flow occlusion and recovery in the thrombosed artery. The carotid artery will be dried and 10% $FeCl_3$ (0.62 M $FeCl_3$ on 0.5 x 0.5-mm filter paper) placed on the artery for 3 minutes, removed, and tissues washed 3 times with warm saline. After injury, blood flow will be continuously monitored as it decreases in concert with vessel occlusion over 5-10 minutes. Mice which do not experience carotid occlusion (flow reduction of more than 25%) will be omitted from sonothrombolysis studies because they did not form defined thrombi. Once the thrombus is formed, nanodroplets will be administered via 27-gauge indwelling tail-vein catheter, and will presumably circulate and collect in the clot. This study will not involve acoustic clot lysis. After 30 minutes, the thrombus will be surgically removed, and the aforementioned nanodroplet ALPHA assay will be conducted. Success will be indicated by at least 2x of the targeted nanodroplets retained in the clot matrix compared to the control non-targeted nanodroplets. ***Studies will verify targeted nanodroplet retention in clots in-vivo and quantify increased retention over non-targeted nanodroplets.***

Rigor and Reproducibility:

Both male and female animals will be utilized for the in-vivo studies, 7 each (14 per group), with the group size based on obtaining sufficient power suggested by data variance in preliminary studies. In vitro-group sizes will consist of at least 9 technical replicates per parameter, also based on preliminary data. Since we are evaluating many parameters (nanodroplet size, x3; clot density, x4; and targeting ligand density, x3), we will test subgroups with one fixed parameter (ie: maxiumum clot density; minimum droplet size) rather than exploring the full matrix of 36 different variables). We can repeat prior studies with expanded parameter ranges if data suggests that might provide improved results.

Expected outcomes, potential problems, and alternative solutions:

We expect these studies will provide us with a parameter range of optimal nanodroplet diameters to penetrate thrombus. We anticipate that smaller droplets will penetrate more easily, but also that larger droplets will generate more cavitation. The largest nanodroplet that can still penetrate the most contracted clots will be our target formulation, although, we acknowledge that one potential problem might be that smaller diameter droplets are required for more denser clots, but those smaller droplets produce less cavitation when converted to microbubbles. If this is the case, we can work with multiple formulations for different types of clots, although contracted clots are our primary target. We anticipate formulating a nanodroplet that will be preferentially retained in thrombus compared to non-targeted droplets. We have only examined non-targeted droplets in preliminary studies.

The largest potential problem will be in the case where our model generates complete occlusion. The proposed approach requires flow to bring the cavitation enhancing agents to the clot. If we end up with complete occlusion, we will modify the model (reduced time of $FeCl_3$ application) to generate a less substantial thrombus. An alternative approach for clinical translation would involve an approach to deliver the nanodroplets directly to

the clot with a catheter, a concept that has been explored for microbubble delivery by other groups [64]. Catheters have also been explored for direct delivery of rt-PA to thrombis in the clinic [10]. Such a catheter based technique would be possible, but more invasive than our proposed transcutaneous technique.

The fibrin-targeting motif was chosen because it has been optimized to bind with high affinity to fibrin polymerization holes and Dr. Brown's prior work has shown high affinity binding of this peptide to fibrin. However, if the proposed peptide does not mediate adhesion effectively, Dr. Brown has experience with other antibody and antibody fragment based methods for targeting fibrin in-vitro and in-vivo, which can be explored here as necessary.

C.2. Research Design and Methods for Aim 2: Evaluate acoustic conditions to optimize lysis, and degradation of the fibrin structure by nanodroplets and select conditions for maximum efficacy with minimum rt-PA

After optimizing nanodroplet penetration and retention into the clot in aim 1, we will evaluate clot disruption in-vitro as a function of acoustic parameters. Although it is likely that greater acoustic pressures result in greater cavitation, we ultimately wish to find a range of acoustic parameters which cause effective sonothrombolsis, but at the same time, do not cause damage to surrounding healthy tissue. It is also worth noting that in clinical use, acoustic energy is attenuated as it propagates through tissue. Our acoustic pressure measurements will be measured at the clot location with a calibrated hydrophone (Onda, Sunnyvale, CA), and can be back calculated to estimate what type of energy would need to provided from a transducer a known distance from the target.

We will test the input parameters of pressure, duty cycle and pulse repetition frequency (PRF) in the range tabulated in the sections below. We will also examine a small range in driving frequency. These parameters affect droplet vaporization, cavitation dose (both inertial and stable), radiation force (droplet translation), and free-field nucleation.

We will assay sonothrombolysis through four assays: (a) weighing clots before and after treatment (b) quantification of the spatial localization vaporizing nanodroplets and resultant microbubbles by acoustically mapping vaporization and cavitation, and (c) visually observing clot erosion after activating nanodroplets and inducing cavitation with ultrasound, and (d) recording changes in flow as the clot volume is reduced.

These methods provide (i) *quantitative measurement of reduction of clot mass and volume as a function of time,* (ii) *quantitative measures of non-enzymatic thrombolytic efficacy*, (iii) *spatial localization of nanodroplets and magnitude of cavitation, and (iv) qualitative imaging of clot dissolution patterns.*

Experimental in-vitro system

For studies of nanodroplet delivery to thrombus, cavitation monitoring, and clot lysis, we will use the same simulated thrombus flow system described in Aim 1. ***The acoustic and optical monitoring systems are also the same as described in Aim 1.***

Variable: Acoustic parameters: This sub-aim is crucial for our long-term goal, since the nanodroplet-ultrasound interaction is theoretically more complicated than ultrasound alone or ultrasound combined with microbubbles. The acoustic pulse timing is important to allow the nanodroplets to penetrate the clot prior to vaporization. If there is an excessively long acoustic cycle without sufficient off time, flowing nanodroplets will be activated before they reach the clot interior, whereas too short of a pulsing cycle may not apply enough energy to activate the droplets and drive radiation force and cavitation to disrupt the clot from the resulting microbubbles. Exessively high pressure can result in damage to healthy surrounding pressure, while low acoustic pressure will not trigger vaporization and pushing the particles into a fibrin network. As such, we will examine the following acoustic conditions: Frequency (500 kHz, 700 kHz, 1 MHz); Acoustic Pressure (500 kPa, 750 kPa, 1 MPa, 1.5 MPa), Duty Cycle (0.1%, 0.5%, 1%, 5%, 10%). Frequency will be varied by changing the HIFU transducer, whereas other variables can be changed with the function generator. These parameter ranges will sweep across ranges optimized for higher mechanical index (cavitation) and higher radiation force (duty cycle) which likely play a substantial role in clot dissolution. ***These studies will evaluate clot lysis as a function of acoustic parameters.***

Variable: Evaluate reduced rt-PA dose: The clinical dose for rt-PA is 90 milligrams in a 100kg patient, infused over 60 minutes, with 10% of the total dose typically administered in an initial bolus. <u>The possibility for fatal adverse reactions with this anticoagulant are significant, and the ability to utilize a substantially reduced dose, and also be able to affect contracted clots, would make a substantial clinical impact</u>. Our preliminary studies demonstrated low-ultrasound energy sonothrombolysis of contracted clots with a rt-PA dose 10x less (1 microgram/mL) than used clinically. In our in-vitro sonothrombolysis studies, we will explore low-dose rt-PA, with a

target of optimizing acoustic study parameters to maintain rt-PA dose at <1/10 of clinical dose, although we will also evaluate much lower doses with nanodroplet sonothrombolysis (0.5 μg/mL, 0.01 μg/mL, and 0.005 μg/mL). ***These studies will evaluate clot lysis as a function of rt-PA dose, with the intent of identifying the lowest possible dose that is effective with acoustic parameters that are safe to use in-vitro.***

rt-PA loaded nanodroplets as an alternative strategy: A number of prior studies have functionalized lipid coated microbubbles, liposomes, and nanoparticles to carry rt-PA. Of particular note is prior microbubble rt-PA loading as Hua et al. have demonstrated that lipid-shelled perfluoropropane microbubbles can be loaded with rt-PA [65,66]. There is a high likelihood that this would translate to nanodroplets loaded with rt-PA as well, and we will explore this possibility as a means for further reducing systemic rt-PA load while maximizing clot load. Although our prior data has utilized co-administration of rt-PA and nanodroplets, it is possible that loaded nanodroplets would provide even further improved clot lysis. We will explore this direction by formulating rt-PA loaded microbubbles as previously described [66], and condensing them into rt-PA loaded nanodroplets. We will test these in our in-vitro clot system.

Variable: Nanodroplet concentration: We will also vary: the concentration of nanodroplets delivered in solution clots to determine optimal parameters for loading clots with nanodroplets. It is possible that too many nanodroplets will actually have a shielding effect and reduce cavitation internally. ***Studies will identify optimal nanodroplet concentrations which optimally results in clot lysis.***

Assessment: Clot lysis rate: Clots will be carefully weighed with a precision balance (Mettler Toledo, Columbus, OH) before and after treatment. Also, downstream carotid flow will be assessed during the procedure with the aforementioned Transonics Doppler flow probe. ***This will establish mass reduction/lysis rate as well as flow return rate as a function of parameters.***

Assessment: Optical clot dissolution monitoring: High-resolution optical assessment of clot dissolution will be recorded by video microscopy as described in Aim 1. ***Studies will enable visualizion of clot erosion trends (ie internal vs external) as well as confirm lysis rate and dissolution time.***

Assessment: Evaluation of clot residue size: One concern during acoustic clot lysis is the particle fragment size, as large fragments could dislodge and cause additional downstream ischemia. To assess clot fragment size, a subset of experiments will be performed where fragments are captured and analyzed. We will do this with two different approaches. In the first approach, a 0.22-micron large surface area filter will placed downstream from the clot during acoustic lysis. Computerized scanning optical microscopy will then be used to image the filter membrane and count particle size and number. The second approach will consist of collecting fluid downstream from the clot, and then analyzing it with our Accusizer 780 optical particle sizer, which is sensitive to particles from 0.5 – 200 microns. ***Studies will identify fragment sizes under sonothrombolysis conditions, and will provide feedback to optimize parameters for safe fragment sizes.***

Assessment: Vaporization and Cavitation Imaging: Quantitation of nanodroplet vaporization and amount and spatial location of cavitation will be assessed with co-aligned linear array transducer and Verasonics 256 channel ultrasound system as described in Aim 1. ***Studies will quantify droplet vaporization and cavitation during sonothrombolysis.***

Rigor and Reproducibility:
There are no animal studies in Aim 2. However, we will be evaluating a large parameter space (acoustic pressure, 4x; duty cycle, 5x; frequency, 3x; rt-PA dose, 3x; nanodroplet concentration, plus drug-loaded droplets; blood pressure, x3). Hence, we will test subgroups with one fixed parameter first (e.g.: middle acoustic pressure; mddle duty cycle) rather than exploring the full matrix of variables). We can repeat prior studies with expanded parameter ranges if data suggests that moving in a certain trend might provide improved results. In vitro-group sizes will consist of at least 9 technical replicates per parameter, also based on preliminary data.

Expected outcomes, potential problems, and alternative solutions:
We expect the aforementioned studies to provide us with best practice combinations of acoustic pressure, duty cycle, and frequency which maximize sonothrombolysis, and data will generate an understanding of the relationship of acoustic parameters to performance. We also anticipate understanding the role of nanodroplet concentration relationship of these variables to minimum required rt-PA concentration.

One potential problem might be that the fastest dissolution might produce clot lysis fragments that could be large enough for downstream embolization. We will work closely with Dr. Stouffer to understand the impacts of fragment size, and if this is a problem, we will explore parameters which generate smaller fragments. This might mean choosing parameters that do not result in the fastest lysis rate.

C.3. Specific Aim 3: Validate efficacy and safety of nanodroplet-assisted sonothrombolysis in well-characterized *in-vivo* mouse models of arterial thrombosis.

The final project milestone is a a series of *in-vivo* studies using nanodroplet-mediated sonothrombolysis. This study plan has two sub-aims using an arterial clot model. The first of these sub-aims is to assess the safety of ultrasound parameters suggested in Aim 2, to insure that these parameters can be used without substantial bioeffects that would compromise the technique. The second sub-aim will evaluate the clot lysis itself in-vivo.

Preliminary Data:
MPI Wolberg has more than 25 years of experience in the physiologic and pathophysiologic mechanisms in hemostasis and thrombosis, and extensive experience with mouse thrombosis and thrombolysis models. In these studies, we will use well-characterized mouse carotid artery thrombosis/thrombolysis models [49,50,67–70].

Arterial occlusion study: This study will be performed with the mouse carotid artery thrombosis model, as described by Dr. Wolberg [49,69–71]. Briefly, 6- to 8-week old male C57Bl/6 will be anesthetized with isoflurane in oxygen. Both male and female mice will be utilized in equal numbers to test sex as a biological variable. Body temperature will be maintained by a temperature controlled heating pad. The the right common carotid artery will be surgically exposed after midline cervical incision. A Doppler flow probe (Transonic Systems) will be applied and connected to a flow meter (model T206; Transonic Systems). The Doppler probe provides a gold standard of flow occlusion and recovery in the thrombosed artery. The carotid artery will be dried and 10% FeCl3 (0.62 M FeCl3 on 0.5 x 0.5-mm filter paper) placed on the artery for 3 minutes, removed, and tissues washed 3 times with warm saline. After injury, blood flow will be continuously monitored, and is expected to be reduced by up to 90% over the next 5 minutes. Mice which do not experience carotid occlusion reducing flow more than 25% will be omitted from sonothrombolysis studies because they did not form defined thrombi.

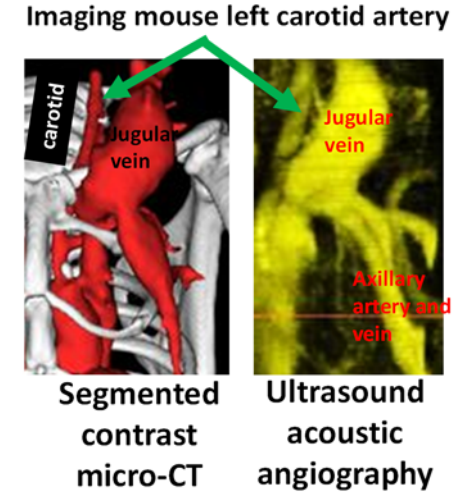

Figure 5. Contrast micro CT as a reference, and live animal 3-D ultrasound angiography with the Vega system, showing visualization of the caudal mouse carotid. F.O.V. ~ 10mm x 5mm, 2-D projection of a volume shown. The Vega system can use image-guidance softare (not shown) to precisely target focused ultrasound to the target site.

For image-guided ultrasound treatment, the aforementioned study will be performed on the imaging platform of a SonoVol Vega image guideded HIFU system. The Vega imaging system provides high-resolution 3-D image guidance co-registered with a programmable HIFU transducer for precise targeting of image guided treatment, and is specifically designed for small rodents. The 'acoustic angiography' mode of the Vega system can clearly resolve the carotid in mice (**Fig. 5)** and will allow image-guided treatment. Ultrasound gel will couple the carotid to the transducer window, and the system software will be used to focus the low-frequency transducer on the carotid clot. Although the standard Vega system utilizes only a 1 MHz HIFU transducer, SonoVol has agreed to modify the system in the Dayton lab so that the HIFU transducer can be changed out between 500 kHz, 700 kHz, or 1 MHz depending on parameter ranges determined in Aims 1-2 (see LOS).

Safety study: Prior to conducting the full sonothrombolysis efficacy study, a subgroup of the arterial occlusion study will be used to specifically focus on safety. For this study arm, sham groups without US or cavitation agents will not be included – instead, four different ultrasound conditions (**Table 1**), each tested in Aims 1-2 and likely to produce the best sonothrombolysis results, will be used to sonicate the carotid artery under thrombolysis conditions. It is likely that several combinations of acoustic pressure and duty cycle will have some sonothrombolysis efficacy, but there is likely to be a balance of minimum heating (lower duty cycle) and lower mechanical index (pressure and frequency) which are least likely to produce peripheral bioeffects. Tissue from

Group	US	rt-PA	Cavitation agent	N
1	Yes (P1)	Yes	Nanodroplet	14
2	Yes (P2)	Yes	Nanodroplet	14
3	Yes (P3)	Yes	Nanodroplet	14
4	Yes (P4)	Yes	Nanodroplet	14

Table. 1. *In-vivo* test groups for safety study. Specific parameters and rt-PA dose will be selected from Aims 1 and 2. Four different pressure and duty cycle conditions which demonstrated efficacy will be evaluated, to establish optimal efficacy with minimal bioeffects.

the treated area will be prepared for histology and provide to the Histology Core to be examined for coagulation, hemorrhage, endothelial damage, and heat shock protein expression. ***Studies will indicate liklihood of bioeffects from higher energy acoustic parameters and indicate parameter ranges that can be used safely.***

In-vivo sonothrombolysis efficacy study: These rodent studies will be the culmination of this project, where the final three most promising parameter sets will be evaluated in in-vivo models.

The Vega system will be used to visualize the carotid thrombus, and the HIFU transducer will be targeted to the thrombus region. The acoustic parameters will be set to the desired range. A 27-gauge indwelling catheter will be used for microbubble or nanodroplet infusion, and infusion will be controlled by a calibrated syringe pump as described previously. Nine different treatment groups will be tested (**Table 2**) with acoustic and nanodroplet parameters estabilished in specific Aims 1 and 2, with the two most promising pressure groups after the bioeffects study arm. rt-PA as well as MB or ND will be continuously administrated over the target time window. ***Clot dissolution will be established by return to flow of at least 90% of baseline flow before thrombus formation, as measured by the calibrated Transonic flowmeter. For each every treatment case, we will analyze 1) restoration starting time, 2) restoration rate, 3) competed restoration time, and 4) restored flow rate.*** We do not expect that the transonic flowmeter will interfere with ultrasound imaging since it will be positioned downstream from the clot study region, however, if it does, we will turn it off during image guidance.

Group	US	rt-PA	Cavitation agent	N
1	No	Yes	No	14
2	Yes (P1)	No	Microbubble	14
3	Yes (P2)	No	Nanodroplet	14
4	Yes (P1)	Yes, co-administered	Microbubble	14
5	Yes (P2)	Yes, co-administered	Nanodroplet	14
6	Yes (P1)	Yes, co-administered	Microbubble	14
7	Yes (P2)	Yes, co-administered	Nanodroplet	14
8	Yes (P1)	rt-PA loaded Nanodroplet		14
9	Yes (P2)	rt-PA loaded Nanodroplet		14

Table. 2. *In-vivo* sonothrombolysis groups. Specific parameters and rt-PA dose will be selected from Aims 1 and 2. Two different pressure conditions which demonstrated efficacy and safety will be evaluated.

Post-treatment histology:
After study conclusion, the carotid and surrounding tissues will be fixed will be dehydrated and paraffin-embedded. Consecutive 5-micron sections will be mounted with vectamount (UNC Lineberger Comprehensive Cancer Center Animal Histopathology Core). Slides will be stained with hematoxylin and eosin to visualize the thrombus and surrounding tissue , and imaged with a Retiga 400R camera (Qimaging) linked to an Olympus Bx 61 microscope with a 20x U Plan FL N, 0.5 NA objective lens. Images will be analyzed with Adobe Photoshop CS (v8.0). For immunohistochemistry, antigen retrieval will be performed in Target Retrieval Solution in a 95°C water bath. Slides were stained with anti-fibrin antibody (59D8, 1:1000), which detects both human and mouse fibrin. ***Histology will evaluate 1) structural integrity of thrombus, and verify 2) absence of ultrasound-induced damage to the vessel or surrounding tissue.***

Rigor and Reproducibility: Both male and female animals will be utilized for the in-vivo studies, 7 each (14 per group), with the group size based on obtaining sufficient power suggested by variance in preliminary studies.

Expected outcomes, potential problems, and alternative solutions: We expect to observe statistically significant differences in 1) restoration rate and 2) restored flow rate by ND+US+reduced rt-PA treatment compared to other test groups. The analysis results will provide in-vivo validation of nanodroplet-enhanced sonothrombolysis. One potential problem is that acoustic parameters that carry over from Aim 2 result in thermal damage or mechanical damage to the vascular endothelium or surrounding tissue. If this happens, we will consult with Dr. Stouffer about what is acceptable. We may reduce acoustic pressure and duty cycle, increase rt-PA dose if needed, and repeat studies at lower energies until a threshold which does not result in bioeffects is reached.

PHS Human Subjects and Clinical Trials Information

OMB Number: 0925-0001

Expiration Date: 02/28/2023

Use of Human Specimens and/or Data

Does any of the proposed research in the application involve human specimens and/or data * ❍ Yes ● No

Provide an explanation for any use of human specimens and/or data not considered to be human subjects research.

Are Human Subjects Involved ● Yes ❍ No

Is the Project Exempt from Federal regulations? ❍ Yes ● No

Exemption Number ❑ 1 ❑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7 ❑ 8

Other Requested Information

Human Subject Studies

Study#	Study Title	Clinical Trial?
1	Novel advances in stroke and arterial thrombus lysis through image guided cavitation with clot penetrating nanodroplets	No

Section 1 - Basic Information (Study 1)

OMB Number: 0925-0001

Expiration Date: 02/28/2023

1.1. Study Title *

Novel advances in stroke and arterial thrombus lysis through image guided cavitation with clot penetrating nanodroplets

1.2. Is this study exempt from Federal Regulations * ❍ Yes ● No

1.3. Exemption Number ❑ 1 ❑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7 ❑ 8

1.4. Clinical Trial Questionnaire *

1.4.a. Does the study involve human participants?	● Yes	❍ No
1.4.b. Are the participants prospectively assigned to an intervention?	❍ Yes	● No
1.4.c. Is the study designed to evaluate the effect of the intervention on the participants?	❍ Yes	● No
1.4.d. Is the effect that will be evaluated a health-related biomedical or behavioral outcome?	❍ Yes	● No

1.5. Provide the ClinicalTrials.gov Identifier (e.g. NCT87654321) for this trial, if applicable

Section 2 - Study Population Characteristics (Study 1)

2.1. Conditions or Focus of Study

- venous thromboembolism

2.2. Eligibility Criteria

Healthy individuals and medically-stable patients, all genders, race, ethnicity, adults and children

2.3. Age Limits	Min Age:	12 Years	Max Age:	99 Years
2.3.a. Inclusion of Individuals Across the Lifespan	Individuals_Across_the_Lifespan1046185232.pdf			
2.4. Inclusion of Women and Minorities	Women_and_Minorities1046094987.pdf			
2.5. Recruitment and Retention Plan	Recruitment_and_Retention_Plan1046185244.pdf			
2.6. Recruitment Status	Not yet recruiting			
2.7. Study Timeline	Study_Timeline1046185215.pdf			
2.8. Enrollment of First Participant	07/01/2021	Anticipated		

INCLUSION OF INDIVIDUALS ACROSS THE LIFESPAN

We will recruit healthy individuals for the proposed studies. Thrombosis risk increases with age, but can affect children. Therefore, children will not be excluded from the proposed research. However, the majority of our donors are adult (greater than 18 years old), so we expect the number of children included in the study will be small (less than 5% of subjects). As with adults, children will be recruited without regard to gender, race, or ethnicity. Since there is not a scientific need to specifically study very young children for this work, we will not draw blood from children under 12 years old.

The study team is trained to consent and draw blood from individuals of all ages, and all members of the team undergo mandatory training in the Protection of Human Subjects and the Health Insurance Portability and Accountability Act (HIPAA). Phlebotomy takes place in a dedicated room located adjacent to the research laboratory, providing privacy and protection for all subjects of all ages. IRB approvals are reviewed and obtained annually and include mechanisms for obtaining consent from both children and parents/legal guardians.

INCLUSION OF WOMEN AND MINORITIES

Targeted/planned distribution of subjects

Women will be recruited for all studies requiring human subjects. No discrimination will be made with respect to gender, race, or ethnicity. See “Targeted/Planned Enrollment Table.”

Rationale for selection of gender, racial, and ethnic group members

Arterial thromboembolism occurs in all racial and ethnic groups. Minorities will be recruited for all studies requiring human subjects. The racial/ethnic composition of the study population will reflect the demographics of our local population. Based on prior experience and the demographic population of Orange County, North Carolina, we anticipate 25% racial minority subjects and 5% Hispanic or Latino subjects. See “Targeted/Planned Enrollment Table.”

Proposed outreach programs

Recruitment will be through UNC Institutional Review Board-approved posters and emails advertising this study. These are displayed in public areas to increase the study population diversity.

Rationale for exclusion

No exclusions; no discrimination will be made with respect to gender, race, or ethnicity.

RECRUITMENT AND RETENTION PLAN

Healthy human subjects will be recruited for the proposed studies. This will include up to 100 healthy individuals. Subjects will be recruited from the local Chapel Hill, Durham, and Raleigh communities.

Subject recruitment activities: Healthy subjects are recruited from our existing cohort of healthy subjects, by flyers advertising the study, and by word-of-mouth. Methods to ensure successful recruitment and appropriate diversity within the subject population have been developed in Dr. Wolberg's laboratory and practiced for over 20 years. The recruited subjects will reflect the diversity of the Chapel Hill, Durham, and Raleigh communities. In the unlikely event we unable to achieve sufficient numbers of subjects, we will increase outreach efforts using social media and interactions with the UNC Clinical and Translational Science Awards Program.

Subject retention activities: Subjects are not required to participate more than once, so retention activities are not applicable.

STUDY TIMELINE

Timeline	Year 1	Year 2	Year 3	Year 4
Nanodroplet parameters and optimization in-vitro (Aim 1)				
Acoustic parameters and optimization in-vitro (Aim 2)				
In-vivo thrombolysis tests using rodent models (Aim 3)				

2.9. Inclusion Enrollment Reports

IER ID#	Enrollment Location Type	Enrollment Location
Study 1, IER 1	Domestic	University

Inclusion Enrollment Report 1

1. Inclusion Enrollment Report Title* : Novel advances in stroke and arterial thrombus lysis through image guided cavitation with clot penetrating nanodroplets

2. Using an Existing Dataset or Resource* : ❍ Yes ● No

3. Enrollment Location Type* : ● Domestic ❍ Foreign

4. Enrollment Country(ies): USA: UNITED STATES

5. Enrollment Location(s): University

6. Comments:

Planned

Racial Categories	Ethnic Categories				Total
	Not Hispanic or Latino		Hispanic or Latino		
	Female	Male	Female	Male	
American Indian/ Alaska Native	2	2	0	0	4
Asian	5	5	0	0	10
Native Hawaiian or Other Pacific Islander	1	1	0	0	2
Black or African American	7	7	0	0	14
White	30	30	5	5	70
More than One Race	0	0	0	0	0
Total	45	45	5	5	100

Cumulative (Actual)

Racial Categories	Ethnic Categories									Total
	Not Hispanic or Latino			Hispanic or Latino			Unknown/Not Reported Ethnicity			
	Female	Male	Unknown/ Not Reported	Female	Male	Unknown/ Not Reported	Female	Male	Unknown/ Not Reported	
American Indian/ Alaska Native	0	0	0	0	0	0	0	0	0	0
Asian	0	0	0	0	0	0	0	0	0	0
Native Hawaiian or Other Pacific Islander	0	0	0	0	0	0	0	0	0	0
Black or African American	0	0	0	0	0	0	0	0	0	0
White	0	0	0	0	0	0	0	0	0	0
More than One Race	0	0	0	0	0	0	0	0	0	0
Unknown or Not Reported	0	0	0	0	0	0	0	0	0	0
Total	0	0	0	0	0	0	0	0	0	0

Section 3 - Protection and Monitoring Plans (Study 1)

3.1. Protection of Human Subjects

Protection_of_Human_Subjects1046094986.pdf

3.2. Is this a multi-site study that will use the same protocol to conduct non-exempt human subjects research at more than one domestic site?

❍ Yes ● No ❍ N/A

If yes, describe the single IRB plan

3.3. Data and Safety Monitoring Plan

3.4. Will a Data and Safety Monitoring Board be appointed for this study?

❍ Yes ● No

3.5. Overall structure of the study team

PROTECTION OF HUMAN SUBJECTS

1. Risks to Human Subjects

a. Human Subjects Involvement Characteristics, and Design: This study will involve blood draws to prepare clots for the described experiments. For these experiments, 10-55 ml of blood will be drawn from a vein in the subject's arm approximately 1, but no more than 2 times per week. Representatives from all genders, racial, and ethnic categories will be recruited. The composition of the study population will mimic the gender and racial/ethnic composition of the local demography and will consist of approximately 100 individuals, approximately 50% men and 50% women, predominantly ages 18-100. Children will not be excluded from participation; however, we will not include very young children (<12 years old) because there is not a specific scientific need to study this population at this time. Subjects will consist of healthy, non-patient volunteers, as well as individuals with factor V Leiden or obesity. These individuals will be determined to be medically stable prior to their participation in this study. Donors will be drawn in dedicated space for phlebotomy adjacent to the PI's laboratory or at the UNC Hemophilia and Thrombophilia Centers. Subjects will be selected without regard to race or gender. No prisoners, institutionalized individuals, or other vulnerable populations will be included in this study.

b. Study Procedures, Materials, and Potential Risks of Materials: Blood will be obtained from participating subjects specifically for this research project. Notes from experiments performed on donated blood will be maintained in laboratory notebooks in a secure database in Dr. Wolberg's laboratory. The material and data obtained from these studies will be used specifically for research purposes. All samples will be coded and a link between the sample and any identifying information will be secured in password-protected files.

c. Potential Risks: This study involves the likely but non-serious possibility of slight bleeding at the site of venipuncture. This study also involves the unlikely possibility of fainting. There are no anticipated psychological, social, or legal risks for participating in this study. There are no anticipated long-term risks.

2. Adequacy of Protection Against Risks

a. Informed Consent and Assent: Recruitment of healthy control subjects will be conducted "word-of-mouth," and by UNC Institutional Review Board-approved posted flyers or emails. Subjects may also be recruited from the UNC Hemophilia or UNC Thrombophilia Centers' patient populations. Consent will be obtained by Dr. Wolberg or someone under Dr. Wolberg's supervision and will be performed prior to phlebotomy. Consent forms used in this study will be approved by the UNC Institutional Review Board, and include consent forms for adults and children, and parents/legal guardians of children. Signed consent forms will be maintained in Dr. Wolberg's laboratory for a period of 7 years, according to Institutional Review Board regulations.

b. Protections Against Risk: Every effort will be taken to protect the identity of the participants in this study. Consent forms and links between samples and identifying information will be maintained in a locked cabinet in the laboratory and in password-protected files. No subjects will be identified in any report or publication of this study or its results. The subjects will be informed that there is no guarantee that the information cannot be obtained by legal process or court order. This study involves only minimal medical risk, as described above in "Potential Risks." Subjects will be instructed to apply pressure to the site of venipuncture in order to minimize this risk. Subjects will be informed that immediate necessary care is available if an individual is injured because of participation in this research project. Subjects will be informed that there is no provision for free medical care or for monetary compensation for such injury and that further information can be obtained from UNC Hospital's Risk Management Office.

c. Vulnerable Subjects: Not applicable

3. Potential Benefits of the Proposed Research to the Subjects and Others

There are no direct benefits to individual subjects for participating in this study. Participants incur only minimal risk by participating (see above, Potential Risks). However, new findings from this study will aid in understanding hemostatic and thrombotic disorders and may aid in improving treatment for patients with coagulopathy.

4. Importance of the Knowledge to be Gained

This study proposes to investigate mechanisms for treatment of arterial thromboembolism. Risks to the subjects included in this study are minimal, involving only the small chance of bleeding at the venipuncture site or fainting. However, information gained from this and other research studies may aid in improving diagnostic or therapeutic strategies for the treatment and prevention of stroke, heart attack, or other arterial thrombosis diseases.

Section 4 - Protocol Synopsis (Study 1)

4.1. Study Design

4.1.a. Detailed Description

4.1.b. Primary Purpose

4.1.c. Interventions

Type	Name	Description

4.1.d. Study Phase

Is this an NIH-defined Phase III Clinical Trial? ❍ Yes ❍ No

4.1.e. Intervention Model

4.1.f. Masking ❍ Yes ❍ No

❑ Participant ❑ Care Provider ❑ Investigator ❑ Outcomes Assessor

4.1.g. Allocation

4.2. Outcome Measures

Type	Name	Time Frame	Brief Description

4.3. Statistical Design and Power

4.4. Subject Participation Duration

4.5. Will the study use an FDA-regulated intervention? ❍ Yes ❍ No

4.5.a. If yes, describe the availability of Investigational Product (IP) and Investigational New Drug (IND)/ Investigational Device Exemption (IDE) status

4.6. Is this an applicable clinical trial under FDAAA? ❍ Yes ❍ No

4.7. Dissemination Plan

Delayed Onset Studies

Delayed Onset Study#	Study Title	Anticipated Clinical Trial?	Justification
The form does not have any delayed onset studies			

VERTEBRATE ANIMALS

DESCRIPTION OF PROCEDURES

Mice will be used for this study. Mice will be on a C57Bl/6 background and will be between 8-14-weeks old. Males and females will be used for the arterial thrombosis models. These mice will be subjected to the FeCl3-induced arterial thrombosis model, and then will be imaged and treated with therapeutic ultrasound to reduce the clot and restore flow. A tail vein catheter will be utilized to introduce cavitation enhancement agents and rt-PA. Blood will be collected via IVC puncture. Prospective power analysis based on published and preliminary data has been used to estimate minimum numbers of mice needed per group for these experiments. We estimate we will need 14 mice per experimental group (7 males and 7 females), and we have budgeted one extra mouse per group in case of complications. Thrombosis studies will be performed by Dr. Wolberg's group at UNC and targeting studies will be performed by Dr. Brown's group at the NC State Veterinary Small Animal Facilities. Faculty in the Joint Department of Biomedical Engineering have full access to facilities at both institutions. We will utilize 13 experimental groups for the thrombosis studies, for a total of 195 mice. For the targeting study, we will use two experimental groups of 14 mice each, and we have budgeted for a total of 32 mice, allowing two extra per group in case of complications. Mice will be purchased instead of purpose bred, thus requiring only minimal caging needs to coordinate experiments.

JUSTIFICATIONS

Previous *in vitro* studies have provided significant information regarding the role(s) of soluble, cellular, and physical (shear) influences in thrombus formation. However, no model system, *in vitro*, *in silico* or other, except a whole animal recapitulates the complex molecular and cellular mechanisms of multiple biologic systems that, when combined, result in thrombosis. For these reasons, it is necessary to use an animal model for these experiments as a prerequisite to trials in humans. The murine system: 1) is amenable to established techniques for protein infusion and genetic manipulation that allow one to define the molecular mechanisms of disease, 2) permits ease of husbandry, and 3) mimics processes that contribute to clot formation in humans. We will conduct these experiments in mice with a C57BL/6 background because they are an established model of arterial and venous thrombosis, and because most genetically-modified strains of mice that we will use in future studies are in this background. Thus, results from studies described in this proposal will provide important data that will be used to design future experiments. *The number of mice selected for study represents the minimum numbers that will permit us to determine a statistically significant difference among experimental groups, derived from statistically relevant power calculations.*

MINIMIZATION OF PAIN AND DISTRESS

At UNC, mice are checked daily by Division of Comparative Medicine staff and also by Dr. Wolberg and Dr. Dayton's laboratory staff. At NC State, mice are checked daily by the College of Veterinary Medicine staff and also by Dr. Brown's laboratory staff. For all procedures, pain and discomfort will be minimized during all surgical and non-surgical procedures (injections and ultrasound treatment). Mice will be anesthetized for surgery and ultrasound treatment by exposure and maintenance with a constant flow of isoflurane (2-3%) mixed with oxygen. After induction, anesthesia will be maintained with 1.5-2% isoflurane mixed with oxygen. This anesthetic allows for complete sedation with rapid recovery. As an alternative, mice are anesthetized by ketamine/xylazine. For surgical procedures, paws will be gently restrained by Velcro attached to the ECG pads and/or by a loop of PE tubing attached to the board. Post-surgical analgesia will be provided by buprenorphine and Tylenol water. Bleeding via the retro-orbital plexus will also take place under complete sedation. After all survival procedures, the mice will be observed until fully recovered and resuming normal behavior before being returned to their animal cubicles. None of the procedures outlined will subject mice to any discomfort, distress, pain, or injury and occur under the most stringent aseptic conditions.

Veterinary Care: All work will be approved by the UNC IACUC, supervised by the Division of Comparative Medicine at UNC, and the NC State IACUC, and will be conducted in accordance with the Public Health Service (PHS) policy on Humane Care and Use of Laboratory Animals, the Amended Animal Welfare Act of 1985, and the regulations of the United States Department of Agriculture. Veterinary care is provided by veterinary staff at both institutions. The veterinary staff is fully accredited and excellent at both institutions. At UNC, the Division of Comparative Medicine is responsible for the care of all vertebrate animals on the UNC-CH campus. The Division of Comparative Medicine is committed to programs of excellence in veterinary care and laboratory animal management practices for all species used in UNC-CH research endeavors. The Division's program is reviewed

semi-annually by the Institutional Animal Care and Use Committee and has been accredited by the Association for the Assessment and Accreditation of Laboratory Animal Care International (AAALAC) since 1973. The Division of Comparative Medicine and the University also meet U. S. Department of Agriculture (registration number 55-R-004) and Public Health Service standards and regulations (assurance number A3410-01). UNC is committed to providing a high-quality program of animal care in compliance with the State and Federal Animal Welfare Acts and the standards and policies of the Department of Health and Human Services, adhering to the NIH documents “Principles for Use of Animals” and “The Guide for the Care and Use of Laboratory Animals.”

METHOD OF EUTHANASIA

Mice will be euthanized under supervision of the Division of Comparative Medicine following the standardized guidelines for humane euthanasia that have been detailed in the June 2007 AVMA Guidelines on Euthanasia. Mice will be euthanized either by cervical dislocation under anesthesia or by carbon dioxide (compressed CO_2 gas) followed by cervical dislocation. Any animal that exhibits serious discomfort, pain or injury during procedures will be euthanized if there are no correctable causes for its suffering.

MULTIPLE PI LEADERSHIP PLAN

This project is greatly strengthened by the Multiple PI approach because it will combine an interdisciplinary team in Dr. Wolberg's extensive experience in the area of thrombosis with Dr. Dayton's experience in biomedical ultrasound and phase change nanodroplets. Both areas of knowledge are crucial to the success of this project. Together, Drs. Wolberg and Dayton will provide leadership and oversight for the project including development of all procedures and policies to be followed by research staff.

Project Management

The PIs will conduct formal monthly meetings with the entire research team to discuss results and plan future experiments. Internal budgeting has already been formalized to account for this breakdown between PI labs' efforts across the aims. The contact PI for this project will be Dr. Dayton, who will be responsible for submission of progress reports and other communications to the NIH in close collaboration with Dr. Wolberg as well as co-investigators Jiang, Brown, and Stouffer.

Conflict Resolution

In the unlikely event of conflicts between the researchers, a panel consisting of three experienced leaders across UNC and NCSU will serve as arbitrators.

- Robert Duronio, Ph.D. – Associate Dean for Research at the UNC School of Medicine
- George Ligler, D.Phil. – Dean's Eminent Professor of the Practice, UNC-NC State Joint Dept of Biomedical Engineering
- Victoria Bautch, Ph.D. – Beverly Long Chapin Distinguished Professor and Co-Director, UNC McAllister Hearth Institute

REFERENCES CITED

[1] M. Muchada, D. Rodriguez-Luna, J. Pagola, A. Flores, E. Sanjuan, P. Meler, S. Boned, J. Alvarez-Sabin, M. Ribo, C.A. Molina, M. Rubiera, Impact of time to treatment on tissue-type plasminogen activator-induced recanalization in acute ischemic stroke, Stroke. 45 (2014) 2734–2738. doi:10.1161/STROKEAHA.114.006222.

[2] Y.D. Kim, H.S. Nam, S.H. Kim, E.Y. Kim, D. Song, I. Kwon, S.H. Yang, K. Lee, J. Yoo, H.S. Lee, J.H. Heo, Time-Dependent Thrombus Resolution after Tissue-Type Plasminogen Activator in Patients with Stroke and Mice, Stroke. 46 (2015) 1877–1882. doi:10.1161/STROKEAHA.114.008247.

[3] A.L. Samson, I. Alwis, J.A.A. Maclean, P. Priyananda, B. Hawkett, S.M. Schoenwaelder, S.P. Jackson, Endogenous fibrinolysis facilitates clot retraction in vivo, Blood. 130 (2017) 2453–2462. doi:10.1182/blood-2017-06-789032.

[4] C.A. Molina, A.D. Barreto, G. Tsivgoulis, P. Sierzenski, M.D. Malkoff, M. Rubiera, N. Gonzales, R. Mikulik, G. Pate, J. Ostrem, W. Singleton, G. Manvelian, E.C. Unger, J.C. Grotta, P.D. Schellinger, A. V Alexandrov, Transcranial ultrasound in clinical sonothrombolysis (TUCSON) trial, Ann. Neurol. 66 (2009) 28–38. doi:10.1002/ana.21723.

[5] A. V. Alexandrov, M. Köhrmann, L. Soinne, G. Tsivgoulis, A.D. Barreto, A.M. Demchuk, V.K. Sharma, R. Mikulik, K.W. Muir, G. Brandt, J. Alleman, J.C. Grotta, C.R. Levi, C.A. Molina, M. Saqqur, D. Mavridis, T. Psaltopoulou, M. Vosko, J.B. Fiebach, P. Mandava, T.A. Kent, A.W. Alexandrov, P.D. Schellinger, Safety and efficacy of sonothrombolysis for acute ischaemic stroke: a multicentre, double-blind, phase 3, randomised controlled trial, Lancet Neurol. 18 (2019) 338–347. doi:10.1016/S1474-4422(19)30026-2.

[6] D. Michael, G. Achim, R. Peter, S. Matthias, S. Ulrich, E. Thomas, S. Oliver, K.W. J., H.M. G., Transcranial Low-Frequency Ultrasound-Mediated Thrombolysis in Brain Ischemia, Stroke. 36 (2005) 1441–1446. doi:10.1161/01.STR.0000170707.86793.1a.

[7] N. Aliona, K.C. E., N. Halvor, Ø. Halvor, L. Nicola, A. Jörg, W.-A. Ulrike, K.K. D., N. Gesche, T. Lars, NOR-SASS (Norwegian Sonothrombolysis in Acute Stroke Study), Stroke. 48 (2017) 335–341. doi:10.1161/STROKEAHA.116.014644.

[8] A. V Alexandrov, G. Tsivgoulis, M. Köhrmann, A.H. Katsanos, L. Soinne, A.D. Barreto, T. Rothlisberger, V.K. Sharma, R. Mikulik, K.W. Muir, C.R. Levi, C.A. Molina, M. Saqqur, D. Mavridis, T. Psaltopoulou, M.R. Vosko, J.B. Fiebach, P. Mandava, T.A. Kent, A.W. Alexandrov, P.D. Schellinger, Endovascular equipoise shift in a phase III randomized clinical trial of sonothrombolysis for acute ischemic stroke, Ther. Adv. Neurol. Disord. 12 (2019) 1756286419860652. doi:10.1177/1756286419860652.

[9] R.P. Engelberger, D. Spirk, T. Willenberg, A. Alatri, D. Do Do, I. Baumgartner, N. Kucher, Ultrasound-Assisted versus conventional catheter-directed thrombolysis for acute iliofemoral deep vein thrombosis, Circ. Cardiovasc. Interv. 8 (2015) 1–10. doi:10.1161/CIRCINTERVENTIONS.114.002027.

[10] S. Vedantham, S.Z. Goldhaber, J.A. Julian, S.R. Kahn, M.R. Jaff, D.J. Cohen, E. Magnuson, M.K. Razavi, A.J. Comerota, H.L. Gornik, T.P. Murphy, L. Lewis, J.R. Duncan, P. Nieters, M.C. Derfler, M. Filion, C.-S. Gu, S. Kee, J. Schneider, N. Saad, M. Blinder, S. Moll, D. Sacks, J. Lin, J. Rundback, M. Garcia, R. Razdan, E. VanderWoude, V. Marques, C. Kearon, Pharmacomechanical Catheter-Directed Thrombolysis for Deep-Vein Thrombosis, N. Engl. J. Med. 377 (2017) 2240–2252. doi:10.1056/NEJMoa1615066.

[11] J.T. Sutton, N.M. Ivancevich, S.R. Perrin, D.C. Vela, C.K. Holland, Clot Retraction Affects the Extent of Ultrasound-Enhanced Thrombolysis in an Ex Vivo Porcine Thrombosis Model, Ultrasound Med. Biol. 39 (2013) 813–824. doi:10.1016/j.ultrasmedbio.2012.12.008.

[12] X. Zhang, G.E. Owens, C.A. Cain, H.S. Gurm, J. Macoskey, Z. Xu, Histotripsy Thrombolysis on Retracted Clots, Ultrasound Med. Biol. 42 (2016) 1903–1918. doi:10.1016/j.ultrasmedbio.2016.03.027.

[13] K. Tachibana, S. Tachibana, Albumin microbubble echo-contrast material as an enhancer for ultrasound accelerated thrombolysis., Circulation. 92 (1995) 1148–1150. doi:10.1161/01.CIR.92.5.1148.

[14] B. Yochai, L. Huai, N. Tomoo, F.M. C., P.T. M., L. Shouping, K. David, P.T. R., S.R. J., Noninvasive In Vivo Clot Dissolution Without a Thrombolytic Drug , Circulation. 97 (1998) 130–134. doi:10.1161/01.CIR.97.2.130.

[15] K.B. Bader, M.J. Gruber, C.K. Holland, Shaken and Stirred: Mechanisms of Ultrasound-Enhanced Thrombolysis, Ultrasound Med. Biol. 41 (2015) 187–196. doi:10.1016/j.ultrasmedbio.2014.08.018.

[16] P.S. Sheeran, V.P. Wong, S. Luois, R.J. McFarland, W.D. Ross, S. Feingold, T.O. Matsunaga, P.A. Dayton, Decafluorobutane as a Phase-Change Contrast Agent for Low-Energy Extravascular Ultrasonic Imaging, Ultrasound Med. Biol. 37 (2011) 1518–1530. doi:10.1016/j.ultrasmedbio.2011.05.021.

[17] P.S. Sheeran, S. Luois, P.A. Dayton, T.O. Matsunaga, Formulation and acoustic studies of a new phase-shift agent for diagnostic and therapeutic ultrasound, Langmuir. 27 (2011) 10412–10420. doi:10.1021/la2013705.

[18] P.S. Sheeran, P.A. Dayton, Phase-change contrast agents for imaging and therapy, Curr. Pharm. Des. 18 (2012) 2152–65. doi:10.2174/138161212800099883.

[19] P.S. Sheeran, S.H. Luois, L.B. Mullin, T.O. Matsunaga, P.A. Dayton, Design of ultrasonically-activatable nanoparticles using low boiling point perfluorocarbons, Biomaterials. 33 (2012) 3262–3269. doi:10.1016/j.biomaterials.2012.01.021.

[20] J. Kim, R.M. DeRuiter, L. Goel, Z. Xu, X. Jiang, P.A. Dayton, A Comparison of Sonothrombolysis in Aged Clots between Low-Boiling-Point Phase-Change Nanodroplets and Microbubbles of the Same Composition, Ultrasound Med. Biol. 46 (2020) 3059–3068. doi:https://doi.org/10.1016/j.ultrasmedbio.2020.07.008.

[21] J. Kim, L. Goel, X. Jiang, Z. Xu, P.A. Dayton, Cavitation-Enhanced High-Pressure Pulsed Sonothrombolysis with Perfluorocarbon Nanodroplets versus Microbubbles in Contracted and Uncontracted Clots, in: IEEE Int. Ultrason. Symp. IUS, 2020.

[22] W. Hacke, M. Kaste, E. Bluhmki, M. Brozman, A. Dávalos, D. Guidetti, V. Larrue, K.R. Lees, Z. Medeghri, T. Machnig, D. Schneider, R. von Kummer, N. Wahlgren, D. Toni, Thrombolysis with Alteplase 3 to 4.5 Hours after Acute Ischemic Stroke, N. Engl. J. Med. 359 (2008) 1317–1329. doi:10.1056/NEJMoa0804656.

[23] N. Wahlgren, N. Ahmed, A. Dávalos, G.A. Ford, M. Grond, W. Hacke, M.G. Hennerici, M. Kaste, S. Kuelkens, V. Larrue, K.R. Lees, R.O. Roine, L. Soinne, D. Toni, G. Vanhooren, Thrombolysis with alteplase for acute ischaemic stroke in the Safe Implementation of Thrombolysis in Stroke-Monitoring Study (SITS-MOST): an observational study, Lancet. 369 (2007) 275–282. doi:https://doi.org/10.1016/S0140-6736(07)60149-4.

[24] G. Micieli, S. Marcheselli, P.A. Tosi, Safety and efficacy of alteplase in the treatment of acute ischemic stroke, Vasc. Health Risk Manag. 5 (2009) 397–409. doi:10.2147/vhrm.s4561.

[25] J.-M. Escoffre, A. Bouakaz, eds., Therapeutic Ultrasound, Springer International Publishing, Cham, Switzerland, 2016. doi:10.2301/neurosonology.16.146.

[26] F. Siddiqi, T.M. Odrljin, P.J. Fay, C. Cox, C.W. Francis, Binding of tissue-plasminogen activator to fibrin: effect of ultrasound., Blood. 91 (1998) 2019–2025.

[27] V. Tutwiler, A.D. Peshkova, G. Le Minh, S. Zaitsev, R.I. Litvinov, D.B. Cines, J.W. Weisel, Blood clot contraction differentially modulates internal and external fibrinolysis, J. Thromb. Haemost. 17 (2019) 361–370. doi:10.1111/jth.14370.

[28] M. Zhang, M.L. Fabiilli, K.J. Haworth, J.B. Fowlkes, O.D. Kripfgans, W.W. Roberts, K.A. Ives, P.L. Carson, Initial Investigation of Acoustic Droplet Vaporization for Occlusion in Canine Kidney, Ultrasound Med. Biol. 36 (2010) 1691–1703. doi:https://doi.org/10.1016/j.ultrasmedbio.2010.06.020.

[29] O.D. Kripfgans, J.B. Fowlkes, D.L. Miller, O.P. Eldevik, P.L. Carson, Acoustic droplet vaporization for therapeutic and diagnostic applications, Ultrasound Med. Biol. 26 (2000) 1177–1189. doi:10.1016/S0301-5629(00)00262-3.

[30] E. Vlaisavljevich, O. Aydin, Y.Y. Durmaz, K.W. Lin, B. Fowlkes, Z. Xu, M.E.H. ElSayed, Effects of Droplet Composition on Nanodroplet-Mediated Histotripsy, Ultrasound Med. Biol. 42 (2016) 931–946. doi:10.1016/j.ultrasmedbio.2015.11.027.

[31] S.M. Fix, B.P. Koppolu, A. Novell, J. Hopkins, T.M. Kierski, D.A. Zaharoff, P.A. Dayton, V. Papadopoulou, Ultrasound-Stimulated Phase-Change Contrast Agents for Transepithelial Delivery of Macromolecules, Toward Gastrointestinal Drug Delivery, Ultrasound Med. Biol. 45 (2019) 1762–1776. doi:10.1016/j.ultrasmedbio.2019.02.004.

[32] L.C. Moyer, K.F. Timbie, P.S. Sheeran, R.J. Price, G.W. Miller, P.A. Dayton, High-intensity focused ultrasound ablation enhancement in vivo via phase-shift nanodroplets compared to microbubbles, J. Ther. Ultrasound. 3 (2015) 7. doi:10.1186/s40349-015-0029-4.

[33] O.D. Kripfgans, M.L. Fabiilli, P.L. Carson, J.B. Fowlkes, On the acoustic vaporization of micrometer-sized droplets, J. Acoust. Soc. Am. 116 (2004) 272–281. doi:10.1121/1.1755236.

[34] O.D. Kripfgans, J.B. Fowlkes, M. Woydt, O.P. Eldevik, P.L. Carson, In vivo droplet vaporization for occlusion therapy and phase aberration correction, IEEE Trans. Ultrason. Ferroelectr. Freq. Control. 49 (2002) 726–738. doi:10.1109/TUFFC.2002.1009331.

[35] S.A. Choudhury, F. Xie, P.A. Dayton, T.R. Porter, Acoustic Behavior of a Reactivated, Commercially Available Ultrasound Contrast Agent, J. Am. Soc. Echocardiogr. 30 (2017) 189–197.

doi:10.1016/j.echo.2016.10.015.

[36] T. Lacour, M. Guédra, T. Valier-Brasier, F. Coulouvrat, A model for acoustic vaporization dynamics of a bubble/droplet system encapsulated within a hyperelastic shell, J. Acoust. Soc. Am. 143 (2018) 23–37. doi:10.1121/1.5019467.

[37] S. Guo, X. Guo, X. Wang, D. Zhou, X. Du, M. Han, Y. Zong, M. Wan, Reduced clot debris size in sonothrombolysis assisted with phase-change nanodroplets, Ultrason. Sonochem. 54 (2019) 183–191. doi:10.1016/j.ultsonch.2019.02.001.

[38] P.S. Sheeran, N. Matsuura, M.A. Borden, R. Williams, T.O. Matsunaga, P.N. Burns, P.A. Dayton, Methods of Generating Submicrometer Phase-Shift Perfluorocarbon Droplets for Applications in Medical Ultrasonography, IEEE Trans. Ultrason. Ferroelectr. Freq. Control. 64 (2017) 252–263. doi:10.1109/TUFFC.2016.2619685.

[39] E.C. Everbach, C.W. Francis, Cavitational mechanisms in ultrasound-accelerated thrombolysis at 1 MHz, Ultrasound Med. Biol. 26 (2000) 1153–1160. doi:10.1016/S0301-5629(00)00250-7.

[40] H.L. Weiss, P. Selvaraj, K. Okita, Y. Matsumoto, A. Voie, T. Hoelscher, A.J. Szeri, Mechanical clot damage from cavitation during sonothrombolysis., J. Acoust. Soc. Am. 133 (2013) 3159–75. doi:10.1121/1.4795774.

[41] A. Soltani, R. Singhal, M. Obtera, R.A. Roy, W.M. Clark, D.R. Hansmann, Potentiating intra-arterial sonothrombolysis for acute ischemic stroke by the addition of the ultrasound contrast agents (Optison™ & SonoVue®), J. Thromb. Thrombolysis. 31 (2011) 71–84. doi:10.1007/s11239-010-0483-3.

[42] P.S. Sheeran, P. a Dayton, Improving the performance of phase-change perfluorocarbon droplets for medical ultrasonography: current progress, challenges, and prospects., Scientifica (Cairo). 2014 (2014) 579684. doi:10.1155/2014/579684.

[43] P.S. Sheeran, J.D. Rojas, C. Puett, J. Hjelmquist, C.B. Arena, P.A. Dayton, Contrast-enhanced ultrasound imaging and in vivo circulatory kinetics with low-boiling-point nanoscale phase-change perfluorocarbon agents, Ultrasound Med. Biol. 41 (2015) 814–831. doi:10.1016/j.ultrasmedbio.2014.10.020.

[44] P.S. Sheeran, K. Yoo, R. Williams, M. Yin, F.S. Foster, P.N. Burns, More Than Bubbles: Creating Phase-Shift Droplets from Commercially Available Ultrasound Contrast Agents, Ultrasound Med. Biol. 43 (2016) 531–540. doi:10.1016/j.ultrasmedbio.2016.09.003.

[45] P.S. Sheeran, J.E. Streeter, L.B. Mullin, T.O. Matsunaga, P.A. Dayton, Toward Ultrasound Molecular Imaging With Phase-Change Contrast Agents: An In Vitro Proof of Principle, Ultrasound Med. Biol. 39 (2013) 893–902. doi:https://doi.org/10.1016/j.ultrasmedbio.2012.11.017.

[46] L. Goel, H. Wu, B. Zhang, J. Kim, P.A. Dayton, Z. Xu, X. Jiang, Nanodroplet Mediated Catheter-Directed Sonothrombolysis of Retracted Blood Clots: an In Vitro Study, Microsystems Nanoeng. Submitted (2020).

[47] D. Suo, Z. Jin, X. Jiang, P.A. Dayton, Y. Jing, Microbubble mediated dual-frequency high intensity focused ultrasound thrombolysis: An *In vitro* study, Appl. Phys. Lett. 110 (2017) 023703. doi:10.1063/1.4973857.

[48] J. Kim, W. Chang, B.D. Lindsey, P.A. Dayton, J.M. Stavas, X. Dai, X. Jiang, Laser-generated-focused ultrasound transducers for microbubble-mediated, dual-excitation sonothrombolysis, in: IEEE Int. Ultrason. Symp. IUS, 2016: pp. 8–11.

[49] K.R. Machlus, J.C. Cardenas, F.C. Church, A.S. Wolberg, Causal relationship between hyperfibrinogenemia, thrombosis, and resistance to thrombolysis in mice, Blood. 117 (2011) 4953–4963. doi:10.1182/blood-2010-11-316885.

[50] Z. Tang, S. Kattula, L.A. Holle, B.C. Cooley, F.-C. Lin, A.S. Wolberg, Factor XIII deficiency does not prevent FeCl3-induced carotid artery thrombus formation in mice, Res. Pract. Thromb. Haemost. 4 (2020) 111–116. doi:10.1002/rth2.12278.

[51] J.D. Rojas, P.A. Dayton, Vaporization Detection Imaging: A Technique for Imaging Low-Boiling-Point Phase-Change Contrast Agents with a High Depth of Penetration and Contrast-to-Tissue Ratio, Ultrasound Med. Biol. 45 (2019) 192–207. doi:10.1016/j.ultrasmedbio.2018.08.017.

[52] J.E. Streeter, R. Gessner, I. Miles, P. a. Dayton, Improving sensitivity in ultrasound molecular imaging by tailoring contrast agent size distribution: In vivo studies, Mol. Imaging. 9 (2010) 87–95. doi:10.2310/7290.2010.00005.

[53] S. Nandi, L. Sommerville, K. Nellenbach, E. Mihalko, M. Erb, D.O. Freytes, M. Hoffman, D. Monroe, A.C. Brown, Platelet-like particles improve fibrin network properties in a hemophilic model of provisional matrix structural defects, J. Colloid Interface Sci. 577 (2020) 406–418. doi:https://doi.org/10.1016/j.jcis.2020.05.088.

[54] S. Nandi, K. Mohanty, K. Nellenbach, M. Erb, M. Muller, A.C. Brown, Ultrasound Enhanced Synthetic

Platelet Therapy for Augmented Wound Repair, ACS Biomater. Sci. Eng. 6 (2020) 3026–3036. doi:10.1021/acsbiomaterials.9b01976.

[55] E. Chee, S. Nandi, K. Nellenbach, E. Mihalko, D.B. Snider, L. Morrill, A. Bond, E. Sproul, J. Sollinger, G. Cruse, M. Hoffman, A.C. Brown, Nanosilver composite pNIPAm microgels for the development of antimicrobial platelet-like particles, J. Biomed. Mater. Res. Part B Appl. Biomater. 108 (2020) 2599–2609. doi:10.1002/jbm.b.34592.

[56] E.P. Sproul, S. Nandi, E. Chee, S. Sivadanam, B.J. Igo, L. Schreck, A.C. Brown, Development of Biomimetic Antimicrobial Platelet-Like Particles Comprised of Microgel Nanogold Composites, Regen. Eng. Transl. Med. 6 (2020) 299–309. doi:10.1007/s40883-019-00121-6.

[57] S. Nandi, E.P. Sproul, K. Nellenbach, M. Erb, L. Gaffney, D.O. Freytes, A.C. Brown, Platelet-like particles dynamically stiffen fibrin matrices and improve wound healing outcomes, Biomater. Sci. 7 (2019) 669–682. doi:10.1039/C8BM01201F.

[58] E.P. Sproul, S. Nandi, C. Roosa, L. Schreck, A.C. Brown, Biomimetic Microgels with Controllable Deformability Improve Healing Outcomes, Adv. Biosyst. 2 (2018) 1800042. doi:10.1002/adbi.201800042.

[59] E. Mihalko, K. Huang, E. Sproul, K. Cheng, A.C. Brown, Targeted Treatment of Ischemic and Fibrotic Complications of Myocardial Infarction Using a Dual-Delivery Microgel Therapeutic, ACS Nano. 12 (2018) 7826–7837. doi:10.1021/acsnano.8b01977.

[60] A.C. Brown, S.R. Baker, A.M. Douglas, M. Keating, M.B. Alvarez-Elizondo, E.L. Botvinick, M. Guthold, T.H. Barker, Molecular interference of fibrin's divalent polymerization mechanism enables modulation of multiscale material properties, Biomaterials. 49 (2015) 27–36. doi:https://doi.org/10.1016/j.biomaterials.2015.01.010.

[61] A.C. Brown, S.E. Stabenfeldt, B. Ahn, R.T. Hannan, K.S. Dhada, E.S. Herman, V. Stefanelli, N. Guzzetta, A. Alexeev, W.A. Lam, L.A. Lyon, T.H. Barker, Ultrasoft microgels displaying emergent platelet-like behaviours, Nat. Mater. 13 (2014) 1108–1114. doi:10.1038/nmat4066.

[62] R.F. Doolittle, G. Spraggon, S.J. Everse, Evolution of Vertebrate Fibrin Formation and the Process of its Dissolution, Ciba Found. Symp. 212 - Plasminogen-Related Growth Factors. (2007) 4–23. doi:doi:10.1002/9780470515457.ch2.

[63] I.N. Chernysh, C. Nagaswami, P.K. Purohit, J.W. Weisel, Fibrin Clots Are Equilibrium Polymers That Can Be Remodeled Without Proteolytic Digestion, Sci. Rep. 2 (2012) 879. doi:10.1038/srep00879.

[64] A.J. Dixon, J. Li, J.M.R. Rickel, A.L. Klibanov, Z. Zuo, J.A. Hossack, Efficacy of Sonothrombolysis Using Microbubbles Produced by a Catheter-Based Microfluidic Device in a Rat Model of Ischemic Stroke, Ann. Biomed. Eng. (2019). doi:10.1007/s10439-019-02209-0.

[65] X. Hua, P. Liu, Y.-H. Gao, K.-B. Tan, L.-N. Zhou, Z. Liu, X. Li, S.-W. Zhou, Y.-J. Gao, Construction of thrombus-targeted microbubbles carrying tissue plasminogen activator and their in vitro thrombolysis efficacy: a primary research, J. Thromb. Thrombolysis. 30 (2010) 29–35. doi:10.1007/s11239-010-0450-z.

[66] X. Hua, L. Zhou, P. Liu, Y. He, K. Tan, Q. Chen, Y. Gao, Y. Gao, In vivo thrombolysis with targeted microbubbles loading tissue plasminogen activator in a rabbit femoral artery thrombus model, J. Thromb. Thrombolysis. 38 (2014) 57–64. doi:10.1007/s11239-014-1071-8.

[67] J.H. Foley, B.L. Walton, M.M. Aleman, A.M. O'Byrne, V. Lei, M. Harrasser, K.A. Foley, A.S. Wolberg, E.M. Conway, Complement Activation in Arterial and Venous Thrombosis is Mediated by Plasmin, EBioMedicine. 5 (2016) 175–182. doi:https://doi.org/10.1016/j.ebiom.2016.02.011.

[68] M.M. Aleman, B.L. Walton, J.R. Byrnes, J.G. Wang, M.J. Heisler, K.R. Machlus, B.C. Cooley, A.S. Wolberg, Elevated prothrombin promotes venous, but not arterial, thrombosis in mice, Arterioscler. Thromb. Vasc. Biol. 33 (2013) 1829–1836. doi:10.1161/ATVBAHA.113.301607.

[69] K.R. Machlus, F.-C. Lin, A.S. Wolberg, Procoagulant activity induced by vascular injury determines contribution of elevated factor VIII to thrombosis and thrombus stability in mice, Blood. 118 (2011) 3960–3968. doi:10.1182/blood-2011-06-362814.

[70] B.L. Walton, M. Lehmann, T. Skorczewski, L.A. Holle, J.D. Beckman, J.A. Cribb, M.J. Mooberry, A.R. Wufsus, B.C. Cooley, J.W. Homeister, R. Pawlinski, M.R. Falvo, N.S. Key, A.L. Fogelson, K.B. Neeves, A.S. Wolberg, Elevated hematocrit enhances platelet accumulation following vascular injury, Blood. 129 (2017) 2537–2546. doi:10.1182/blood-2016-10-746479.

[71] R. Marchi, B.L. Walton, C.S. McGary, F.-C. Lin, A.D. Ma, R. Pawlinski, N. Mackman, R.A. Campbell, J. Di Paola, A.S. Wolberg, Dysregulated coagulation associated with hypofibrinogenaemia and plasma hypercoagulability: Implications for identifying coagulopathic mechanisms in humans, Thromb Haemost. 108 (2012) 516–526. doi:10.1160/TH12-05-0355.

Microvascular Therapeutics, LLC

October 1, 2020

Paul Dayton, PhD

William R. Kenan Jr. Distinguished Professor
Joint Department of Biomedical Engineering
UNC Chapel Hill

Mary Ellen Jones 9018A
109 Mason Farm Road
Chapel Hill, NC 27599

Dear Dr. Dayton,

As Executive Chairman of Microvascular Therapeutics, I am very enthusiastic about your project "Novel advances in stroke and arterial thrombus lysis through image guided cavitation with clot penetrating nanodroplets." Your preliminary data recently published in Ultrasound in Medicine and Biology and presented at the 2020 IEEE Ultrasonics Symposium has demonstrated that low-boiling point phase change nanodroplets mediate a highly novel mechanism of clot disruption, cavitation internally to the clot. This suggests a way to overcome prior challenges faced by microbubble mediated sonothrombolysis, as microbubbles have been challenged to penetrate the dense fibrin matrix of older clots. If this technique continues to show promise as suggested in preliminary studies, it could have a major impact in our ability to treat thrombosis non-invasively. As you know, Microvascular Therapeutics is very interested in clinical translation and commercialization of low-boiling point phase change nanodroplets for both imaging and therapeutic applications, and we have licensed your nanodroplet patents with this intent. The potential use of nanodroplet enhanced sonothrombolysis as a novel treatment for arterial thrombus dissolution is of great interest to us. Microvascular Therapeutics now has capability for large-scale manufacture of GMP nanodroplets, and we would be an eager collaborator for your future studies in large animals and humans. We wish you the best of luck on this exciting and potentially highly impactful proposal.

Sincerely,

Evan Unger

Evan Unger, MD, FACR
Executive Chairman
Microvascular Therapeutics
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Tucson, AZ 85715
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October 8, 2020

Re: Sonothrombolysis project

Dear Paul,

SonoVol is enthusiastic that you will be using our Vega imaging platform for your project "*Novel advances in stroke and arterial thrombus lysis through image guided cavitation with clot penetrating nanodroplets*." We understand that a crucial part of this research is delivering the low-frequency ultrasound directly to the clot region in the mouse carotid artery or saphenous vein. As you know, the focused ultrasound (FUS) enabled Vega, or TherUS, enables image-guided FUS and is specifically designed for such an application. It is well suited to address the needs of your project.

Our high resolution 35 MHz volumetric scanning system captures whole-body data of mice in 3D, and allows registration between the imaging transducer and the multi-axis focusable high intensity focused ultrasound (HIFU) transducer. This enables the user to sonicate a target with FUS with extremely high precision. Our benchmark studies have shown that we can produce HIFU lesions with repeatability errors <0.5 mm when hitting the same target over multiple trials and with multiple users. I understand that as part of your study, you will be evaluating the effect of frequency and other acoustic parameters in sonothrombolysis. As the TherUS system typically utilizes a 1 MHz transducer, we want to assure you that we have flexibility to accommodate your treatment ranges. SonoVol will be willing to swap out the 1MHz with a 500 kHz, 700 kHz, or 2 MHz transducer as you desire for your studies. The HIFU transducer is completely programmable, and you already have full control over peak pressure, pulse length, and duty cycle for your studies, so operating the system at low power ranges is completely possible as well if you wish.

Thanks for your interest in continuing to use the Vega system for your studies, and we look forward to continuing to work with you.

Regards,

Ryan Gessner, Ph.D.
CEO
SonoVol, Inc.

AUTHENTICATION OF KEY BIOLOGICAL AND/OR CHEMICAL RESOURCES

The proposed work will use mouse models of thrombosis and thrombolysis.

ELISAs and Antibody Specificity: An ELISA kit for fibrin degradation products (D-dimer) will be purchased commercially. We will include plasma samples from mice with genetic deficiency of fibrinogen as a control for specificity.

Mice: Wild-type C57BL/6 mice will be purchased from Jackson Laboratories.

Chemicals: Chemicals will be purchased commercially, and will be of the highest grade available.